

# An Experimental Study of the Fluid-Structure Interactions of Water Entry of Compliant Structures

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## **ABSTRACT**

Water entry of compliant structures is a major area of interest within different fields of engineering. In the case of highly flexible panels, an application for this topic is on drag reduction due to shape reconfiguration of panels near the free surface to support further development of undulatory propulsors. Moreover, it has been an important concept in the study of the slamming of small high-speed craft with flexible bottom structures, such as those made of composites. In this work, this fluid-structure interaction problem is experimentally investigated in different stages. In Stage I, free-falling water entry experiments are conducted on wedges that have bottom panels with different flexural rigidities. Kinematics, hydrodynamics, spray root propagation, and structural response of the model are measured during the experiments. Results are interpreted to evaluate the effect of flexural rigidity on the slamming characteristics. The comparison between the rigid and flexible wedges shows that the evolution of the spray root on a flexible wedge is influenced due to fluid-structure interaction. In Stage II, a hybrid approach is proposed that incorporates spray root measurements with the existing analytical models in order to estimate the hydrodynamic loads in water entry of wedges with different boundary conditions. The validity of this approach is evaluated using a case study of a flexible wedge drop experiment. The results of this analysis show that the proposed approach can reasonably predict the wedge kinematics and hydrodynamic pressure due to impact. Future components of this study will further develop this tool to be used for highly-flexible structures, where it is not easy to install traditional

pressure sensors. Stage III of this work is on analysis of a tow-tank test of a rigid composite planing-hull model performed at the U.S. Naval Academy. Experiments conducted in regular waves were examined in terms of their kinematics and pressure loads. The goal of this analysis is to begin planning of the towing-tank tests that will be conducted at the VT Advanced Towing Tank Facility. These future VT experiments will combine the flexible composite panel with the hull form and motions, which are analyzed in the tow-tank study to investigate the fluid-structure interaction in the slamming of a flexible-planing hull. In stage IV, The findings of experimental investigations on wedge water entry are utilized in a 2D+t method to predict the hydrodynamics and motions of a prismatic planing craft. In this approach, the hydrodynamic loading on each V-type section of the vessel is calculated employing wedge water entry experiments (Stage I) and existing theoretical models (Stage II). A modified strip theory, also known as 2D+t, is then implemented to use these data and solve for the hydrodynamics and motion of the high-speed craft in calm water. Results show a good agreement with that of Savitsky prediction method and existing towing tank measurements.

# **An Experimental Study of the Fluid-Structure Interactions of Water Entry of Compliant Structures**

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## **GENERAL AUDIENCE ABSTRACT**

Water entry of compliant structures is a major area of interest within different fields of engineering. One application is to study the motion of highly-flexible plates near the free-surface. This is inspired by the manta rays that change the stiffness of their flapping fins during swimming in the ocean in order to have a more smooth motion. Another application is related to small high-speed craft that repeatedly become airborne and impact the water surface in waves. These individual impacts, called slamming, can adversely influence the maneuverability, cause failure to the structure or injure the crew on board. Thus, it is crucial to study and understand this phenomenon in order to mitigate its negative effects. The problem becomes more challenging when the vessel structure, such as those made of composites, can endure some deflections in order to dampen the slams. The interaction between this deflection and the water impact can deviate the slamming characteristics from the traditional theoretical predictions. Therefore, more investigation is needed to study this fluid-structure interaction. In this work, this problem is experimentally investigated in different stages. In Stage I, water entry of a wedge is analyzed, where the wedge represents a section of the high-speed craft. Different bottom panels, including a composite panel, are studied and results are compared in order to understand the effects of deflection of the panel on the slamming characteristics. In Stage II, a new approach is proposed to estimate the hydrodynamic loading in water entry of objects with arbitrary shapes. This approach is developed to estimate the slamming loads during the impact without using any sensors.

In fact, the high-speed videos of the water contact around the object during the impact is used to predict the slamming characteristics on the model. This technique can be so useful for the water entry of the objects that have very thin panels, which restrict mounting the pressure sensors. In Stage III, an analysis is conducted on the tow-tank tests of a composite planing-hull model. In these experiments, a small model is tested in regular waves and slamming is examined in terms of the kinematics and pressure loads on the model. The future experiments in VT Advanced Towing Tank will combine the flexible composite panel with the hull form and motions, which are analyzed in the tow-tank study to evaluate the fluid-structure interaction in the slamming of a flexible-planing hull. In Stage IV, the findings of experimental investigations on wedge water entry are utilized to predict the hydrodynamics and motions of a high-speed craft. In this approach, the hydrodynamic loading on each V-type cross section of the vessel is estimated employing wedge water entry measurements (Stage I) and existing theoretical models (Stage II). The calculated hydrodynamic loading on the vessel sections are then used to solve for the hydrodynamics and motion of the entire vessel.

*Dedicated to my mother*

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# Chapter 1

## Introduction

Water entry has different applications in engineering. The entry of birds into the water, shape reconfiguration of flapping fins during swimming of manta rays, the impact of sea-planes with the water surface upon landing, recovering space capsules, slamming of marine vessels, wave impacts onto offshore structures and the impact in the water turbine blades are some examples of this phenomenon.

Recently, attention has been drawn to some aquatic creatures, which instinctively change the shape and stiffness of the fins to reduce the drag or enhance the maneuverability during the swimming ([1, 2]). As an instance, manta rays alter the surrounding flow field with shape reconfiguration of their fins in order to reduce the drag during the motion. To further study this phenomena, the scaling laws should be developed to explain the reduction in drag coefficient due to the reconfiguration of the plate in the proximity of the free surface. This can be evaluated by studying flow field changes due to the shape reconfiguration of a highly flexible plate near an air-water interface during water entry. Inspired by nature, this shape reconfiguration could be actively implemented to reduce the drag on undulatory propulsors and vessels.

Regarding the marine applications, nowadays, high-speed planing craft are widely used in commercial, recreational, and naval applications. As the growth in utilization of these vessels is observed, naval architects strive to improve the overall performance while the safety metrics are adequately maintained. One of the major concerns that challenges both the performance

and structural strength of small high-speed craft is hull slamming. Once the vessel is subject to incoming waves, the hull repeatedly becomes airborne and then impacts the water surface. These slams cause operators to reduce the speed, and the maneuverability of the vessel is also influenced. Additionally, slamming can lead to the serious injury of sailors in rough sea conditions. Severe motion of the vessel due to impact may also adversely affect the operability of equipment on board, meaning that autonomous vessels are still vulnerable to these types of loading events. As a result, it is crucial to study and understand the slamming in high-speed craft in order to mitigate its negative effects.

In recent years, there has been increased use of advanced materials such as composites in the construction of high-speed planing craft. The current design criteria for design of high-speed planing craft are mostly based on the models that are established for the rigid structures. It has been shown that the slamming characteristics of the panels with lower flexural rigidity could be tangibly different than those of a rigid panel [3, 4]. Therefore, designing the recent composite high-speed craft (that could have flexible bottom structures) based on the existing design criteria might not lead to an optimum design. As a result, efforts have been made to update the design criteria for such vessels so that the flexibility effects could also be taken into account. To this end, computational, analytical and experimental approaches have been conducted by many researchers, yet there are some challenges in understanding the behavior of these vessels against slamming when structural response becomes nonnegligible.

The slamming of high-speed craft is sometimes reduced to the study of a wedge that vertically enters calm water. In this case, the wedge represents a section of the high-speed craft that vertically enters the water after becoming partially airborne. There are some other applications for wedge water entry which make this problem interesting for scientists. Predicting hydrodynamics of high-speed craft using the 2D+t theory is an example of such applications (Zarnick (1978) [5], Faltinsen (2005) [6]).

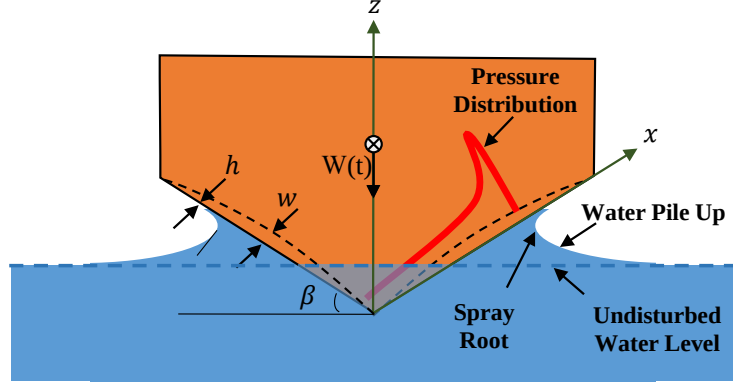


Figure 1.1: A schematic depicting the vertical water entry of a flexible wedge.

As a wedge model with the deadrise angle  $\beta$ , passes through a still-water surface, the resulting disturbance forms water pile-up and a spray jet, as shown in Fig. 4.1. The intersection between the spray jet and water pile-up is called the spray-root. The peak pressure lags behind the location of the spray-root, and if the velocity of the wedge is constant, the distance between the peak pressure and the zero pressure point is constant. When the velocity is nonconstant, the distance between the zero pressure and peak pressure scales with the instantaneous velocity (Vorus (1996) [7] and Judge et al. (2004) [8]). For water entry of a wedge with a deformable bottom, the hydrodynamic loading is also dependent on the structural response of the wedge because the deformed surface alters the fluid dynamics. Because of this dependence, the flexible-wedge-drop problem becomes more challenging to predict with classical hydrodynamic models. This challenge has drawn the attentions to the recent experimental studies on fluid-structure interaction of flexible panels.

## 1.1 Literature Review

The problem was first investigated in the design of seaplane structures, which were aimed to endure the impact loads upon the landing on the water surface. The first theoretical solution

for wedge water entry was provided by von Karman (1929) [9]. This model was developed based on the momentum theory and the water-added mass. A flat plate approximation was considered in this model to estimate the maximal pressure on the hull of landing seaplanes. Neglecting the water surface elevation and gravity, this model underestimates the impact loads, especially for the small deadrise angles.

Wagner (1932) [10] employed asymptotic theory to develop the von Karman model. With this assumption, the 2-D wedge-water entry problem is simplified into dropping an expanding flat plate. The length of the flat plate is defined as the horizontal distance between the water-body intersection (also known as spray root, see Fig. 4.1) at both sides of the wedge. Assuming small deadrise angle and taking into account the water pile-up, the problem is solved for the outer domain, which is the area outside the jet (over the length of the flat plate). Wagner model predicts reasonably better impact pressure compared to the von Karman theory. However, there exists a singularity at the spray root at each time step. Wagner model does not solve for the pressure data at and after the spray root (inner domain).

Further developed asymptotic models such as Generalized Wagner Model [11], Original and Modified Logvinovich models [12, 13] were then introduced by implementing different assumptions for handling the hydrodynamic terms in the governing equations. Even though these models result in different levels of accuracy, the solution was still singular at the location of spray root. Several attempts were then made by researchers to handle the singularity problem at the location of spray root [14, 15, 16]. These studies considered a piece-wise solution for the outer domain (outside the jet) and the inner domain (inside the jet). These two formulations are linked together by matching the jet thickness on both sides of the singularity. Thus, the hydrodynamic pressure at the spray root and in the jet area (inner domain) could also be estimated with the use of these models. Some of the models mentioned here, that are all based on the linearized hydrodynamic equations, will be elaborated more

in Chapter 3.

Another approach to resolve the singularity problem was sought in accounting for the nonlinear hydrodynamic terms. Vorus (1996) employed a flat cylinder theory to avoid the singularity by compromising between direct numerical inversion of the governing equations and the simplified asymptotic theories [7]. It was achieved by retaining a single solution field and hydrodynamic nonlinearity of the governing equations. Vorus model could also take into account the separation of the flow near the intersection points. Xu (1998) showed that the numerical results obtained within the Vorus model have a good agreement with the experimental measurements [17]. He extended this theory to solve for asymmetric vessel impact. Another implementation of the nonlinear hydrodynamics was conducted by Zhao et al. [11, 18] that uses the nonlinear Bernoulli equation but with a slightly different approach than that proposed by Vorus.

Several experimental investigations were also carried out to study the water entry of wedge-shaped bodies. Bisplinghoff and Doherty (1952) conducted a series of drop tests for wedges with different deadrise angles [19]. They also developed a mathematical model based on Wagner model to predict the acceleration and the water pile-up during the entry. Accelerations and water pile-up were measured and compared with their proposed theory. The comparison showed that the theory overestimates the pile-up water surface.

Chuang (1966) carried out several experiments on the wedges with low deadrise angles, in particular less than  $15^\circ$  [20]. The pressure was measured at and away from the keel using pressure gages. These tests were intended to provide a set of empirical relations for estimating the maximum pressure on the rigid body slamming of the wedges. He also concluded that for the wedges with the deadrise angle of  $3^\circ$  and lower, the effect of trapped air needs to be taken into account. Chuang further developed his study and presented his modified correction factors to increase the accuracy of the Wagner model in predicting the maximum

exerted hydrodynamic pressure [21, 22].

Lin and Ho (1994) investigated the effect of water depth on the maximum hydrodynamic pressure during the wedge-water entry [23]. They repeated the experiments for different water depths and compared the recorded maximum pressure with their numerical approach. Several wedges with deadrise angles ranging from  $6^\circ$  to  $60^\circ$  were investigated. They showed that the shallow water can increase the maximum measured peak pressure for a certain wedge. In fact, with the decrease in the water depth from 0.2 m, the maximum measured pressure was drastically increasing. Therefore, they concluded that any water depth more than 0.2 m could be considered as the deep water for wedge-water entry.

Greenhow and Lin (1983) conducted a set of experiments to visualize the free-surface elevation in wedge-water entry experiments [24]. They employed a high-speed camera to capture the free-surface for some small wedges with different deadrise angles. To this date, the photography taken at these experiments are among the limited resources for visualization of the free-surface, that is used by other researchers for validation purposes.

The entry of wedges with small deadrise angles was more analyzed by Okada and Sumi (2000) [25]. They investigated free-falling wedges with deadrise angles smaller than  $4^\circ$  to study the effect of the air entrapment on the hydrodynamic impact pressure. Their analysis revealed that the presence of the air cushion underneath the wedge during the impact may affect the magnitude and pattern of the pressure distribution over the body. They also proposed an empirical relation for evaluating maximum peak pressure for flat plate entry.

Engle and Lewis (2003) conducted a series of experiments on the symmetric entry of wedges to compare and validate the accuracy of several numerical methods [26]. They studied entry of wedges with  $10^\circ$  and  $20^\circ$  deadrise angles, which were dropped from different heights. The comparison showed that their experimental results were in good agreement with the Chuang

(1970) [21] empirical prediction relations.

Wu et al. (2004) experimentally studied entry of wedges with  $20^\circ$  and  $45^\circ$  deadrise angles to validate their proposed numerical boundary element method [27]. They reported the experimental instantaneous acceleration and compared them with those of their numerical approach. Although, they found an acceptable agreement between the experiment and numerical simulation for the wedge with  $45^\circ$  deadrise angle, some deviation were discernible for  $20^\circ$  deadrise angle wedge.

Yettou et al. (2006) conducted a parametric study on different wedges falling into water [28]. They studied the effects of drop height, deadrise angle and the wedge mass on the impact pressure. Considering the influence on the amount of pressure coefficient and its distribution, their experiments showed that the drop height and the wedge weight are less important than the wedge deadrise angle. Lewis et al. (2010) performed some experiments on entry of a  $25^\circ$  deadrise angle [29]. Pressure was measured by three transducers at six locations across the wedge surface with 10 kHz sampling rate. The repeatability of the tests and the deceleration time history were also reported in their research.

There are also many studies in which the wedge speed is almost constant. Tveitnes et al. (2008) investigated entry and exit of wedge sections at near constant velocity [30]. Entry force and the velocity of penetration were measured. Wetting factor, added mass, chine immersion and flow momentum drag coefficient were well studied in this research. They showed that the measured pressure in a constant velocity entry is much higher than what recorded in the same impact velocity of a free falling impact. Moreover, some experimental researches exist in which the wedges are not rigid or the entry is not symmetric; however, these cases are out of the scope of this paper. Seddon and Moatamedi (2006) [31] and Abrate (2011) [32] did a comprehensive review on this topic.

Panciroli et al. (2013) carried on experiments on a compliant wedge to evaluate the hydroelastic effects of flexible panels during water entry [33]. In their experiments, the plates were cantilevered from the keel and free at the chine. They implemented particle image velocimetry (PIV) and considered different drop heights to assess the effects of plate flexural rigidity on hydrodynamic loading, location of maximum and minimum pressure as well as deflection of the tip of the panels. They collected the experimental results to compare with their semi-analytical model. Panciroli et al. (2015) used particle image velocimetry and high-speed video recording to assess the effects of flexural rigidity on hydrodynamic loading, location of maximum and minimum pressure, as well as deflection of the tip of the open wedge [34]. They compared their experimental results with their semi-analytical model. This model incorporates the experimental loading with the Kirchhoff-Love linear plate model to study the multimodal vibration.

Shams et al. (2017) also experimentally investigated water entry and exit of a flexible wedge [35]. They used PIV to compute the velocity and pressure fields around the wedge. Free surface elevation was also studied in their experiments. Using experimental results for kinematics, hydrodynamic loading, and water pile-up, they validated parts of their semi-analytical model for slamming, which they used to numerically investigate the role of flexibility on the pile-up, hydrodynamic loading, and impact force.

The interaction between fluid pressure and structural response of a flexible wedge was experimentally studied by Ren et al. (2018) [36]. They implemented Stereoscopic-Digital Image Correlation (S-DIC) to measure the displacement and out-of-plane deflection of the wedge during the impact. They also proposed a semi-theoretical model that used the Vorus model as the loading condition on a dynamic, linear finite element plate model. Predicted pressure and deflection data were compared with experiments to evaluate the effect of hydroelasticity in wedge water entry. They concluded that rigid wedge based water entry theories such as

Wagner cannot be one-way coupled to a structural solver for flexible wedges since they may over-predict hydrodynamic loadings on flexible structures. When using the experimentally measured pressures in the structural solver, the deformation response more closely matched the experimental observations (Ren et al. (2019) [37]).

Duan et al. experimentally studied the slamming pressure in water entry of rigid and elastic wedges with varying deadrise angles from  $0^\circ$  to  $45^\circ$  [38]. They investigated the spacial variation of the slamming pressure during these tests to evaluate the propagation of the slamming pressure. Duen et al. showed that the ratio between experimental speed of pressure propagation and Wagner's theoretical speed is 0.8. They also reported that the flexural rigidity of the panel influences the slamming pressure for small deadrise angles in that the duration of the slamming pressure increases with the decrease in the flexural rigidity. As far as the pressure coefficient is concerned, they showed that the pressure coefficient might increase or decrease depending on the ratio between pressure duration and natural period of the structure.

Although all the numerical and theoretical models utilize the water-body intersection as the key element of the analysis, there are very limited studies in which the spray root propagation is adequately investigated due to challenges in numerical modeling and experimental measurements. This was the first motivation for the authors to conduct some experiments and visualize the spray root propagation. Having experimental measurements for the spray root, current analytical formulations for the propagation of spray root can be studied and validated. Additionally, in the fluid-structure interaction problem of a flexible plate, the spray root represents the crucial role of in the fluid loading on the plate. Thus, it is beneficial to study the interaction between the spray root, kinematics and structural response of the model in order to shed light on understanding the physics of slamming of the flexible plates. To the knowledge of the author at the time of writing this dissertation, there are not

enough studies, in which this interaction between these key elements is sufficiently investigated. In this body of work, the effect of flexural rigidity on the propagation of spray root, hydrodynamics and structural response of the flexible plates is experimentally investigated. To this goal, an experimental setup is designed and manufactured to study water entry of different models. Different models with different flexural rigidity are considered to study the effect of flexural rigidity on the slamming characteristics. These results fill a gap in current experimental studies of the water entry of flexible structures. The outcome can also be used to validate computational studies, enhance theoretical models or provide further guidance for future experiments.

For another thing, it was discussed that the experimental investigation is a powerful means to evaluate the slamming characteristics of flexible structures. As far as these experiments are concerned, the hydrodynamic pressure is often measured using point pressure sensors. Although these pressure sensors function very well, there is a limit in their use for highly flexible panels. In fact, mounting pressure sensors on the panel would add to the stiffness of the plate as the sensors act like stiffeners. This would change and reinforce the mechanical properties of the investigating plate. This challenge was another motivation for the authors to look for alternative ways for measuring hydrodynamic pressure in slamming of highly flexible structures. In this thesis, a novel hybrid approach is introduced that employs experimental spray root information along with the existing theoretical models to estimate the pressure distribution during the experiments without using any pressure sensor.

## 1.2 Dissertation Structure

This dissertation is structured in the “manuscript” format (“multi-paper” format), in which each of the main chapters represent a peer-reviewed journal paper. Regarding these papers,

Chapter 2 was a peer-reviewed conference paper at SNAME Maritime Convention (SMC) that has been slightly modified and has been submitted to the Journal of Ship Research. At the time of this dissertation, this paper is under review at the Journal of Ship Research. Chapter 3 is a draft journal paper that is ready for submission to the Applied Ocean Research. Chapter 4 is a paper that has been published in the Proceedings of the Symposium on Naval Hydrodynamics, which has a peer review process. Chapter 5 is a paper that is submitted to the International conference on FAST Sea Technology.

In this chapter, the slamming problem of high-speed craft was introduced and the key elements in this problem were elaborated. A literature review containing the substantial previous efforts was reported to outline the available gap in the field and the main motivation of the author to perform this study. It should be noted that two thirds of the dissertation is funded by Office of Naval Research and a third is funded by National Science Foundation.

In Chapter 2, the free-fall water entry of a flexible wedge with a 20-degree deadrise angle is experimentally investigated with a goal of understanding the role of structural deformation on the spray root propagation. As the wedge penetrates the calm water surface, kinematics and hydrodynamic loading are measured using appropriate transducers. High-speed video recording is conducted 1) to measure the deflection on the bottom of the wedge using Stereoscopic-digital Image Correlation technique and 2) to visualize the spray root propagation on the wedge. Given the fully synchronized measurements, the instantaneous correlation between impact loading, structural response and spray root propagation is carefully investigated. Experiments are carried out on a rigid wedge of similar dimensions and mass for comparison. To evaluate the effect of impact velocity, experiments are repeated from different drop heights. Finally, the experimental results are compared with an in-house code that combines Wagner's theoretical model with a linear finite element solver.

In Chapter 3, a hybrid approach to estimate hydrodynamic pressures by taking spray root

position measurements and existing theoretical models is introduced. This paper shows the validity of this approach using a case study of a flexible wedge drop experiment. This approach is aimed to be developed as a powerful tool to estimate the pressure on the flexible panels in the design process. To this goal, the analytical models are re-derived to account for the loss of velocity so that they are comparable with the free-fall experimental results. Future components of this work will further develop this tool to be used for highly-flexible structures and also in-situ pressure measurements at sea. Results of this chapter show that the proposed approach can reasonably predict the hydrodynamic pressure distribution on the bottom panel of the wedge. The hydrodynamic force and kinematics of the model can also be adequately estimated using this approach. This approach will also be used to study the flow field changes due to shape reconfiguration of highly-flexible plates near the free-surface in order to reduce the drag on undulatory propulsors and vessels.

The slamming of composite high-speed craft is experimentally investigated in Chapter 4. This study is conducted in two phases. In the first phase, nearly free-falling water entry experiments are conducted on a wedge with flexible bottom panels. Kinematics, hydrodynamics, structural deformation, and spray-root propagation on the bottom of the wedge are measured and analyzed. The models include a wedge with a composite bottom panel. The second phase of this paper is on analysis of a tow-tank test of a rigid composite planing-hull model performed at the U.S. Naval Academy. Experiments conducted in regular waves were examined in terms of their kinematics and pressure loads. The goal of this analysis is to begin planning of non-traditional towing-tank tests that will be conducted at the VT Advanced Towing Tank Facility. These future VT experiments, Phase III, will combine the flexible panel of Phase I with the hull form and motions, which are analyzed in Phase II to study the fluid-structure interaction in the slamming of a flexible-planing hull.

In Chapter 5, a modified strip theory, also known as 2D+t approach, is introduced to predict

the hydrodynamics and motion of a planing craft in calm water. In this Chapter, the experimental measurements of Chapter 2 along with the novel approach discussed in Chapter 3 are utilized to estimate the hydrodynamic loading on the V-type cross sections of a prismatic planing hull. Results are then used as the input in a 2D+t approach to predict the resistance, wetted length and trim of a prismatic planing hull in calm water. The analysis is repeated for different cruising speeds to study the effect of Froude number on the results. To verify the proposed approach, outcomes are compared with the well-known Savitsky prediction method and available towing tank experiments.

Finally, the last chapter concludes this dissertation by highlighting the key findings and further perspectives for future work.

## 1.3 Contribution to Co-Authored Papers

As mentioned earlier, three journal papers were prepared in collaboration with other co-authors. The author contributions for each of the following papers are listed as follows:

1. Chapter 2 (Authors: M. Javad Javaherian, Zhongshu Ren, and Christine Gilbert).
  - Took major part in planning the paper.
  - Took part in setting up the experiment.
  - Took part in conducting the experiment.
  - Took major part in setting up the elements of the setup regarding the visualization study.
  - Conducted the image processing on the videos and related analysis of the paper.
  - Wrote 90% of the paper.

2. Chapter 3 (Authors: M. Javad Javaherian and Christine Gilbert).
  - Took major part in planning the paper.
  - Made the analytical re-derivations.
  - Performed the analysis.
  - Wrote 90% of the paper.
3. Chapter 4 (Authors: M. Javad Javaherian, Zhongshu Ren, Carolyn Q. Judge and Christine Gilbert).
  - Took major part in planning the paper.
  - Took part in setting up the phase I experiments: rigid-body kinematic motions, hydrodynamic loading, and structural response.
  - Took major part in setting up the elements of the setup regarding the visualization study in phase I experiments.
  - Took part in conducting the phase I experiments.
  - Conducted the visualization analysis of phase I.
  - Conducted the analysis on the phase II experiments.
  - Wrote 90% of the paper.
4. Chapter 5 (Authors: M. Javad Javaherian, Richard Royce, Raju Datla and Christine Gilbert).
  - Took major part in planning the paper.
  - Conducted the analysis of the proposed 2D+t approach.
  - Implemented the Savitsky prediction method.
  - Performed the analysis and comparison.

- Wrote 95% of the paper.

## Chapter 2

# Flow Visualization, Hydrodynamics, and Structural Response of a Flexible Wedge in Water Entry Experiments

(Reprinted with permission from “Flow Visualization, Hydrodynamics, and Structural Response of a Flexible Wedge in Water Entry Experiments” by M. Javad Javaherian, Zhongshu Ren, and Christine Gilbert, 2020. *Journal of Ship Research*, Under review at the time of the dissertation.)

M. Javad Javaherian, Zhongshu Ren, and Christine Gilbert

(ABSTRACT) In this paper, a free-falling flexible wedge into calm water is experimentally studied to understand the relationship between the spray root, peak pressure, and structural response. High-speed cameras are employed to record the spray root propagation, while hydrodynamic loading is measured with an array of pressure transducers. Stereoscopic Digital Image Correlation (S-DIC) is used to measure deflection on the bottom of the wedge during the impact. Experiments are conducted from different drop heights to study the effect of impact velocity. Results are interpreted in light of an experimental data set of a rigid wedge of comparable dimensions. The comparison between the rigid and flexible wedges shows that due to fluid-structure interaction, the evolution of the spray root on a flexible wedge is

slightly delayed compared to the rigid one.

## 2.1 Introduction

Nowadays, high-speed planing craft are widely used in commercial, recreational, and naval applications. As the growth in utilization of these vessels is observed, naval architects strive to improve the overall performance while the safety metrics are adequately maintained. One of the major concerns that challenges both the performance and structural strength of small high-speed craft is hull slamming. Once the vessel is subject to incoming waves, the hull repeatedly becomes airborne and then impacts the water surface. These slams cause operators to reduce the speed, and the maneuverability of the vessel is also influenced. Additionally, slamming can lead to the serious injury of sailors in rough sea conditions. Severe motion of the vessel due to impact may also adversely affect the operability of equipment on board, meaning that autonomous vessels are still vulnerable to these types of loading events. As a result, it is crucial to study and understand the slamming in high-speed craft in order to mitigate its negative effects.

The slamming of high-speed craft is sometimes reduced to the study of a wedge that vertically enters calm water. In this case, the wedge represents a section of the high-speed craft that vertically enters the water after becoming partially airborne. There are some other applications for wedge water entry which make this problem interesting for scientists. Predicting hydrodynamics of high-speed craft using the 2D+T theory is an example of such applications (Faltinsen [6]).

The wedge water entry problem was first initiated for seaplane structures, which were designed to land on the water surface. The quest for knowledge on this topic was started with the research of Von Karman [9], whose mathematical model presented a good macroscopic

description of the problem. Wagner developed the von Karman model further by adding asymptotic theory [10]. The Wagner model evaluates a reasonable impact pressure for a wide range of deadrise angles. Due to the linear nature of the hydrodynamic term in the governing equation, there exists a singularity at the spray root. Further development of the Wagner model solved the non-linear hydrodynamic problem and are now utilized to study pressure distribution (Korobkin [13]), spray and jet formation (Vorus [7]), oblique and asymmetric impacts (Xu et al. [39], Judge et al. [8]) as well as hydroelasticity (Qin and Batra [40]). In these newer models, the singularity no longer exists because the full nonlinear hydrodynamic terms are retained in the governing equation.

As the wedge model, with the deadrise angle  $\beta$ , breaks through the undisturbed water level, the resulting disturbance initiates the water pile-up and spray root, as shown in Fig. 4.1. There is a direct relationship between the spray root propagation on the bottom of the wedge and peak pressure coefficient (Vorus [7] and Judge et al. [8]). When considering the water entry of a wedge with a deformable bottom, the hydrodynamic loading (i.e. the pressure distribution) is dependent on the structural response of the wedge. In other words, the impact loading causes the bottom of the wedge to deflect, which influences the fluid flow and therefore, the instantaneous pressure loading on the structure. Because of this dependence, the flexible wedge drop problem becomes more challenging to predict with models.

Among the experimental investigations on wedge water entry, Lewis et al. conducted some experiments on entry of a rigid wedge with a deadrise angle of 25 degrees [29]. Pressure was measured by three transducers at six locations across the wedge surface with a 10 kHz sampling rate. Lewis also tried to investigate the jet formation, its evolution and eventual breakdown to correlate it with the pressure distribution. Panciroli carried on experiments on a compliant wedge to evaluate the hydroelastic effects of flexible panels during water entry [33]. In their experiments, the plates were cantilevered from the keel and free at the

chine. Panciroli implemented particle image velocimetry (PIV) and considered different drop heights to assess the effects of flexibility on hydrodynamic loading, location of maximum and minimum pressure as well as deflection of the tip of the panels. They collected the experimental results to compare with their semi-analytical model. Shams et al. also experimentally investigated water entry and exit of a flexible wedge [35]. They used PIV to compute the velocity and pressure fields around the wedge. Free surface elevation was also studied in their experiments. Using experimental results for kinematics, hydrodynamic loading, and water pile-up, they validated parts of their semi-analytical model for slamming, which they used to numerically investigate the role of flexibility on the pile-up, hydrodynamic loading, and impact force.

The interaction between fluid pressure and structural response of a flexible wedge was experimentally studied by Ren et al. [36]. They implemented Stereoscopic-Digital Image Correlation (S-DIC) to measure the displacement and out-of-plane deflection of the wedge during the impact. They also proposed a semi-theoretical model that used the Vorus model as the loading condition on a dynamic, linear finite element plate model. Predicted pressure and deflection data were compared with experiments to evaluate the effect of hydroelasticity in wedge water entry. They concluded that rigid wedge based water entry theories such as Wagner cannot be one-way coupled to a structural solver for flexible wedges since they may over-predict hydrodynamic loadings on flexible structures. When using the experimentally measured pressures in the structural solver, the deformation response more closely matched the experimental observations (Ren et al. [37]).

In this paper, the free-fall water entry of a flexible wedge with a 20-degree deadrise angle is experimentally investigated with a goal of understanding the role of structural deformation on the spray root propagation. It is noted that due to the friction in the system, there is a deviation up to 4% from ideal free-fall acceleration. As the wedge penetrates the calm water

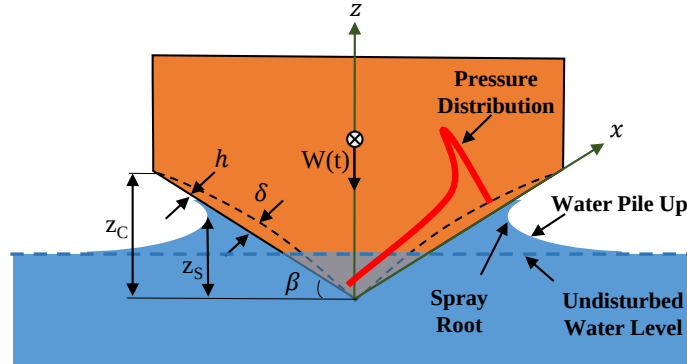


Figure 2.1: A schematic depicting the vertical water entry of a flexible wedge including water pile-up, spray root, pressure distribution, deflection and definitions of some variables.

surface, kinematics and hydrodynamic loading are measured using appropriate transducers, to be discussed in the following section. Two high-speed cameras are utilized to measure the deflection on the bottom of the wedge using S-DIC. Another high-speed camera is employed to record the spray root propagation on the wedge. Given the fully synchronized measurements, the instantaneous correlation between impact loading, structural response and spray root propagation is carefully investigated. Experiments are carried out on a rigid wedge of similar dimensions and mass for comparison. To evaluate the effect of impact velocity, experiments are repeated from different drop heights. Finally, the experimental results are compared with an in-house code that combines Wagner's theoretical model with a linear finite element solver.

## 2.2 Experiments

### 2.2.1 Experimental Setup

The experiments were conducted in the Hydroelasticity Laboratory housed in the Kevin T. Crofton Department of Aerospace and Ocean Engineering at Virginia Tech. Fig. 4.2 shows a

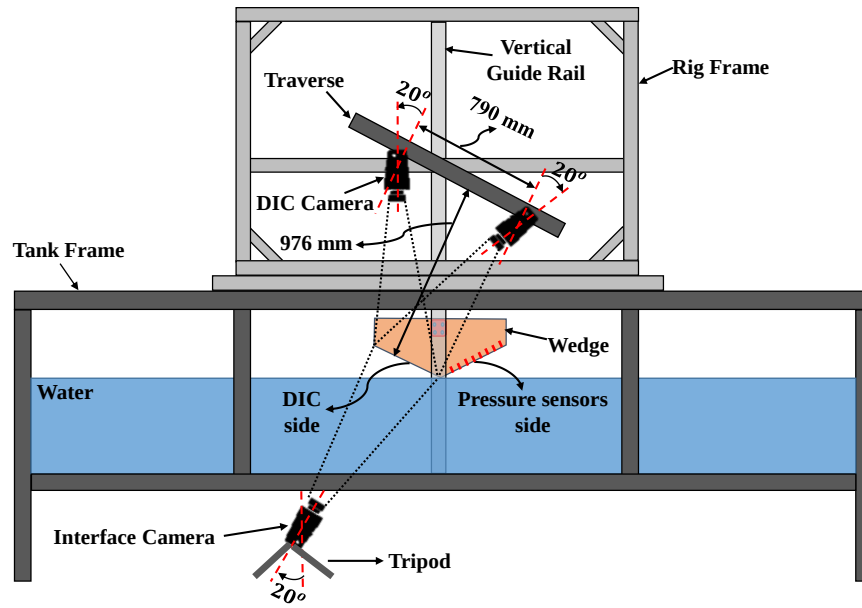


Figure 2.2: Schematic of the experimental setup showing reinforced transparent tank, rig frame, wedge and location of high-speed cameras. Two cameras are on the rig and used in the DIC measurements. The camera below the tank is used to take video of the spray root from below.

schematic of the experimental setup. The water tank is a reinforced, transparent tank made by acrylic sheets and is 4.4-m long, 2.4-m wide, and 1.2-m deep, with a nominal water depth of 0.83 m. The tank is raised off the ground by 0.7 m to allow for optical access to the tank bottom. A rig, composed of 80/20 T-slot aluminum, is mounted on top of the water tank to support the wedge model and measurement instruments. The rig includes two vertical guide rails through which the model can freely fall into the water with only a small amount of friction. Linear bearings are used on both guide rails to constrain the motion of the model in the vertical direction during the fall. Electromagnets are used as part of a quick release system to hold and release the wedge from arbitrary drop heights below 51 cm, where the drop heights are measured from the keel to the calm water surface.

In this study, one flexible and one rigid wedge model are examined. These models are manufactured from aluminum alloy 6061-T6, and are of comparable size, mass, and deadrise

angle. Having deadrise angles of 20 degrees, each wedge is 63.5 cm long, 57.27 cm wide, and 40.89 cm high. For the rigid wedge, all side and bottom plates have a thickness of 12.7 mm. This thickness ensures adequate stiffness for the rigid wedge. On the other hand, while the side plates on the flexible wedge are 12.7 mm thick, the bottom plates have a thickness of only 3.17 mm. Therefore, the flexible wedge has a lower structural rigidity than the rigid wedge. The flexible model has a keel stiffener, which is a 3.17 mm x 63.5 mm flat bar with the same length of the model. Due to the stiff plates welded to the flexible plate, the boundary conditions are expected to be fixed on all edges. In order to maintain comparable mass between the two wedges, additional weights are added to the flexible wedge to compensate for the change in volume of material and make both wedges identical in mass and equal to 40.6 kg.

Fig. 4.4 displays a schematic of the front view of the mid-plane section of the wedge. Pressure sensors are installed on the right side, whereas the S-DIC measurement is covered on the left side. The string potentiometer is attached to the keel at the center of the keel line. AC accelerometers are also mounted on top of the side plates. The top view of the wedge along with detailed dimensions are shown in Fig. 5.2. It is a rotated view compared to the front view, where the  $y$ -direction that runs along the keel is horizontal. The vertical red dashed line represents the characteristic line ( $x$ ), which is surrounded by S-DIC measurement area. The DC accelerometer,  $A_{DC}$ , is also located next to one of the AC accelerometers,  $A_1$ , along the chine on the top of Fig. 5.2. The DC accelerometer is implemented to record the free-fall acceleration, while the AC accelerometers are used to capture the impact acceleration.

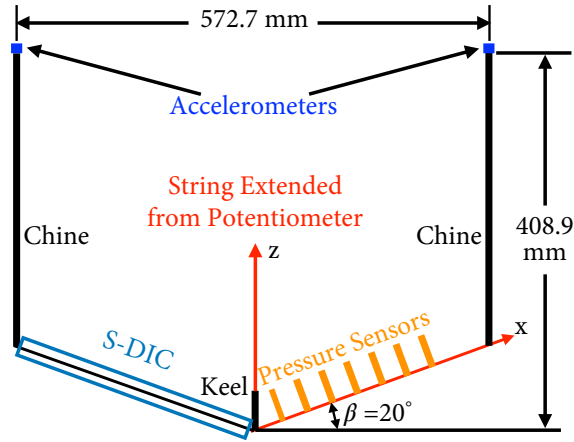


Figure 2.3: Front view of the flexible wedge showing the mid-plane section along with dimensions, mounted instruments and S-DIC measurement area (left side bottom).

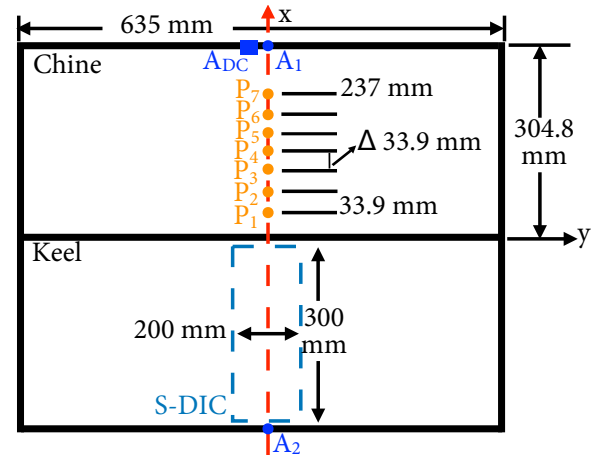


Figure 2.4: Top view of the flexible wedge and the dimensions along with details of mounted instruments and S-DIC measurement area (bottom middle).

### 2.2.2 Instruments

Seven PCB-CA102B18 pressure transducers are mounted on the wedge to measure the hydrodynamic loading due to the slam on the bottom of the wedge. The sensors measure the dynamic pressure with the resolution of 0.007 kPa under non-linearity of less than 1.0% Full Scale. This sensor is also treated with a thermal ablative coating, which makes it resistant against thermal shock in the measurement range of 344.7 kPa. The first sensor,  $P_1$ , is located 33.9 mm away from the keel along the characteristic line, as seen in Figs. 4.4 and 5.2. The other pressure sensors are installed at increments of 33.9 mm also along the characteristic line. These pressure sensors are used to record the instantaneous impact pressure with a sampling rate of 200 kHz and are powered, amplified, and conditioned using two 482C05 signal conditioners with 4 channels per unit. Raw measured data were filtered using a Low-pass Butterworth filter with a cutoff frequency of 2 kHz. The filtered data were accurately checked to ensure that the physical peak pressures are not lost. In fact, filtering should not influence the recorded data since the sampling rate was chosen to be high enough for the experiments (Zeraatgar et al. [41]).

Vertical impact acceleration of the wedge is measured using two miniature PCB-352A24 accelerometers. These accelerometers,  $A_1$  and  $A_2$ , are located on top of the chine at both sides of the wedge (Figs. 4.4 and 5.2). The Measurement range for these accelerometers is  $\pm 50$  g with the non-linearity of less than 1.0% Full Scale. A 482C05 signal conditioner amplifies the signal of these accelerometers. Moreover, a DC response accelerometer PCB-3741B122G is also utilized to record the low frequency acceleration caused by the free-fall. The DC sensor has a measurement range of  $\pm 2g$  with the non-linearity of less than 1.0% Full Scale. The sampling rate of all accelerometers is 2 kHz.

The position-time history of the wedge is recorded by an SGD-120-3 Resistive Linear Po-

sition String Potentiometer by Measurement Specialties. As shown in Fig. 4.4, the string of the potentiometer is connected to the keel of the wedge to track the location of the keel during experimental runs. This sensor has accuracy of 0.35% Full Scale with the range of measurement up to 305 cm.

### 2.2.3 Stereoscopic-Digital Image Correlation

The structural response of the bottom of the wedge is measured using the Stereoscopic-Digital Image Correlation (S-DIC) technique. This system consists of two high-speed Phantom VEO 710S cameras that are mounted on top of the rig (as shown in Fig. 4.2) to record overlapping images from the bottom of the wedge at a sampling rate of 2000 Hz. It is noted that the two cameras are stationary while the wedge is in motion. Two powerful GSVITEC LED lights assist the recording by illuminating the wedge bottom plating, which is coated by a high-density speckle pattern (as in [37]). Using a two dimensional cross correlation technique of the random speckle pattern from two mapped camera views, the displacement, full-field out-of-plane deflection, and strain on the bottom plate of the wedge can be measured and computed. These measurements are taken on the S-DIC area shown in Figs. 4.4 and 5.2 using LaVision DaVis 8.0 software. The setup is optimized so that the displacement accuracy of 0.02 pixel is achieved. A static test was also conducted for the wedge and the displacement noise floor was measured to be 0.01 mm.

### 2.2.4 Spray Root Visualization

As the wedge penetrates into the water, a spray root is formed on either side of the wedge bottom and propagates from the keel to the chine. The spray root location at a given time is highly correlated with the peak pressure coefficient, and therefore, is important to

the structural response of the wedge. As previously discussed, there has not been much experimental research on the spray root evolution, as it is difficult to measure. The studies that have been performed have been mostly focused on rigid wedges (Lewis et al. [29]). In order to measure the spray root evolution on a flexible wedge, another Phantom VEO 710S high-speed camera is located underneath the tank (as shown in Fig. 4.2) to visualize the interface and evolution of the spray root on the bottom of the wedge during the impact. Video is recorded at 2000 fps with resolution of 1280 x 800 pixels. The orientation of this camera is set so that it looks perpendicular at the bottom of the wedge. This configuration facilitates the image processing. Images from this camera synchronized with the data from other instruments and cameras assist to study the instantaneous correlation between spray root propagation, structural response, and hydrodynamics of the wedge.

### 2.2.5 Data Acquisition

Data Acquisition is conducted using a National Instruments PXIe-8821 Controller DAQ. All instruments are connected to this device through different modules, each of which acquires samples simultaneously. A pulse generator, Berkley Nucleonics 577 Digital Delay, is also used to send the trigger signal and synchronize all the instruments and cameras with the electromagnets. In fact, once the run button is pressed, the pulse generator cuts the power of electromagnets and triggers all instruments (through the DAQ) and cameras at the same time. It is worth mentioning that there is a very short delay between the trigger and the actual time in which the electromagnets release the wedge. Captured by an oscilloscope, this delay is also recorded and considered to define the precise release moment of the wedge. This guarantees an accurate synchronization of the system.

Table 2.1: Experimental Matrix

| Case# | Wedge    | Drop Height [cm] |
|-------|----------|------------------|
| 1     | Flexible | 7.9              |
| 2     |          | 25.4             |
| 3     |          | 50.8             |
| 4     | Rigid    | 7.9              |
| 5     |          | 25.4             |
| 6     |          | 50.8             |

### 2.2.6 Experimental Matrix

Table 5.1 shows the experimental matrix for the experiments on the rigid and flexible wedges. Three repeated runs are conducted for each experimental condition to ensure the appropriate repeatability of results. In each run, the wedge is lifted by a winch and is attached to the electromagnets. Once held by the electromagnets, the wedge is unhooked so that it is only held by the electromagnets. By the trigger signal from the pulse generator, the wedge is released while all the instruments and cameras acquire data simultaneously. After each run, the bottom of the wedge is dried carefully and the water surface is allowed to calm as recommended by Van Nuffel et al. [42].

## 2.3 Results

Following the experimental matrix reported in Table 5.1, experiments were carried out and results were recorded. In this section, results of kinematics, hydrodynamic pressure, structural response, and visualization tests will be presented and discussed. The wedge kinematics are also compared with the results of an in-house code that was developed to simulate a flexible wedge during the impact (Fisher et al. [43]).

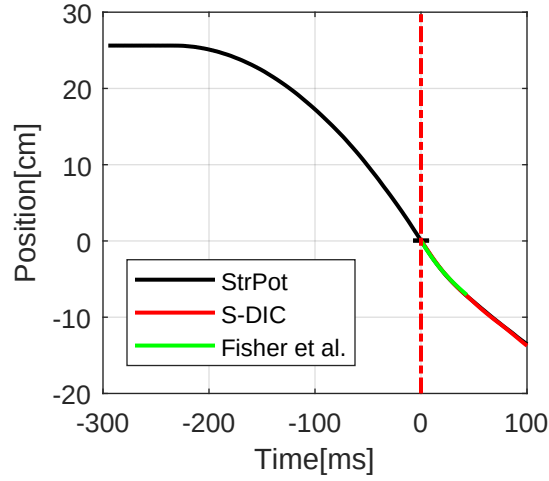


Figure 2.5: Mean position-time history of three repeated runs for a drop height of  $H = 25.4$  cm along with the corresponding position data as measured by S-DIC.

### 2.3.1 Kinematics and Hydrodynamics

Figs. 2.5 to 2.8, to be discussed in subsequent paragraphs, show kinematics and hydrodynamics of the flexible aluminum wedge, falling from height of 25.4 cm. In all figures, the time  $t=0$  shows the impact moment. Each curve represents the mean data of three repeated runs. It is noted that this is a representative test case and all others have similar qualitative characteristics and similar run-to-run variability.

The position-time history of the wedge measured by the potentiometer is shown in Fig. 2.5. Negative position data show displacements lower than the undisturbed water level. Results of the displacement of the wedge measured by the S-DIC method are also plotted in this figure, which shows very good agreement with the data from the potentiometer. The comparison also depicts the results of the in-house code, Fisher et al. [43], which predict the vertical displacement of the wedge very well.

Fig. 2.6 depicts the acceleration measured by the three accelerometers that are mounted on the chines. The wedge falls with the acceleration of nearly  $-g$ , which is standard gravita-

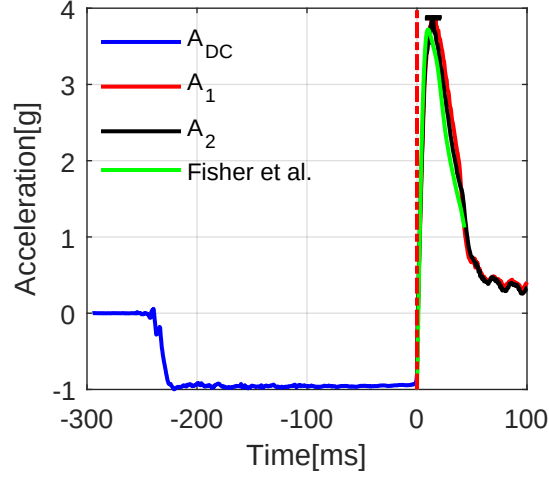


Figure 2.6: Mean acceleration-time history of three repeated runs for a drop height of  $H = 25.4$  cm measured by the three accelerometers.

tional acceleration. Once the impact occurs, the wedge experiences a large peak of positive acceleration (upward) in a very short period of time. The maximum acceleration is measured to be  $3.9\text{ g}$  and is recorded at  $t=14$  ms after impact for this experimental condition of a  $25.4$  cm drop height. The deviation between peaks of  $A_1$  and  $A_2$  is only  $2.6\%$ . The reasonable agreement between the data from these accelerometers indicates the symmetry of the wedge motion about the keel in the experiments. Moreover, the small error bars, showing standard deviation of three repeated runs, reveal good repeatability of the experiments. The in-house code results of fisher et al. [43] also show very good agreement with experimental results. The maximum acceleration predicted by the code is only  $4\%$  lower than the maximum value recorded during the tests. This is reasonable considering the code is 2-D and the experiment is 3-D.

Having position-time history from potentiometer, one can calculate the vertical velocity of the wedge by differentiating the position data. To this end, the position data are first filtered using Savitsky-Golai filter with frame length of 33. This frame length is chosen so that the filtering removes the electric 60 Hz noise. The velocity of the wedge can also be

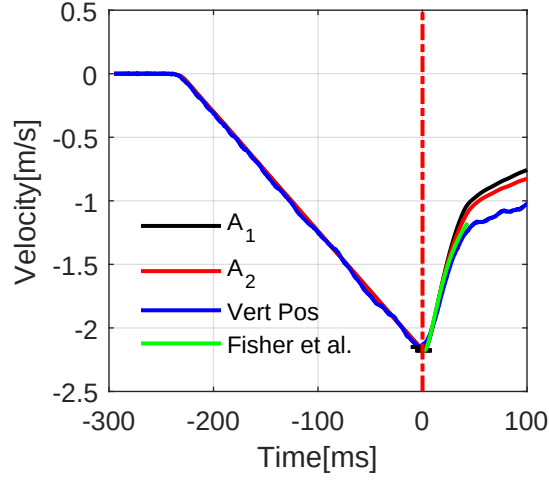


Figure 2.7: Mean vertical velocity of three repeated runs for a drop height of  $H = 25.4$  cm. The velocity is calculated by both differentiating the position curve and integrating the acceleration curves.

obtained by integrating the acceleration data that are recorded by accelerometers. These results are plotted in Fig. 2.7. It can be shown that the curves have good agreement for the falling period before the impact moment. However, the deviation becomes discernible after impact. This is due to this fact that the velocity is calculated using two different methods, i.e. integration and differentiation. As a matter of fact, in the process of integrating the acceleration data, the small perturbations in acceleration are also taken into account and integrated over the time. Since these perturbations do not have zero mean, the integration uncertainty is accumulated over the period of integration. Therefore, the velocity given from acceleration gradually overestimates the wedge speed. However, the impact velocity from position data is equal to 2.18 m/s that shows only 1% deviation from the velocity derived from acceleration. Hereafter, the velocity from position data is considered as the reference for future analysis. Comparison between experimental and Fisher et al. [43] results shows that the simulation predicts the velocity very well at the early stages of penetration. The velocity curve from the code after  $t=23$  ms falls between those from the experiment that were computed from the acceleration and position data.

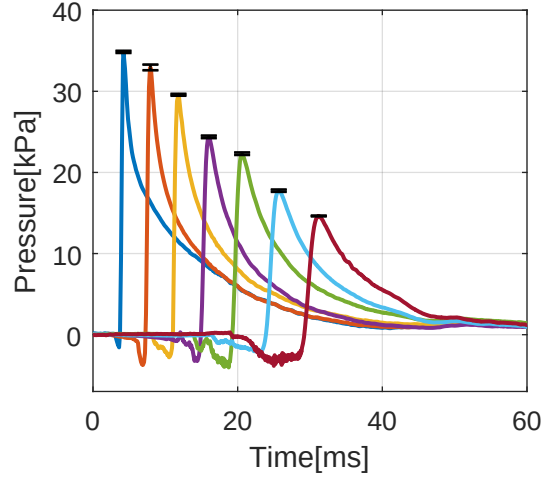


Figure 2.8: Mean pressure-time histories seven pressure sensors for three repeated runs and a drop height of  $H = 25.4$  cm. The pressure curves are measured and plotted from  $P_1$  to  $P_7$ , consecutively.

Pressure-time histories due to the 25.4-cm drop are illustrated in Fig. 2.8. Pressure signals from sensors  $P_1$  to  $P_7$  are plotted from left to right, consecutively. As shown, each sensor records a sharp rise in pressure and reaches the peak in a very short period of time. The maximum peak pressure is measured by  $P_1$  and is equal to 34.86 kPa. The peak pressure gradually decreases for following sensors, where the last sensor records the peak pressure of 29.83 kPa. The vertical error bars are plotted at the peak location to show the standard deviation in peak pressure magnitude for the repeated runs. These small error bars confirm the good repeatability of experiments. It should be noted that each pressure sensor experiences a negative pressure a few milliseconds before the region of rapidly rising pressure. High-speed video analysis shows that these negative pressures are caused by the water spray that detach from the water surface and reach the sensor before the spray root. The rise time corresponds to the time in which the spray root reaches the sensor. Since the spray propagation speed is greater than that of the spray root, the period in which each pressure sensor records the negative pressure increases from  $P_1$  to  $P_7$ .

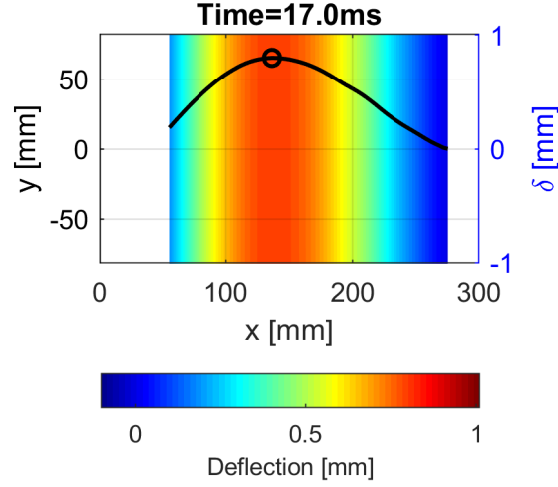


Figure 2.9: Deflection contour at  $t = 17$  ms after impact as measured by the S-DIC. The contour corresponds to the measurement area shown in Fig. 5.2.

### 2.3.2 Structural Response

Fig. 2.9 shows a deflection contour of the DIC measurement area for a drop height of  $H = 25.4$  cm and a time  $t = 17$  ms after the impact moment. This is the moment in which the peak pressure reaches the fourth sensor,  $P_4$ . As expected, the mid-plane area is experiencing close to its maximum deflection while the area close to the sides are remained almost without any deflection. This is due to the rigid side plates of the wedge that provide a fixed boundary condition for the bottom plate in the area in proximity to the sides. The curve, shown on the contour plot, is of the deflection of the characteristic line (which was the red dashed line in Fig. 5.2). The maximum deflection at this time is  $\delta = 0.80$  mm and occurs at  $x = 136$  mm away from the keel. The resulting deflection on characteristic line is further analyzed in following sections.

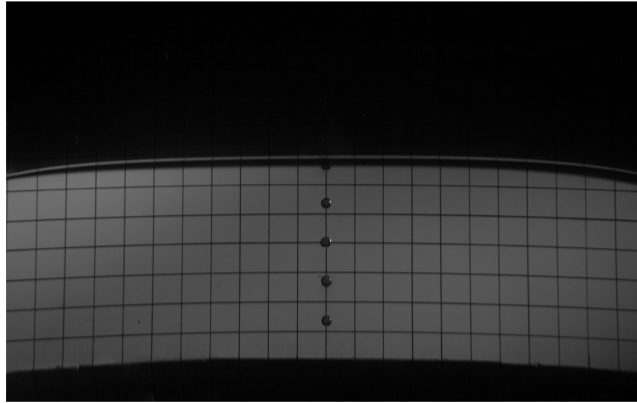


Figure 2.10: A typical image, with adjusted contrast, recorded by the interface camera showing the submerged part of the bottom of the wedge. The lines drawn on the bottom of the wedge are used as the reference for calibration process.

### 2.3.3 Flow Visualization

A typical image, with adjusted contrast, recorded by the bottom-view camera is shown in Fig. 2.10. The image shows the part of the bottom of the wedge that is submerged into water. The lower visible part is the keel line of the wedge. As can be seen, there are some lines drawn on the bottom of the wedge that were used as reference for the calibration process during image post-processing. The image post-processing is conducted using a gradient-based edge detection technique. The canny method is used to detect the interface along with the calibration lines. It should be noted that there are some bubbles above the upper line that are formed during the penetration of the wedge. These bubbles are removed during the image processing because they are not of interest in the current work. The lines are then extracted as depicted in Fig. 2.11. The upper line represents the interface line or the hypothesized spray root line. It is extracted and calibrated for further analysis.

Fig. 2.12 shows a representative set of images used to check the accuracy of the spray root derived from the interface line (extracted from the bottom view camera). In this test, the two cameras are synchronized to capture the impact of the wedge from the side and bottom

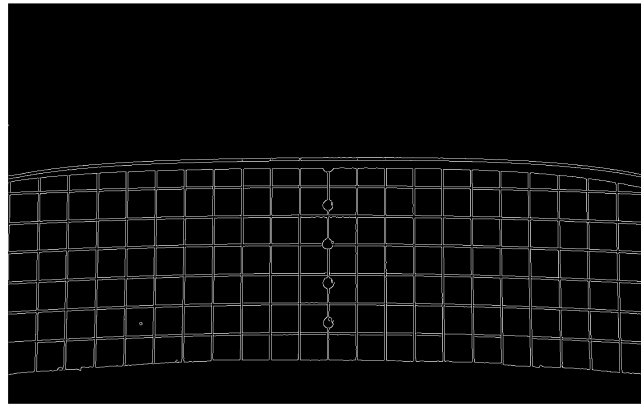
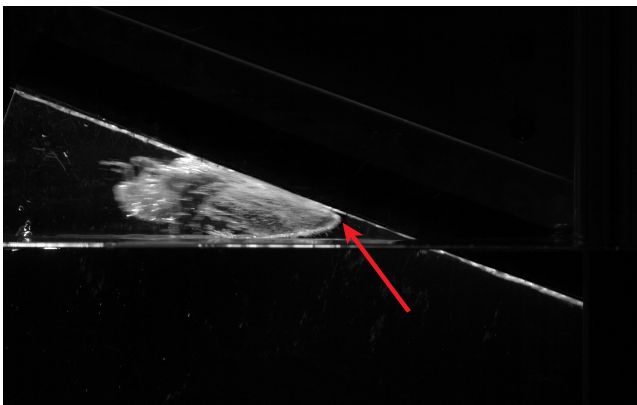


Figure 2.11: The results of the gradient-based edge detection algorithm executed on Fig. 2.10 showing the edges extracted. The upper-most line is the interface line.



(a)



(b)

Figure 2.12: The (a) side and (b) bottom view of the wedge recorded during a test that is meant to check the accuracy of interface line (from the bottom view) to the spray root (marked by the red arrow in the side view) location.

view. Marked by the red arrow, the spray root is referred to the point in which the water surface has the maximum curvature ([10], [7] and [8]).

In the current experimental setup, it is possible to use up to three high-speed cameras at one time. Since the waterline is subjected to 3-D effects (as seen in Fig. 2.12 (b)), the side view recording (Fig. 2.12 (a)) will not allow the measurement of the waterline at the midspan. Therefore, the camera view below the tank is used to measure the interface line at the midspan of the wedge. As a result, a comparison between the bottom view and side view were conducted to determine whether or not the spray root was visible in the bottom view images.

The interface line location from bottom view and the spray root location from side view are extracted and plotted together in Fig. 2.13 for the drop height of  $H = 15.9$  cm. Horizontal dashed lines show the boundaries of the pressure sensors ( $P_4$  to  $P_7$  from bottom to the top, respectively). It is noted that the far right edge of Fig. 2.12(b) was correlated with the spray root recorded from Fig. 2.12(a). The side view (Fig. 2.12(a)) was viewed from the far right side of the wedge as seen in Fig. 2.12(b). Results show that the extracted points show good agreement. In fact, the deviation between the spray root and the interface line is less than  $\pm 1.5\%$  as illustrated in Fig. 2.14. As a result, the interface line extracted from the bottom view can be approximated as the spray root. Hereafter, all spray root results are presented based on the image processing on the bottom view images.

## 2.4 Discussion

Discussion and further analysis of the results are presented in this section. The three nondimensional parameters that are used in this section are  $t^*$ ,  $z^*$ , and  $\delta^*$ , which are the nondimensional time,  $z$ -position, and deflection, respectively. The non-dimensional time, which

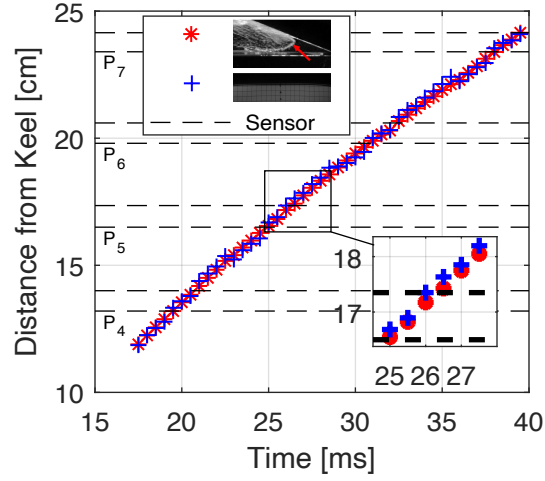


Figure 2.13: The interface line extracted from the bottom view video (blue +) and the spray root extracted from the side view (red circles) for the drop height of  $H = 15.9$  cm. The horizontal dashed lines show the boundaries of the pressure sensor locations.

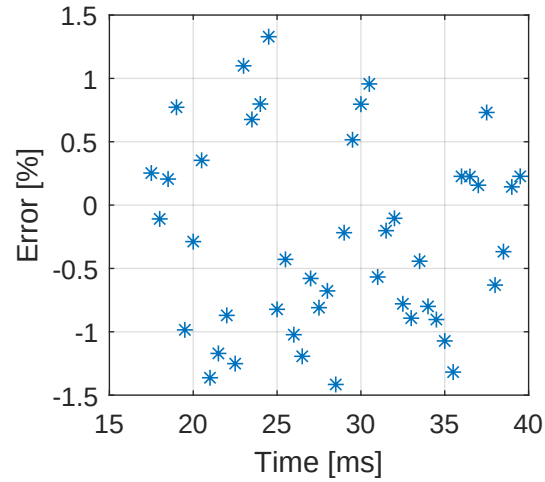


Figure 2.14: The percent difference between the spray root location from the side view and from bottom view for drop height of  $H = 15.9$  cm.

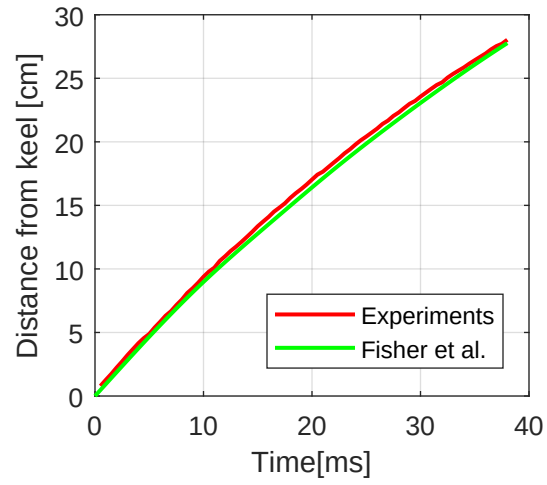


Figure 2.15: The spray root location (distance from the keel) for the flexible wedge dropped from a height of  $H = 25.4$  cm.

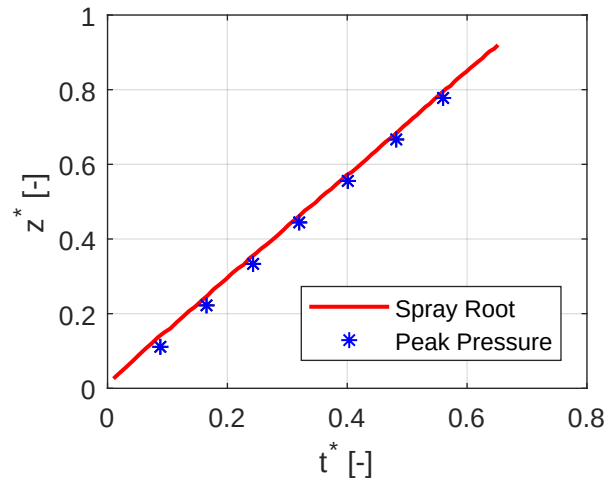


Figure 2.16: Nondimensionalized spray root and peak pressure locations versus time for the flexible wedge dropped from  $H = 25.4$  cm.

represents the time it takes for the wedge to pass the calm water line and submerge up to the chine, is defined as:

$$t^* = \frac{\int W(t)dt}{z_C}, \quad (2.1)$$

where  $W$  is the wedge instantaneous vertical velocity and  $z_C$  is the vertical distance of the chine with respect to the keel as shown in Fig. 4.1. Non-dimensional spray root position is also defined as the spray root position normalized by the total distance from the keel to the chine ( $z_C$ ), or:

$$z^* = \frac{z_S}{z_C} \quad (2.2)$$

where  $z_S$  is the vertical distance of spray root with respect to the keel. Finally, the non-dimensional deflection is defined as follows:

$$\delta^* = \frac{\delta}{h} \quad (2.3)$$

where  $\delta$  and  $h$  are deflection and thickness of the bottom plate of the wedge, respectively (See Fig. 4.1).

### 2.4.1 Spray Root Propagation

The spray root propagation for the flexible wedge dropped from  $H = 25.4$  cm is summarized in Fig. 2.15. The spray root data are extracted from characteristic line for this analysis. Results show that the spray root propagation has a linear speed for the early stages of penetration. The propagation speed, or the slope of the curve, is then reduced due to

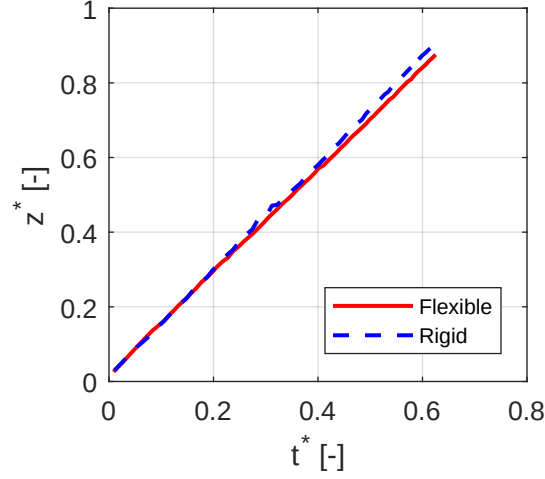


Figure 2.17: Nondimensional spray root versus time for both the flexible and rigid wedges dropped from a height of  $H = 25.4$  cm.

deceleration of the wedge. The distance between the spray root and the keel line is 9.38, 17.03 and 23.56 cm at times 10, 20 and 30 ms, respectively. As it is shown in Fig. 2.15, the measured spray root location versus time agrees quite well with the predictions of the in-house code (fisher et al. [43]).

Fig. 2.16 shows the nondimensional position of the spray root and peak pressure for the drop height of  $H = 25.4$  cm. Results show that the peak pressure propagates almost with the same speed as the spray root while it is always lagging behind the spray root propagation. This is consistent with that reported by Vorus water entry theory ([7]).

### 2.4.2 Effect of Flexibility

To study the effect of flexibility of the structure, the spray root propagation for the flexible wedge is compared with that of the rigid wedge for the same drop height,  $H = 25.4$  cm. Although the kinematics for both wedges are very similar (as discussed in Ren et al. [44]), non-dimensional analysis is utilized to rule out even the minor deviations. Such a comparison,

exhibited in Fig. 2.17, reveals that the spray root propagation is almost identical at the initial stages of penetration until  $t^*$  values of 0.2. However, after this moment, the spray root propagation for flexible wedge lags behind the rigid one. The key to interpret this behavior is the structural response of the flexible wedge. In fact, since the deflection causes a curvature in the bottom plating of the wedge, the spray root attached to the flexible wedge has to travel longer compared to the rigid one. The non-dimensional speed for propagation of the spray root on the flexible wedge is reduced by up to 4% due to flexibility. A similar behaviour for flexible structures was reported by Qin and Batra [40].

### 2.4.3 Effect of Impact Velocity

To evaluate the effect of impact velocity on the spray root propagation, experiments have been conducted with different drop heights and results of the spray root propagation are displayed in Fig. 2.18. As expected, the increase in impact velocity is highly correlated with the spray root propagation speed. Considering the slope of the curves for the initial stage of penetration, the spray root speed increases from 5.4 m/s for  $H = 7.9$  cm to 10.14 m/s for  $H = 25.4$  cm, and 13.74 m/s for  $H = 50.8$  cm. On the other hand, the reduction of speed of the spray root due to deceleration of the wedge is higher for higher impact velocities. For example, in the case of drop height of 50.8 cm, the spray root loses almost 50% of its speed after 25 ms. For a similar time period (namely,  $t = 0$  to  $t = 25$  ms), the spray root speed experiences a reduction of 35% for  $H = 25.4$  cm while it stays nearly constant for  $H = 7.9$  cm.

Fig. 2.19 shows the nondimensional spray root propagation for the flexible wedge for different drop heights. For the initial stages of penetration, the non-dimensional speed for all drop heights seem to be almost identical and equal to 1.24. However, the curves gradually deviate

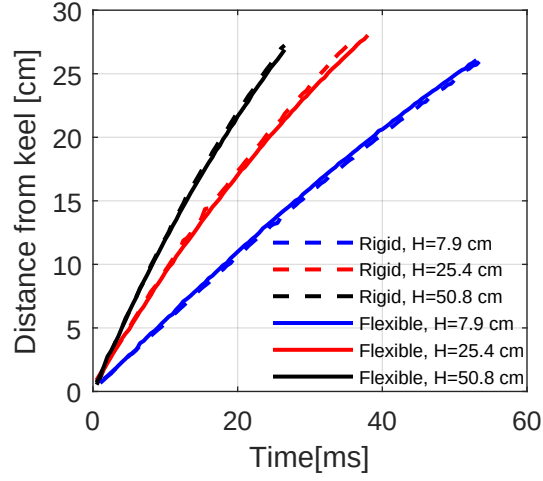


Figure 2.18: Spray root propagation for both the flexible and rigid wedges dropped from different drop heights to show the effect of impact velocity.

from each other such that the spray root propagation for the condition  $H = 7.9$  cm maintains a higher nondimensional speed. As the deflection becomes larger for higher drop heights, the nondimensional speed of the spray root propagation experiences higher reduction. In fact, the nondimensional speed of the spray root is lower for  $H = 50.8$  cm compared to  $H = 25.4$  cm.

The effect of impact velocity on nondimensional spray root propagation for the rigid wedge is shown in Fig. 2.20. Since the deflection is negligible for the rigid wedge, the non-dimensional spray root propagation curves for different drop heights nearly collapse onto one curve. Therefore, for the rigid wedge, the impact velocity has almost no effect on the nondimensional spray root propagation. As shown earlier in Fig. 2.18, the higher impact velocity increases the speed of dimensional spray root propagation, though.

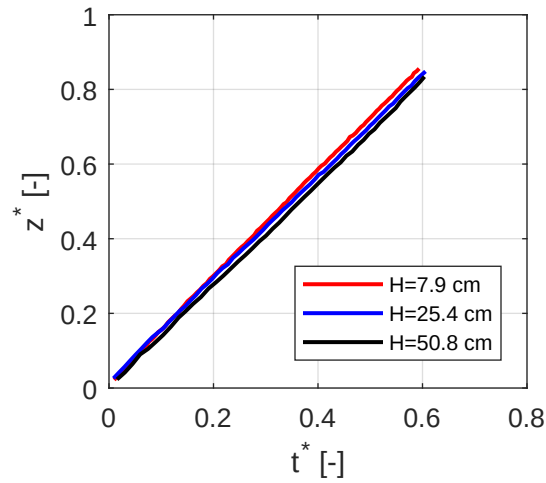


Figure 2.19: Nondimensional spray root propagation for the flexible wedge dropped from different drop heights.

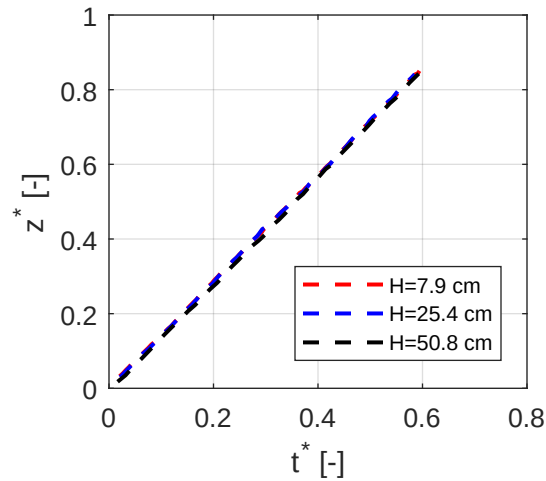


Figure 2.20: Nondimensional spray root propagation for the rigid wedge dropped from different drop heights.

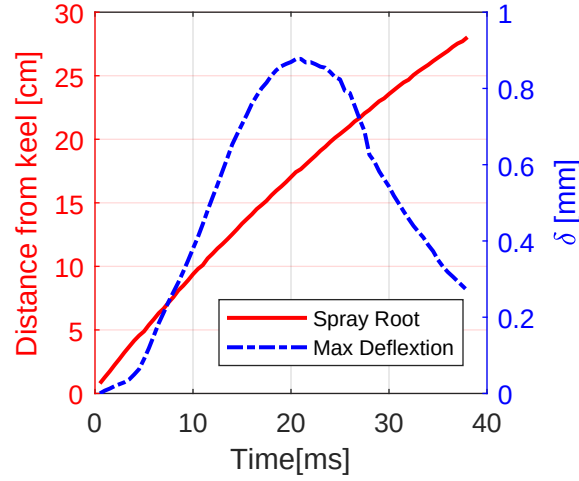


Figure 2.21: Temporal correlation between the spray root and maximum deflection for the flexible wedge dropped from  $H = 25.4$  cm.

#### 2.4.4 Instantaneous Correlation

Having structural response from S-DIC, the correlation between the spray root evolution and deflection magnitudes can be examined. Fig. 2.21 depicts the maximum deflection of the characteristic line along with the spray root propagation for the flexible wedge with drop height of  $H = 25.4$  cm. It can be seen that there is a lag between the spray root evolution and the deflection of the bottom plate at early stages of penetration ( $t = 0$  to  $t = 7$  ms). Once the structural response becomes discernible, the maximum deflection of the plate increases until it reaches a maximum value of  $\delta = 0.88$  mm at  $t = 21$  ms. This moment corresponds to the time in which the spray root has already passed the mid-plane of the wedge ( $z = 17.65$  cm). The maximum deflection then again decreases due to reduction of hydrodynamic loading.

Fig. 2.22 shows the correlation between the spray root propagation, hydrodynamic impact load and maximum deflection of the wedge bottom plating. Again, the phase between structural response of the wedge and hydrodynamic loading is seen at the initial stages of

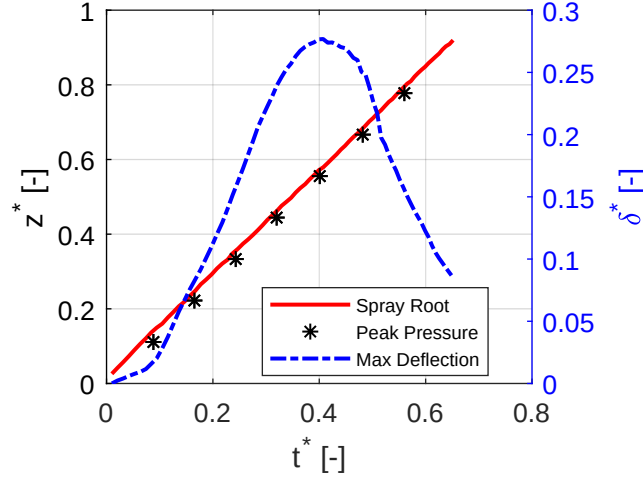


Figure 2.22: Temporal correlation between the spray root, hydrodynamic pressure and maximum deflection for the flexible wedge dropped from a height of  $H = 25.4$  cm.

impact where the deflection of the wedge is still negligible ( $t^* < 0.1$ ). The deflection then increases to reach the maximum,  $\delta^* = 0.28$  at  $t^* = 0.41$ . This is the time in which the peak pressure has just reached the fifth sensor,  $P_5$ , and the spray root has already passed the mid-plane ( $z^* = 0.58$ ). After this point, the maximum deflection decreases after the peak pressure has passed the  $P_6$  (at  $t^* = 0.48$ ). By the time that peak pressure reaches  $P_7$  (at  $t^* = 0.56$ ), the max deflection of the wedge has decreased by 43% and is equal to  $\delta^* = 0.16$ .

The interaction of the location of maximum deflection, spray root, and peak pressure is examined more closely in Figs. 2.23 to 2.25. The peak pressure location was resolved using a spatial pressure reconstruction as detailed in Ren et al. [37]. Fig. 2.23 shows the location of the spray root, maximum deflection, and peak pressure location of the wedge at  $t = 10$  ms. At this time, the spray root is at  $x$  location  $x = 93.8$  cm and has not reached the location of maximum deflection of the plate, which is at  $x = 122.8$  cm. The maximum pressure of the wedge is close to the spray root. The deflection of the point corresponding to the spray root is  $\delta = 0.33$  mm which is 13% lower than the maximum deflection,  $\delta = 0.38$  mm at this specific time.

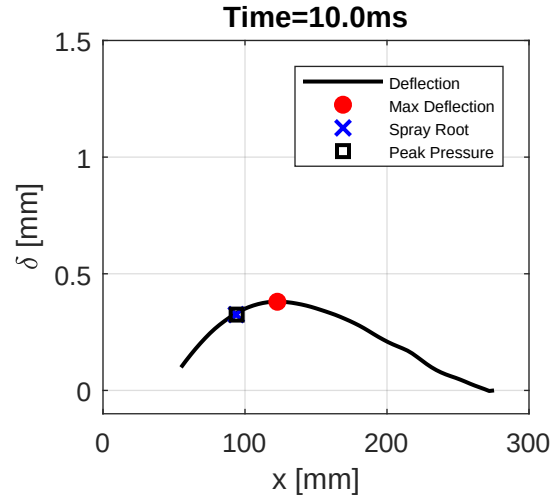


Figure 2.23: Deflection distribution along the characteristic line superimposed with the location of the spray root (blue x) and peak pressure (black square) locations at time  $t = 10.0$  ms after impact. The red circle shows the maximum deflection.

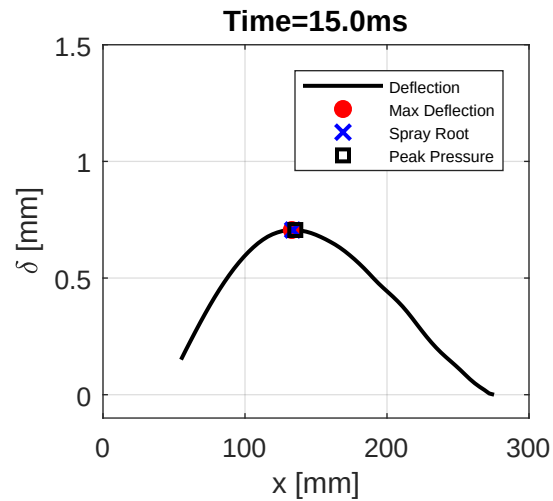


Figure 2.24: Deflection distribution along the characteristic line superimposed with the location of the spray root (blue x) and peak pressure (black square) locations at time  $t = 15.0$  ms after impact. The red circle shows the maximum deflection.

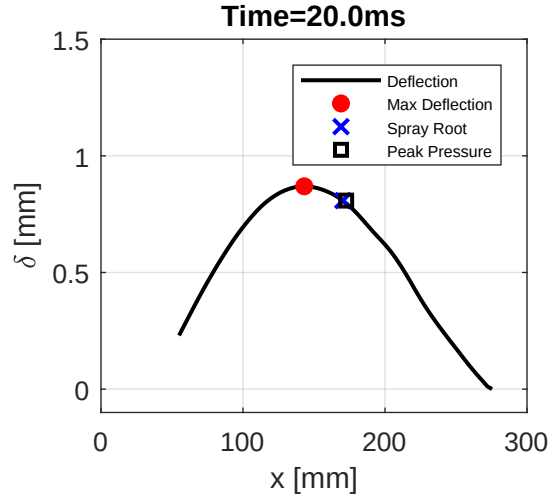


Figure 2.25: Deflection distribution along the characteristic line superimposed with the location of the spray root (blue x) and peak pressure (black square) locations at time  $t = 20.0$  ms after impact. The red circle shows the maximum deflection.

As shown in Fig. 2.24, the spray root and peak pressure coincide with the point of maximum deflection at  $t = 15.0$  ms. At this moment, the deflection is  $\delta = 0.71$  mm and occurs at  $x = 133.6$  cm.

Fig. 2.25 shows a moment,  $t = 20.0$  ms, when the spray root and peak pressure have passed the point of maximum deflection. At this time, maximum deflection is measured to be  $\delta = 0.87$  mm, that is 7% greater than the deflection corresponding to the spray root,  $\delta = 0.81$  mm.

The location of the spray root and corresponding location of the maximum deflection of the wedge can also be analyzed, as shown in Fig. 2.26. This figure shows that the location of the maximum deflection of the characteristic line, at different times, varies in the range of 11.26 cm to 16.35 cm with respect to the keel. For times less than 15 ms, the spray root is behind the location of maximum pressure (i.e. Fig. 2.23). The time  $t = 15$  ms, shown by the black dashed line, is the moment in which the location of the spray root coincides the location of maximum deflection of the wedge (i.e. Fig. 2.24). At this time, both points are

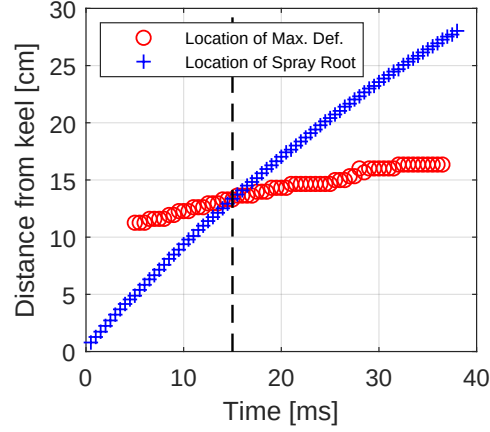


Figure 2.26: Temporal locations of maximum deflection and spray root locations for the flexible wedge for the condition of drop height  $H = 25.4$  cm.

located at the distance of 13.36 cm from the keel. At times greater than 15 ms, the spray root surpasses the location of maximum deflection (i.e. Fig. 2.25).

To further analyze the correlation of spray root and structural response of the wedge, maximum deflection of the plate and the deflection of the point corresponding to the spray root are plotted in Fig. 2.27. This figure shows  $t = 15$  ms, shown by black dashed line, is the moment that the maximum deflection is equal to the deflection at spray root. As shown earlier, this is due to this fact that at this time, spray root point coincides the point of maximum deflection (Figs. 2.24 and 2.26).

Times prior to  $t = 15$  ms; however, represent the situation in which the spray root is behind the maximum deflection point (i.e. see Fig. 2.23). Thus, the deflection at spray root is smaller than the maximum deflection. On the other hand, times after 15 ms correspond to the situation in which the spray root has surpassed the location of maximum deflection (i.e. see Fig. 2.25). Consequently, the deflection at the spray root is again smaller than the maximum deflection. Since the spray root propagation and the maximum deflection propagation are not in phase, the deviation in the second period ( $t > 15$  ms) is larger than

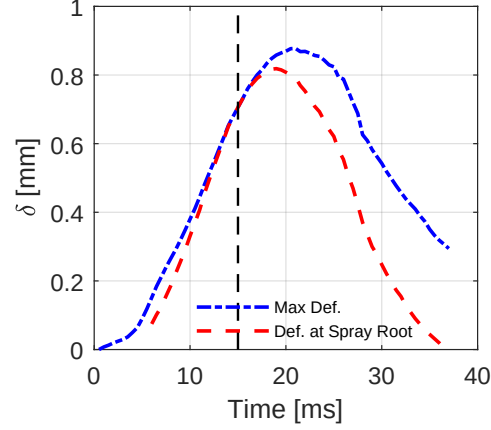


Figure 2.27: Comparison of the maximum deflection with the deflection of the spray root location at different time for the flexible wedge dropped from  $H = 25.4$  cm.

the first period ( $t < 15$  ms). The decay of deflection in the second region proceeds slower than the period of forming of deflection. It is speculated that one possible reason might be this fact that in the second period, the wedge bottom is backed up by water, which hinders the decay process of deflection.

Fig. 2.28 shows a comparison between the nondimensional maximum deflection with the nondimensional deflection of the corresponding point of the spray root at different time steps for the flexible wedge when  $H = 25.4$  cm. Both deflections become identical at  $t^* = 0.30$ , represented by black dashed line, where the non-dimensional deflection is  $\delta^* = 0.22$ . The maximum deflection of the point corresponding to the spray root is  $\delta^* = 0.26$  and occurs at  $t^* = 0.38$ . The maximum deflection of the wedge; however, is measured at  $t^* = 0.41$  and is  $\delta^* = 0.28$  which is 7% larger than the maximum deflection corresponding to the spray root.

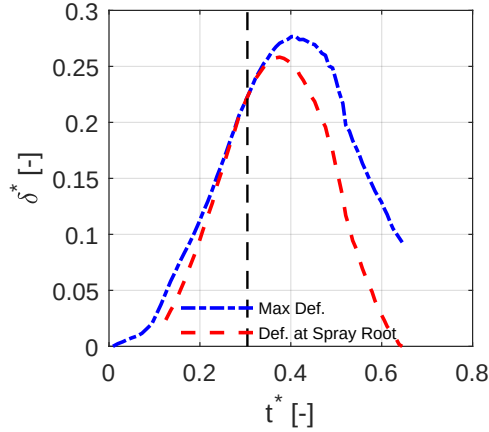


Figure 2.28: Comparison of the nondimensional maximum deflection with the nondimensional deflection of the spray root location at different time for the flexible wedge dropped from  $H = 25.4$  cm.

## 2.5 Conclusions and Future Work

In this paper, the effects of fluid-structure interaction on the water entry of a flexible wedge were experimentally investigated. Free-fall water entry experiments were conducted and kinematics were measured. An array of pressure sensors were installed to the wedge bottom to record hydrodynamic pressure during impact. Structural response of the wedge bottom was measured through the use of the Stereoscopic-Digital Image Correlation Technique, or S-DIC. Additionally, the spray root propagation during the impact was visualized with high-speed cameras. Given a fully synchronized measurement, instantaneous correlation between peak pressure, spray root propagation, and structural response of the wedge was studied. To assess the effect of hydroelasticity, experiments were repeated for a rigid wedge of similar dimensions. Finally, experiments with varying different drop heights were taken into account to gauge the role of impact velocity.

Results show that there is a delay between the spray root and the peak pressure propagation. The distance between these two remains almost constant and equal to 6% of the height of

the chine,  $z_C$ . More specifically, the peak reaches each pressure sensor when the spray root has already passed that sensor as predicted by theory.

Despite being nearly identical in early stages of penetration, the evolution of the spray root on a flexible wedge lags compared to the rigid one at a critical time when the deformation becomes large enough. In fact, once the deformation becomes important, the spray root of the flexible wedge has to travel a longer distance due to the curvature in the bottom of the wedge. The additional distance seemingly hinders the spray root propagation. The larger the deflection, the longer the delay. Results show that for drop height of  $H = 25.4$  cm, the non-dimensional speed of spray root propagation is reduced up to 4%.

Regarding the dimensional spray root propagation, higher impact velocity increases the speed of propagation of spray root for both rigid and flexible wedge, as expected. However, nondimensional analysis reveals that for the rigid wedge, impact velocity has no significant effect on non-dimensional spray root propagation. On the other hand, as far as the flexible wedge is concerned, impact velocity slightly reduces the nondimensional speed of spray root propagation. In other words, higher impact velocity leads to higher hydroelasticity effects that influence the nondimensional spray root propagation at later times during the impact event.

Instantaneous correlation for the flexible wedge of drop height  $H = 25.4$  cm shows that the maximum deflection on the wedge occurs when the peak pressure reaches the fifth sensor,  $P_5$ . At this moment, the spray root has already passed the location of maximum deflection by 20%.

When evaluating the structural response of the wedge, it is seen that the deflection increases at a higher rate than it decreases. One possible reason is when the structure returns to its original shape, the wedge bottom is backed up by the water that was not there when the

deformation was first occurring.

Finally, results were compared with that of an in-house code that integrates the Wagner inner-outer solution with a linear finite element solver for wedge water entry problem. The kinematics and propagation of the wetted length predicted by the code agree well with the experimental data, despite the code being a low-fidelity 2-D solver compared to a 3-D experiment.

In the future, experiments will be continued on composite models with different flexibility to assess a better understanding of the effects of fluid-structure interaction on wedge water entry metrics. Ultimately, a similar study will be carried out on a planing hull in VT Advanced Towing Tank facility. Combining all these results will allow for implementation of new, physics-based design criteria for high-speed craft with flexible structures.

## 2.6 ACKNOWLEDGEMENTS

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# Chapter 3

## Predicting Kinematics and Hydrodynamic Loading Using Spray Root Propagation in Wedge Water Entry

(This is the draft of a paper that is intended to be submitted to Applied Ocean Research.)

M. Javad Javaherian and Christine Gilbert

(ABSTRACT) Water entry has a wide range of applications in different engineering fields including marine, aerospace and mechanical engineerings. The design of high-speed planing craft is one of these applications in marine engineering that is often studied considering wedge water entry. There are a number of theoretical formulations that have been developed in an effort to simplify the process of estimating design loads for vessel design. The work presented in this paper is the first step toward using a hybrid approach to estimate hydrodynamic pressures by taking spray root position measurements and existing theoretical models. This paper shows the validity of this approach using a case study of a flexible wedge drop experiment. To have a better comparison, the analytical models are re-derived to account for the loss of velocity so that they are comparable with the free-fall experimental results.

Future components of this work will further develop this tool to be used for highly-flexible structures and also in-situ pressure measurements at sea. The results of this paper show that the proposed approach using re-derived Armand & Cointe model can reasonably predict the hydrodynamic pressure distribution on the bottom panel of the wedge. The hydrodynamic force and kinematics of the model can also be adequately estimated using this approach. This approach will be used to study the flow field changes due to shape reconfiguration of highly-flexible plates near the free-surface in order to reduce the drag on undulatory propulsors and vessels.

### 3.1 Introduction

Water entry has different applications in engineering. The entry of birds into the water, shape reconfiguration of flapping fins during swimming of manta rays, the impact of sea-planes with the water surface upon landing, recovering space capsules, slamming of marine vessels, wave impacts onto offshore structures and the impact in the water turbine blades are some examples of this phenomenon.

Inspired by the nature, researchers have been studying some aquatic creatures, which instinctively change the shape and stiffness of the fins to reduce the drag or enhance the maneuverability during the swimming ([1, 2]). As an instance, manta rays alter the surrounding flow field with shape reconfiguration of their fins in order to reduce the drag during the motion. To further study this phenomena, the scaling laws should be developed to explain the reduction in drag coefficient due to the reconfiguration of the plate in the proximity of the free surface. This can be evaluated by studying flow field changes due to the shape reconfiguration of a highly flexible plate near an air-water interface during water entry. Inspired by nature, this shape reconfiguration could be actively implemented to reduce the drag on

undulatory propulsors and vessels.

Regarding the marine applications, water entry is one of the well-known problems for naval architects, especially those who work on the design of high-speed vessels. The problem is often simplified to water entry of wedge-shaped bodies, where the wedge represents a section of the high-speed craft. Over the years, many researchers have developed models to simulate a prismatic wedge impacting the water surface, which can be applied to a vessel traveling in waves. Since slamming can tangibly influence the structural strength, overall performance and crew functionality on board, accurate prediction of slamming loads is an indispensable step in the design process of planing vessels. Moreover, there have been attempts to implement the wedge water entry results into predicting the motions of the planing hulls, using 2D+t method (Zarnick [5], Faltinsen [6]). This method relates the problem of the motion of a prismatic planing vessel into problem of a 2D wedge that vertically enters into the water. Once the hydrodynamic loading problem of the 2D wedge is solved, results can be used in order to predict the motion and loads on high-speed planing vessel

It is shown in the literature that as the wedge with a deadrise angle of  $\beta$  penetrates into a calm water, i.e. in Fig. 3.1, the disturbance generates water pile up and the spray root, which is the point of maximum curvature on the water surface. Researchers have studied the relationship between spray root, kinematics and hydrodynamic loading. The problem was first introduced by von Karman [9] for the application of the design of the seaplane landing on a water surface. A more justified model was then suggested by Wagner [10], where he employed asymptotic theory to relate the wedge water entry problem to an expanding flat plate. Even though his model predicts better results for the impact pressure compared to the von Karman [9] model, there exists a singularity in the governing equation for the hydrodynamic loading at the location of the spray root. Further researchers tried to develop new models so that they could solve the singularity problem and estimate more accurate

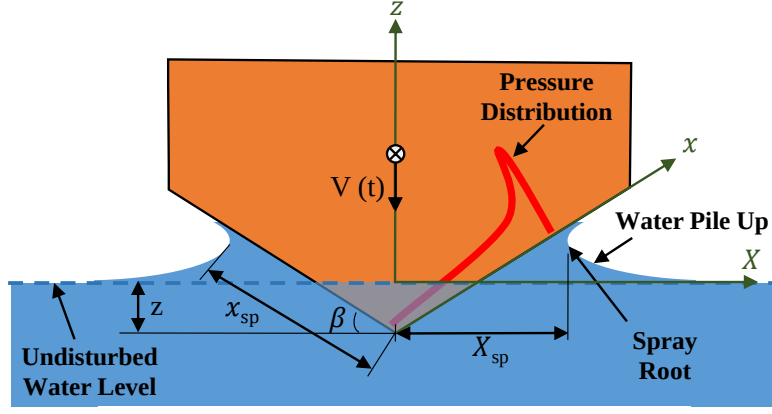


Figure 3.1: A schematic depicting the vertical water entry of a wedge.

results for a wide range of deadrise angles. The researches of [12, 14, 45, 46] and [8] are some examples for these studies. Wagner [10], Generalized Wagner, Logvinovich [12], and Armand & Cointe [14] will be adapted and used in this study and will be discussed in more detail in the next section.

The wedge water entry problem has also been examined experimentally in addition to analytically and numerically. Zhao and Faltinsen proposed a numerical model and carried out a series of experiments to validate their model [18]. These tests covered a wedge and also a bow flare section. They compared free surface profile, pressure coefficient, slamming force, and penetration speed between the experiment and model predictions. Yettou et al. conducted a series of experiments to parametrically investigate the wedge free fall water entry [28]. Effects of drop height, deadrise angle, and wedge mass on the impact pressure were studied. In addition, some non-dimensional analysis was also discussed. Lewis et al. performed some experiments on water entry of a  $25^\circ$  deadrise-angle wedge and measured the kinematics and pressure-time history of the wedge during the impact [29]. They also made attempts to visualize the fluid flow around the wedge to study the effect of the spray root on the exerted hydrodynamic pressure.

The studies mentioned to this point are on rigid wedge impact. Attention is now given to studies of flexible wedge water entry. Different researchers developed separate models to study the water entry of a flexible wedge [34, 40, 47]. In their models, they investigated the effect of the hydrodynamic loading on the structural response of the model.

Experimentally, water entry of a compliant wedge was studied by Panciroli et al. [34]. They used particle image velocimetry (PIV) to evaluate the effects of flexibility on the hydrodynamic loading and structural response of the flexible panels. They compared the experimental results with their semi-analytical model. Shams et al. also experimentally investigated the water entry and exit of a flexible wedge [35]. They used PIV measurements to obtain the velocity and pressure field around the wedge. They also studied free surface elevation to compare these experimental results with their semi-analytical model for slamming.

Ren et al. conducted several experiments to investigate the interaction between hydrodynamic loading and the structural response of a flexible wedge [36, 44, 48]. They implemented Stereoscopic-Digital Image Correlation (S-DIC) to measure the displacement and also out-of-plane deflection of the wedge during the impact. Javaherian et al. introduced a novel method for tracking and measuring the spray root propagation along the bottom of a flexible wedge [48]. Having fully synchronized measurements, they studied the interaction between the spray root, hydrodynamic loading, and structural response of the wedge. Fisher et al. proposed a two-way coupled, low fidelity, semi-analytical approach to simulate water entry of a flexible wedge [43]. The coupling relates the spray root location in the fluid to the deflection on the wedge panel in contact with the water. Results showed a very good agreement with experimental results presented by [48] and [44].

Due to the expanded use of composite materials in high-speed vessels as well as the need to increase speed and decrease vessel weight, the vessel structure can no longer be considered rigid. As previously mentioned, it is important to have an accurate estimation of the design

loads a high-speed vessel will encounter to increase the operational envelope. Use of the established theoretical models for rigid wedges in this context results in uncertainty for flexible wedges since they were not designed to take into account these physics. The approach of this paper is to allow for a semi-empirical prediction of the pressure distribution on the bottom of a wedge due to free-fall by inputting the spray root into existing theoretical models. This approach is important for two reasons: (1) for experiments on highly flexible structures, it is not possible to use traditional pressure sensors as they cause stiffening of the bottom plate that would not normally be present, and (2) an approach that allows for in-situ measurements in the field can be desirable. Even though the current study will be presented with an emphasis on the marine engineering applications, the approach will be elaborated in a generalized form so it can also be utilized for any other applications that was mentioned earlier in this section.

In this paper, a new hybrid technique is proposed to estimate the wedge kinematics and exerted hydrodynamic pressure on the wedge water entry problem using the spray root time-history as an input into existing theoretical models. A case study will be presented in this paper, wherein the proposed hybrid approach will be used for a flexible wedge experiment. The estimated results from this method will be compared to both the full experimental measurements as well as the full theoretical prediction. The methodology developed and presented in this paper is the first step to estimate the hydrodynamic pressure and structural response in water entry of highly flexible structures in the context of both the water entry problem and a vessel traveling through waves.

The rest of the paper is organized as follows. In section 3.2, some of the existing theoretical models, which are derived based on asymptotic theory for the wedge water entry are introduced. The approach that has been utilized for this study is elaborated in section 3.3. This includes the re-derivation of the introduced models, methodology of the proposed approach

and the experimental study that has been considered for the implementation and verification process of this study. In section 3.4, results of the present study including pressure distribution, pressure-time history, force and kinematics are reported and analyzed. The accuracy of the theoretical models regarding the estimation of the spray root location is also evaluated in this section. Finally, some conclusions and future work are outlined in section 3.6.

## 3.2 Theoretical Models

In this section, some of the well-known theoretical models for wedge water-entry are introduced. These models are all based on the asymptotic theory, neglect the effects of viscosity and gravity, and are developed to predict the hydrodynamic pressure on the bottom of the wedge during the impact. The models are: (1) Wagner [10], (2) Generalized Wagner as presented in [11], (3) and (4) Logvinovich and modified Logvinovich as presented in [13], and (5) Armand & Cointe [14]. Of these five models presented, all but Wagner [10] are formulated for constant entry velocity. Wagner [10] was formulated for non-constant entry velocity.

### 3.2.1 Wagner Model

Wagner used asymptotic theory to collapse the 2-D wedge problem to an expanding line-boundary value problem that must be solved at each timestep [10]. The extent of the line, for a symmetric wedge, is  $-X_{sp}(t)$  to  $X_{sp}(t)$ , where  $X_{sp}(t)$  is the instantaneous location of the intersection of the body with the free surface, also known as spray root (see Fig. 3.1). Outside of this domain, the free surface kinematic boundary condition states that all particles on the

free surface must remain on the free surface, so the velocity potential,  $\phi$ , at all  $X$  locations outside of the line must be equal to zero. In this model, the body remains fixed, and the fluid flows upward. The velocity of the fluid is the magnitude of the velocity of the wedge, but in this frame of reference travels upward. The remaining models that will be discussed in this paper have a similar basis to this model. In all these models, the relationship between the vertical position of the wedge and the spray root location is estimated as:

$$X_{sp} = \frac{\pi}{2} \frac{z}{\tan \beta}. \quad (3.1)$$

The accuracy of the Eq. 3.1 will be evaluated later in section 3.5. As mentioned earlier, this equation is common among all the models in order to estimate the location of the spray root. The differences between each subsequent theoretical model and the present one will be noted in the following subsections.

Using potential flow theory, the function for the velocity potential can be solved, and the pressure distribution can be obtained using Euler's equation with the assumption that the fluid acceleration is the dominant term in Euler's equation. Finally, there was a small angle assumption made for the deadrise angle. The final result for the hydrodynamic pressure is:

$$p(X) = \frac{1}{2} \rho V^2 \left[ \frac{\pi}{\beta} \frac{X_{sp}}{\sqrt{X_{sp}^2 - X^2}} - \frac{X^2}{X_{sp}^2 - X^2} + \frac{2\ddot{z}}{V^2} \sqrt{X_{sp}^2 - X^2} \right]. \quad (3.2)$$

The first term here denotes the slamming pressure, while the last term corresponds to the added mass. It is noted that there exists a singularity at the location of the spray root,  $X = X_{sp}$ , in the formulation as observed in Eq. 3.2. The pressure equation is only valid from the range  $-X_{sp}(t) < X < X_{sp}$ , and this model cannot predict the pressure at and beyond the spray root location.

It is worthwhile to mention that once the pressure data are predicted on the  $X$ -axis using these formulations, the results are then transformed back to the  $x$ -axis so they can be comparable with the experimental pressure measurements (see Fig. 3.1). The process of transformation is repeated for all other theoretical models that are covered in this study.

### 3.2.2 Generalized Wagner Model

The Generalized Wagner method was presented by Zhao et al. [11]. The main differences between the Generalized Wagner and original Wagner [10] include: (1) all other terms in Bernoulli's equation were retained except for the hydrostatic pressure, (2) the deadrise angle was allowed to be larger (no small angle approximation was used), and (3) the velocity was assumed to be constant. The hydrodynamic pressure distribution in this model is summarized as:

$$p(X) = \frac{1}{2}\rho V^2 \left[ \frac{\pi}{\tan \beta} \frac{X_{sp}}{\sqrt{X_{sp}^2 - X^2}} - \cos^2 \beta \frac{X_{sp}^2}{X_{sp}^2 - X^2} - \sin^2 \beta + 2 - \pi \right]. \quad (3.3)$$

The first two terms are very similar to those of the Wagner model, except that the restriction of the small angle assumption is removed. The added mass term is not seen in the formulation due to the constant-entry assumption. The last three terms are derived from the additional terms, which were restored in the velocity potential function from asymptotic analysis compared to the velocity potential that was used by Wagner. These terms will be explained later in this section. Again, it can be seen that there exists a singularity at the location of the spray root,  $X = X_{sp}$ , in the formulation as observed in Eq. 3.3, and the formulation is only valid up to the spray root location.

### 3.2.3 Original and Modified Logvinovich Models

The original Logvinovich model [12] is similar to Wagner [10], except that the small deadrise angle assumption is not used in the first term, and the entry velocity is assumed constant (yet the derivation approach is slightly different). The simplified version of the Logvinovich formulation for the hydrodynamic pressure in the contact region is presented by Korobkin [13]:

$$p(X) = \frac{1}{2}\rho V^2 \left[ \frac{\pi}{\tan \beta} \frac{X_{sp}}{\sqrt{X_{sp}^2 - X^2}} - \frac{X_{sp}^2}{X_{sp}^2 - X^2} \right]. \quad (3.4)$$

As can be seen, the higher order terms with respect to wedge deadrise angle,  $\beta$ , and the added mass term are not seen due to the small deadrise angle and constant-entry assumptions, respectively. If the second order terms with respect to  $\beta$  are retained so that the small deadrise angle assumption is not applied, then the Modified Logvinovich model is obtained ([13]) as shown below:

$$p(X) = \frac{1}{2}\rho V^2 \left[ \frac{\pi}{\tan \beta} \frac{X_{sp}}{\sqrt{X_{sp}^2 - X^2}} - \cos^2 \beta \frac{X_{sp}^2}{X_{sp}^2 - X^2} - \sin^2 \beta \right]. \quad (3.5)$$

Again, one can notice from the Eqs. 5.10 to 3.5 that both of the Original and Modified Logvinovich models have singularity at the location of the spray root,  $X = X_{sp}$ , and the formulation can only be used to compute the hydrodynamic pressure inside the wetted area of the body.

### 3.2.4 Armand & Cointe Model

Armand & Cointe [14] resolved the singularity issue in the hydrodynamic pressure prediction by using a piecewise formulation for the pressure. This involves matching the inner domain (inside the jet) solution with the outer domain (outside the jet) solution. These two formu-

lations are linked together by matching the jet thickness on both sides of the singularity. The pressure distribution from Armand & Cointe model [14] in the outer domain,  $X < X_{sp}$ , is written out by Zhao and Faltinsen [18] through using the Bernoulli equation as:

$$p_{out} - p_0 = \rho V X_{sp} \frac{dX_{sp}}{dt} (X_{sp}^2 - X^2)^{-\frac{1}{2}} - \rho V X_{sp} \frac{dX_{sp}}{dt} [2X_{sp}(X_{sp} - X)]^{-\frac{1}{2}} + 2\rho \left( \frac{dX_{sp}}{dt} \right)^2 |\tau|^{\frac{1}{2}} (1 + |\tau|^{\frac{1}{2}})^{-2}. \quad (3.6)$$

For the inner domain,  $X \geq X_{sp}$ , the pressure can be computed using:

$$p_{in} - p_0 = 2\rho [dX_{sp}/dt]^2 |\tau|^{\frac{1}{2}} (1 + |\tau|^{\frac{1}{2}})^{-2}, \quad (3.7)$$

where  $|\tau|$  should be computed at each time step using the transcendental relationship:

$$X - X_{sp} = (\delta/\pi)(-\ln |\tau| - 4|\tau|^{\frac{1}{2}} - |\tau| + 5), \quad (3.8)$$

where  $\delta$ , the jet thickness, can be obtained from:

$$\delta = \pi V^2 2X_{sp} [4dX_{sp}/dt]^{-2}. \quad (3.9)$$

Once the  $\delta$  is known for each time step from Eq. 3.9,  $\tau$  has to be computed numerically from Eq. 3.8. Finally, the hydrodynamic pressure can be determined using both Eqs. 3.6 and 3.7. The entry velocity is still assumed constant for the derivation of this model. This might cause some uncertainty to the analysis, where the solution is considered for a free-fall water entry problem. Therefore, an attempt has been made to re-derive these analytical models so that they could be comparable with the results of free-fall wedge water entry.

### 3.3 Approach

In this body of work, existing theoretical models that have been developed to predict pressures for vertical flexible wedge water entry will be adapted to take the spray root position at each time step to output the resulting pressure on the wedge. The intent of the original theoretical models was to use an impact velocity and acceleration to determine the hydrodynamic pressure distribution and wedge kinematics at all subsequent times. The two long-term goals of this project are to (a) incorporate spray root measurements for a highly-flexible wedge, and (b) use spray root measurements on a high-speed planing vessel in waves to compute accurate hydrodynamic pressure distributions when traditional pressure sensors cannot be used. The work in this paper will establish this method for a flexible wedge by using all the models presented in section 3.2. The spray root position to be used in the current study are measured in the free-fall wedge water entry experiments. The experimental pressure measurements regarding these experiments are also used to verify the outcome of the proposed approach of this study. Since most of the theoretical models introduced in the last section are based on the non-constant entry velocity assumption, the models (2) to (5) are further developed and re-derived to consider non-constant entry velocity. Thus, the results of the proposed approach will be comparable with the experimental measurements. Although the approach will be verified for a wedge water-entry with the application of the high-speed craft, as will be explained later, the approach can be generalized to be used for all other related applications from bio-mechanics to engineering problems.

#### 3.3.1 Re-derivation of the Theoretical Models

As mentioned earlier, all the above-mentioned models except that for the Wagner are based on constant-entry assumption. In this part, the analytical models (2) to (5) are re-derived so

that the formulation could be obtained for the free-fall water entry. To account for the loss of the velocity during the free-fall, the models are re-derived from the governing equation presented by Korobkin [13]:

$$p(X, t) = -\rho \left[ \phi_t + \frac{f_x \dot{z}(t)}{1 + f_x^2} \phi_x + \frac{1}{2} \frac{\phi_x^2 - \dot{z}^2(t)}{1 + f_x^2} \right], \quad (3.10)$$

where  $f(X)$  is the function of the body shape and  $\phi$  is the velocity potential of the fluid around the body. The velocity potential is approximated using asymptotic analysis [13] and is presented as:

$$\phi(X, t) \approx -\dot{z}(t) \sqrt{X_{sp}^2(t) - X^2} - \dot{z}(t) [f(X) - z(t) - d(t)], \quad (3.11)$$

where  $d(t)$  is the splash-up height, at which the linearization is conducted. The velocity potential used by Wagner [10] is limited only to the first term of Eq. 3.11, since he didn't consider the shape of the body (small deadrise angle assumption) and conducted the linearization at the water level. All other theoretical models consider Eq. 3.11 as the velocity potential function with different assumptions. Substituting Eq. 3.11 in Eq. 3.10 will result in the general form of the pressure distribution (please see Appendix A for more details):

$$p(X, t) = \rho \dot{z}^2(t) \left[ \frac{\dot{X}_{sp}(t)}{\dot{z}(t)} \frac{X_{sp}(t)}{\sqrt{X_{sp}^2(t) - X^2}} - \frac{1}{2} \frac{X_{sp}^2(t)}{X_{sp}^2(t) - X^2} \frac{1}{1 + f_x^2} - \frac{1}{2} \frac{f_x^2}{1 + f_x^2} - \frac{\dot{d}(t)}{\dot{z}(t)} + \frac{\ddot{z}(t)}{\dot{z}^2(t)} \left( \sqrt{X_{sp}^2(t) - X^2} + f(X) - z(t) - d(t) \right) \right]. \quad (3.12)$$

The re-derived formulation for analytical models (2) to (5) can be obtained using Eq. A.11 and implementing the original assumptions of each model. The body shape function,  $f(X)$ , can be generalized for all other shapes in this formulation. In other words, this formulation can be utilized for the water entry of any object, including those applications mentioned

in the introduction, to solve for the hydrodynamic pressure during the water entry of the object. This can include a wide variety of objects from water entry of the birds to slamming of the vessels with the complicated body shape. In the following section, the formulations are derived for the wedge bodies to conduct a comparison with the available experimental measurements on wedge-water entry. However, the formulations can be easily adjusted for any specific application by updating the body shape function,  $f(X)$ .

### Re-derived Generalized Wagner

The re-derived formulation for the Generalized Wagner model can be obtained by taking into account the body shape of the wedge,  $f(X) = X \tan(\beta)$ . In this model, the linearization is conducted around the splash up level,  $d = f(X_{sp} - z)$ . Therefore, the re-derived formulation for Generalized Wagner Model will be:

$$p(X) = \frac{1}{2}\rho V^2 \left[ \frac{2\dot{X}_{sp}}{V} \frac{X_{sp}}{\sqrt{X_{sp}^2 - X^2}} - \cos^2 \beta \frac{X_{sp}^2}{X_{sp}^2 - X^2} - \sin^2 \beta + 2 - \frac{2\dot{X}_{sp} \tan \beta}{V} + \frac{2\ddot{z}}{V^2} [\sqrt{X_{sp}^2 - X^2} + (X - X_{sp}) \tan \beta] \right]. \quad (3.13)$$

The last term signifies the added mass pressure, which is now restored after the re-derivation process. With the constant-entry assumption and accounting for Eq. 3.1, the re-derived formulation would be simplified to the original formulation reported in Eq. 3.3.

### Re-derived Original and Modified Logvinovich Model

The original Logvinovich model neglects the body shape,  $f(X) = 0$ , and linearizes the problem around the initial water level,  $d = 0$ . Therefore, the re-derived formulation for the

original Logvinovich model from Eq. A.11 can be seen as:

$$p(X) = \frac{1}{2}\rho V^2 \left[ \frac{2\dot{X}_{sp}}{V} \frac{X_{sp}}{\sqrt{X_{sp}^2 - X^2}} - \frac{X_{sp}^2}{X_{sp}^2 - X^2} + \frac{2\ddot{z}}{V^2} (\sqrt{X_{sp}^2 - X^2} - z) \right]. \quad (3.14)$$

If the body shape is taken into account,  $f(X) = X \tan(\beta)$ , and the linearization is performed at the initial water level,  $d = 0$ , the re-derived formulation for the modified Logvinovich model can be shown as Eq. 3.15:

$$p(X) = \frac{1}{2}\rho V^2 \left[ \frac{2\dot{X}_{sp}}{V} \frac{X_{sp}}{\sqrt{X_{sp}^2 - X^2}} - \cos^2 \beta \frac{X_{sp}^2}{X_{sp}^2 - X^2} - \sin^2 \beta + \frac{2\ddot{z}}{V^2} (\sqrt{X_{sp}^2 - X^2} + X \tan \beta - z) \right]. \quad (3.15)$$

The last term in both Eq. 3.14 and Eq. 3.15 correspond to the added mass pressure that is restored after the re-derivation. Considering constant-entry assumption and Eq. 3.1, the re-derived formulations of Eq. 3.14 and Eq. 3.15 would be simplified to Eq. 5.10 and Eq. 3.5, respectively. Although all these analytical models are re-derived to account for the acceleration terms in the pressure formulation, the singularity problem in these models still exists.

### Re-derived Armand & Cointe Model

The outer solution for the Armand & Cointe model is obtained from the original Wagner model and therefore, the basic assumptions of this solution are similar to that of the Wagner model. In other words, the acceleration terms in the re-derived formulation of the outer solution from Armand & Cointe model will be similar to the term that exists in the original Wagner model. Thus, the re-derived formulation of the outer domain,  $X < X_{sp}$  of the

Armand & Cointe model can be re-written as Eq. 3.16,

$$p_{out} - p_0 = \rho V X_{sp} \frac{dX_{sp}}{dt} (X_{sp}^2 - X^2)^{-\frac{1}{2}} - \rho V X_{sp} \frac{dX_{sp}}{dt} [2X_{sp}(X_{sp} - X)]^{-\frac{1}{2}} + 2\rho \left( \frac{dX_{sp}}{dt} \right)^2 |\tau|^{\frac{1}{2}} (1 + |\tau|^{\frac{1}{2}})^{-2} + \rho \ddot{z} \sqrt{X_{sp}^2 - X^2}. \quad (3.16)$$

The pressure formulation for the inner domain,  $X \geq X_{sp}$ , and the matching conditions will be similar to the original Armand & Cointe model (Eqs. 3.7 to 3.9). These relationships are brought here again:

$$p_{in} - p_0 = 2\rho [dX_{sp}/dt]^2 |\tau|^{\frac{1}{2}} (1 + |\tau|^{\frac{1}{2}})^{-2}, \quad (3.17)$$

$$X - X_{sp} = (\delta/\pi)(-\ln |\tau| - 4|\tau|^{\frac{1}{2}} - |\tau| + 5), \quad (3.18)$$

$$\delta = \pi V^2 2X_{sp} [4dX_{sp}/dt]^{-2}. \quad (3.19)$$

Hereafter, the re-derived formulations, Eqs. 3.13 to 3.19, are used for the analytical models (2) to (5). For the original Wagner model, Eq. 3.2 is considered, since this model already accounts for the acceleration term. This formulation is brought here again:

$$p(X) = \frac{1}{2} \rho V^2 \left[ \frac{\pi}{\beta} \frac{X_{sp}}{\sqrt{X_{sp}^2 - X^2}} - \frac{X^2}{X_{sp}^2 - X^2} + \frac{2\ddot{z}}{V^2} \sqrt{X_{sp}^2 - X^2} \right]. \quad (3.20)$$

### 3.3.2 Methodology

When using the theoretical formulations described in the previous subsections, the spray root location from Eq. 3.1, and the impact velocity,  $V$ , is specified. The pressure data is then predicted using the equations for pressure, given as Eqs. 3.13 to 3.20. Although these methods provide a good estimation for the pressure distribution, there exists some discrepancies

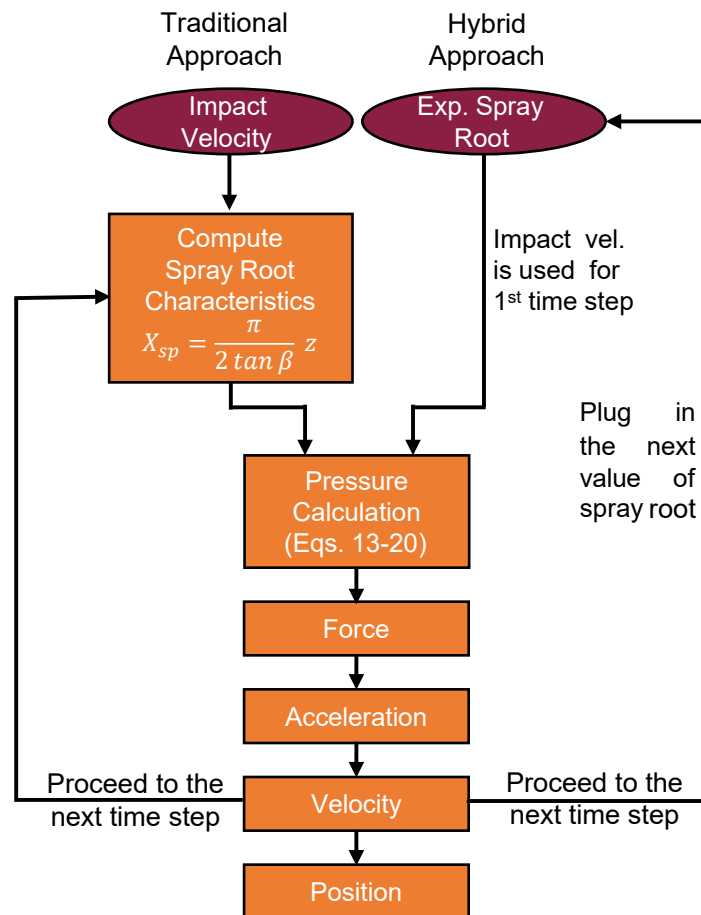


Figure 3.2: Flowchart demonstrating the traditional approach for following the theoretical formulations (left side) and the proposed hybrid approach (right side).

between the predicted values and the actual measurements due to simplifications that were made when deriving the theoretical equations. Once the pressure distribution is computed and integrated to determine the total force on the body, the rigid body kinematics can be computed.

In this study, a modified hybrid approach of adapting these theoretical models is implemented. The experimental spray root position is used as an input into the equations for hydrodynamic pressure (Eqs. 3.13 to 3.20). Once the pressure distribution is computed, then the kinematics of the wedge can also be computed. The differences in the calculation method are shown in Fig. 3.2, where the traditional method is shown on the left and the proposed method is shown on the right. In the traditional method, the calculation is looped through time using updated velocity information. Using the proposed hybrid method, a new spray root position and velocity is taken to estimate pressure at each time. The goal of this new approach is that the method be extended to provide accurate predictions of pressure on highly flexible structures and vessels operating in waves.

### 3.3.3 Experiments

To show a demonstration of the new proposed method, a case study is presented using experimental data from Javaherian et al. [4]. In this experimental study, spray root, hydrodynamic pressure, and rigid body kinematics were measured. For the work presented in this paper, the spray root information is used as the input into the theoretical models, as shown in Fig. 3.2. Since the wedge kinematics and hydrodynamic pressure were also measured, these experimental measurements can then be compared to the results of the new hybrid approach. Finally, the theoretical model predictions for hydrodynamic pressure (shown as the traditional method in Fig. 3.2) will also be compared with the experimental measurements

and the new approach.

The exact details of the experiment from this case study can be found in [44, 48]. The flexible aluminum wedge model with a deadrise angle of  $\beta = 20^\circ$  is used in this case study. Fig. 5.2 shows the locations of all the sensors used in the experiment as seen from the top view. A string potentiometer was used to measure the position time-history of the wedge during the experiments. The vertical impact acceleration of the wedge was recorded by three accelerometers that were mounted on both sides of the wedge ( $A_1$ ,  $A_2$  and  $A_{DC}$ ). An array of seven pressure transducers were mounted on the bottom of the wedge to measure the hydrodynamic loading during the water impact. All sensors were located within increments of 33.9 mm at one side of the wedge bottom, as it is shown in Fig. 5.2. The  $x$ -direction is called the characteristic line. The pressure measurements were sampled at a sampling rate of 200 kHz. Spray root measurements were taken using high-speed photography with 2000 fps and digital image processing to quantify the spray root evolution.

Fig. 3.4a shows the spray root propagation on the wedge bottom for different drop heights of  $H = 7.9, 25.4$  and  $50.8$  cm (corresponding to impact velocity of  $V_0 = 1.219, 2.175$  and  $3.057$  m/s, respectively). The spray root propagation speed for these drop heights is also plotted in Fig. 3.4b. The spray root propagation speed was computed by taking the time-derivative of the spray root position-time history. It can be seen that the higher impact velocities result in faster propagation of spray root. This is due to the fact that the spray root speed is directly related to the wedge speed. As the wedge penetrates into the water, the hydrodynamic force decelerates the wedge and therefore, the spray root also experiences a loss in the propagation speed. These results are taken from Javaherian et al. [4] and are the inputs that are taken in the proposed method.

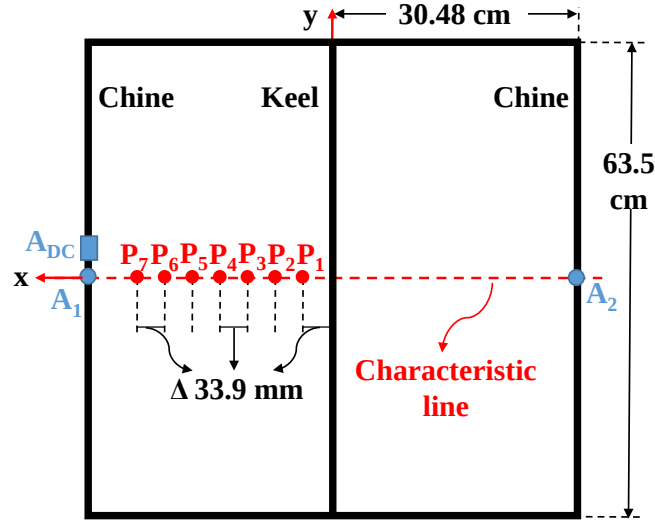


Figure 3.3: Plan view of the wedge showing the dimensions of the wedge and locations of the sensors.

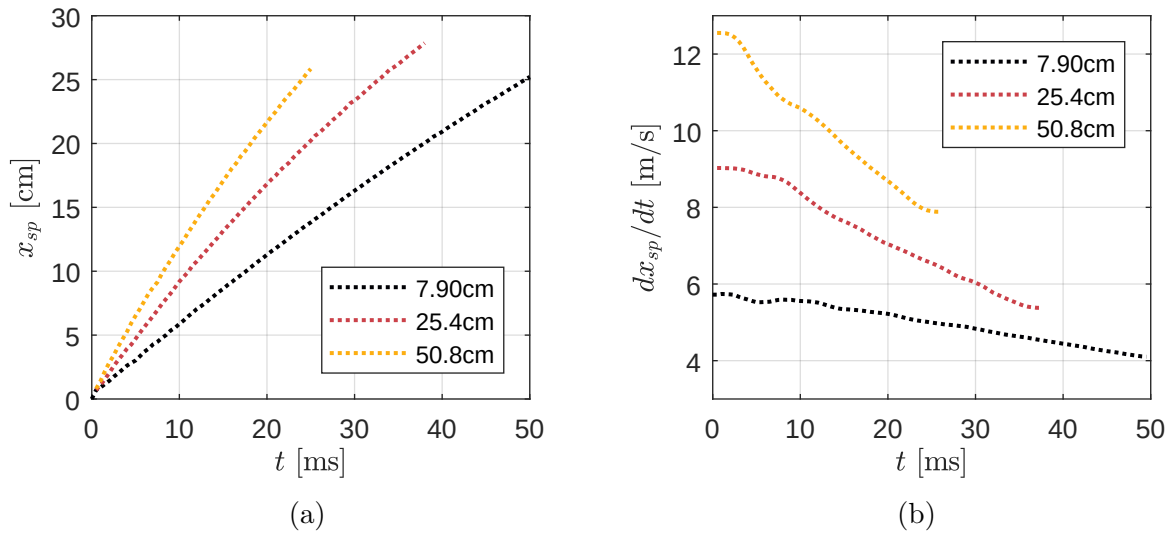


Figure 3.4: The (a) position and (b) speed of the spray root on the bottom of the wedge for different drop heights.

## 3.4 Results and Analysis

In this section, results of the hybrid approach will be reported and compared to pure theory and pure experiment. First, the hydrodynamic pressure distribution will be presented and discussed. Following this will be the presentation and discussion of the pressure time histories compared at experimental pressure sensor locations. Finally, the kinematics estimated from the hybrid approach will be presented and discussed. It is noted that because the kinematics are dependent upon the hydrodynamic pressure, the focus of this section shall be more on the hydrodynamic pressure.

### 3.4.1 Pressure Distribution

Pressure data estimated with the hybrid approach via different theoretical models using experimental spray root data for  $V_0 = 2.175$  m/s are depicted in Fig. 3.5. Results are shown as spatial distribution in different time steps that correspond to the time, in which sensors  $P_1$  to  $P_7$  reach their peak. At each time step, the peak of pressure occurs at the close proximity to the spray root, by which the high pressure wave is carried on. The pressure decays as the distance becomes farther from the spray root towards the keel in the inner domain ( $x < x_{sp}$ ). As far as the area greater than the outer domain is concerned ( $x \gg x_{sp}$ ), the hydrodynamic pressure is zero since the pressure is equal to the atmospheric pressure. Results show that the estimated pressure data from Armand & Cointe and also Logvinovich models predict better results compared to the other models. These models can reasonably predict the peak location and magnitude as well as the residual pressure. The comparison reveals that the results of the hybrid approach in conjunction with the Armand & Cointe match very well with the experiments. This agreement is noticeable especially for the area after the peak pressure. This is due to the fact that the Armand & Cointe model is the only model that

can estimate the pressure data at and after the spray root thanks to its piece-wise solution. Fig. 3.6 quantitatively compares the magnitude and the location of the peak pressure reconstructed from each model with the experimental results. These values are plotted versus location of the sensors for  $V_0 = 2.175$  m/s. The comparison in Fig. 3.6a shows that Armand & Cointe model can predict the location of the peak very well. The error for the peak location is mostly less than 2% for this method. As far as the magnitude of the peaks is concerned, Fig. 3.6b shows the Armand & Cointe model predicts the magnitudes to within 7%. Comparison also shows that after the Armand & Cointe method, the Logvinovich model results in better reconstructed pressure data compared to the remaining methods. The Generalized Wagner, Modified Logvinovich and original Wagner method predict the results with slightly higher deviations.

### 3.4.2 Comparison Between Traditional and the Hybrid Approach

In Fig. 3.2, the hybrid approach that was used in this study was also compared with the traditional approach. To verify the hybrid approach, the study has been repeated using the traditional approach for all the above-mentioned theoretical models. Results are compared to the experimental pressure to gauge the validity of the hybrid approach and evaluate the effectiveness of using this semi-empirical approach. Figs. 3.7 and 3.8 show a comparison of the pressure distribution from pure theory (traditional approach), the hybrid approach (subject of this paper), and pure experiments. These highlight the improved accuracy of these models. This is especially evident for the first two sensors,  $P_1$  and  $P_2$  in all models. Comparing the peaks resulted by the Generalized Wagner theory in Fig. 3.7, one can observe that the peak of sensor  $P_1$  is computed to be 35% higher than the experimental peak. For the hybrid approach, this deviation is reduced to 10%. This improvement can also be observed

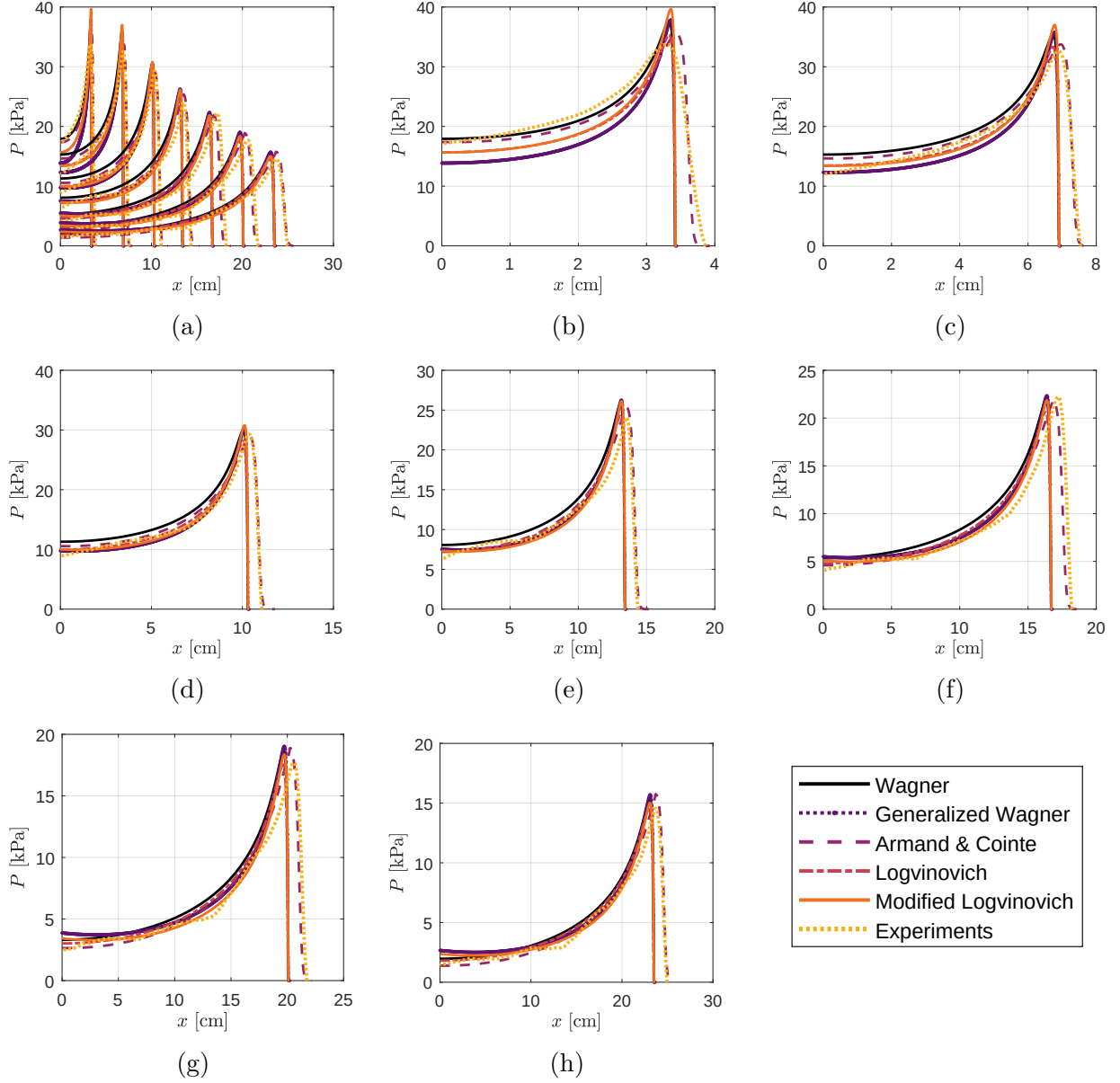


Figure 3.5: Pressure distribution estimated from the hybrid approach from different models at the time-steps in which the peak reaches sensors  $P_1$  to  $P_7$ , respectively. These time-steps are (a) all, (b)  $t = 3.5$  ms, (c)  $t = 7.5$  ms, (d)  $t = 11.5$  ms, (e)  $t = 15.5$  ms, (f)  $t = 20$  ms, (g)  $t = 25$  ms and (h)  $t = 30.5$  ms for  $V_0 = 2.175$  m/s.

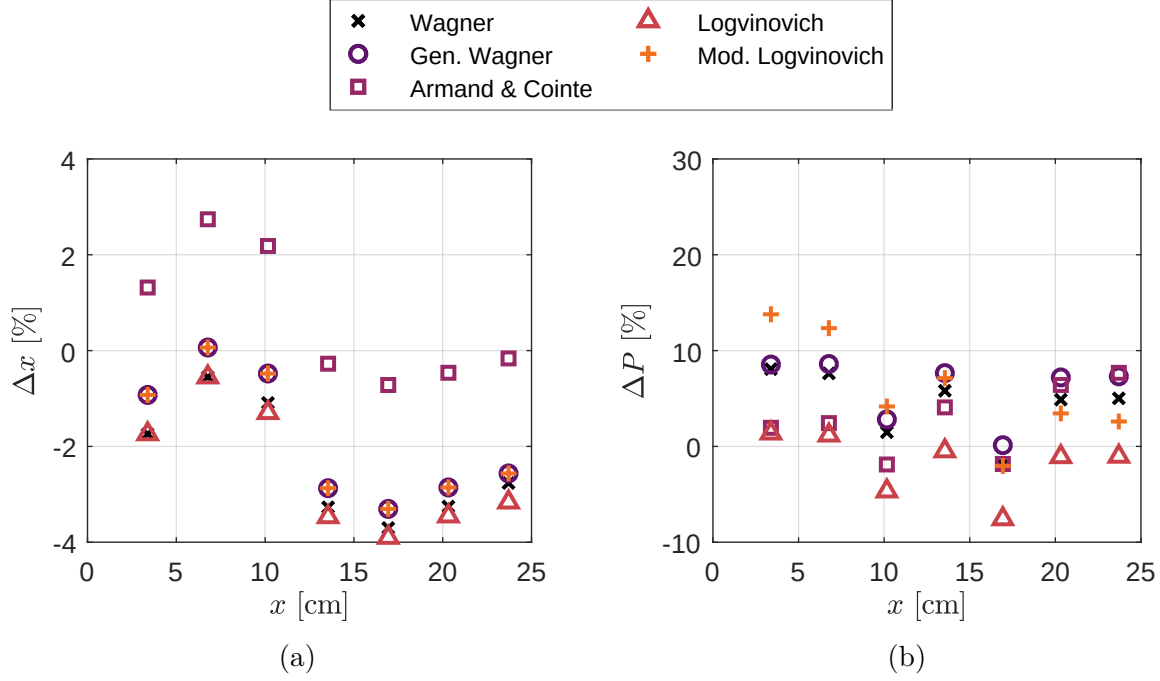


Figure 3.6: Comparison between (a) location and (b) magnitude of the peak pressure predicted from different models with the experimental results for  $V_0 = 2.175$  m/s

in all other models (Fig. 3.8).

It can be observed that for some of the models, the traditional approach can predict the location of the peaks slightly better than the hybrid approach. However, as shown earlier in Fig. 3.6a, the error of estimating the location of the peak pressure for the hybrid approach stays within 3% for most cases.

### 3.4.3 Re-derivation Improvement

It was shown earlier that the original models are re-derived to account for the loss of the velocity during the free-fall period. In this section, the effect of restoring the acceleration term in the pressure formulations is investigated. Fig. 3.9 shows a comparison between the pressure distribution estimated from the hybrid approach using original and re-derived

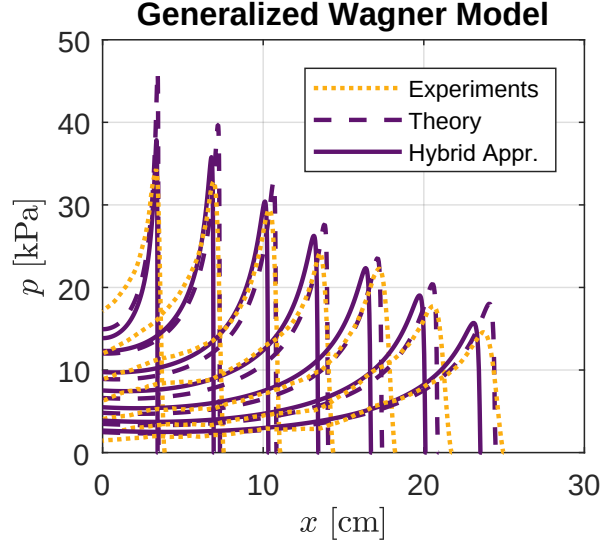


Figure 3.7: Spatial pressure distribution resulted from the traditional (theory) and the hybrid approach using the Generalized Wagner model for  $V_0 = 2.175$  m/s. Results are compared with the experimental pressure distribution. The curves correspond to the time steps that were presented in the Fig. 3.5.

Armand & Cointe models. The acceleration term is restored in the re-derived formulation, while it was neglected in the original model. The comparison shows that while the magnitude and the location of the peaks are almost unchanged between original and re-derived models, the re-derived Armand & Cointe model estimates the residual pressure (tail of the pressure curve) more accurately when compared to the experimental observations. For instance, Fig. 3.9b reveals that the deviation between estimated and experimental pressure at  $x = 2$  cm reduces from 80% for the original model to less than 2% at  $t = 15.5$  ms. This time corresponds to the moment, in which the peak pressure reaches the sensor  $P_4$ .

### 3.4.4 Pressure Time-History

The hydrodynamic pressure measured in the Javaherian et al. [4] experiment were at single points. The experimental pressure-time history for the impact velocity of  $V_0 = 2.175$  m/s

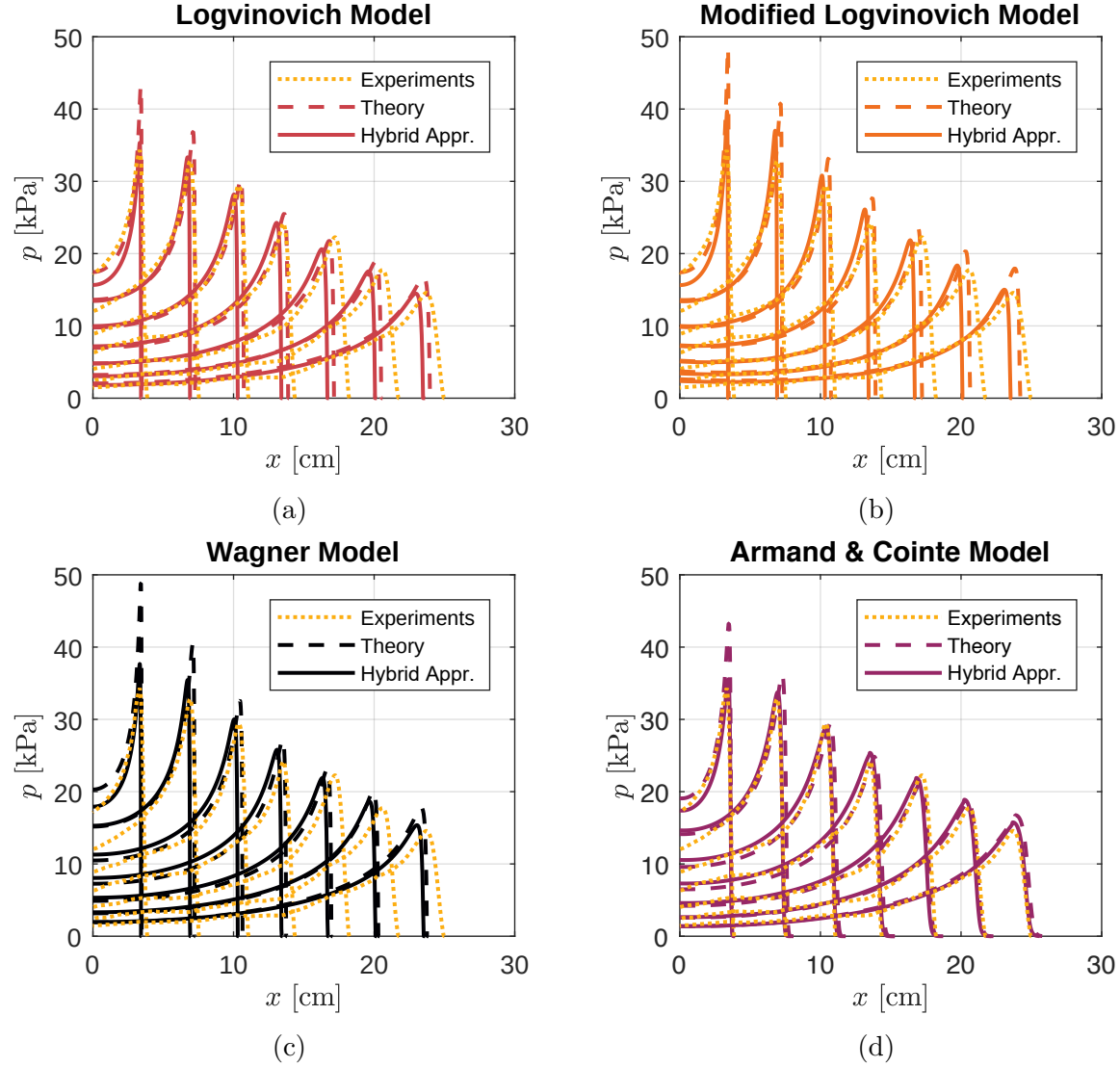


Figure 3.8: Spatial pressure distribution resulted from the traditional (theory) and the hybrid approach using (a) Logvinovich, (b) Modified Logvinovich, (c) Wagner and (d) Armand & Cointe model for  $V_0 = 2.175$  m/s. Results are compared with the experimental pressure distribution. The curves in each figure correspond to the time steps that were presented in the Fig. 3.5.

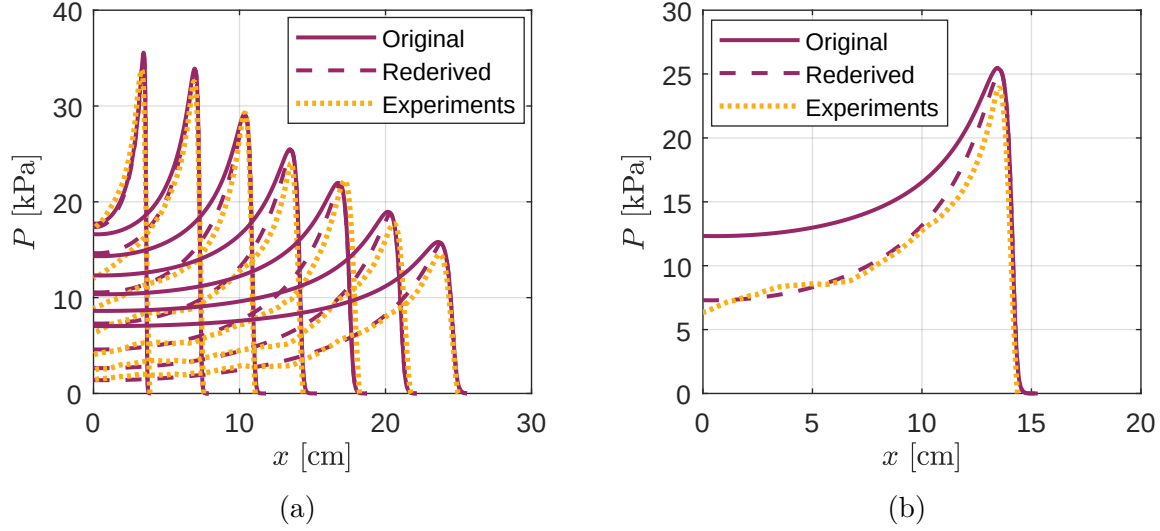


Figure 3.9: Comparison between the pressure distribution from the hybrid approach using original and re-derived Armand & Cointe model at (a) all time steps reported in the Fig. 3.5 and (b)  $t = 15.5$  ms, for the impact velocity of  $V_0 = 2.175$  m/s.

is depicted in Fig 3.10 along with the estimated values from the hybrid approach using the re-derived Armand & Cointe model. The curves from left to the right denote the pressure history at the location of the sensors  $P_1$  to  $P_7$ , respectively (see Fig. 5.2). The comparison indicates that the pressure time-history estimated from hybrid approach matches very well with the experimental measurements.

Figs. 3.11 and 3.12 show a similar comparison for the impact velocity of  $V_0 = 1.219$  and  $V_0 = 3.057$  m/s, respectively. The comparison shows that the pressure-history for these drop heights are also predicted well using the proposed approach. Results from the hybrid approach are plotted until the chine-wetted moment. That's why they cover a shorter range compared with the experiments.

### 3.4.5 Force and Kinematics

To estimate the kinematics of the wedge, the force that exerts on the body during water

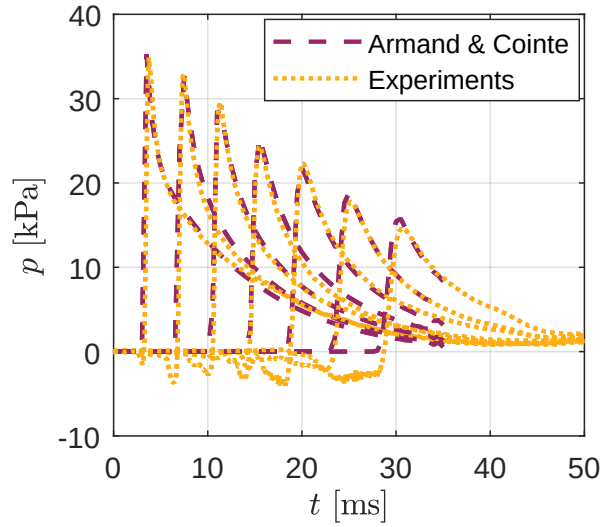


Figure 3.10: Pressure-time history estimated by the hybrid approach from re-derived Armand & Cointe model compared with the experimental results for  $H = 25.4$  cm,  $V_0 = 2.175$  m/s.

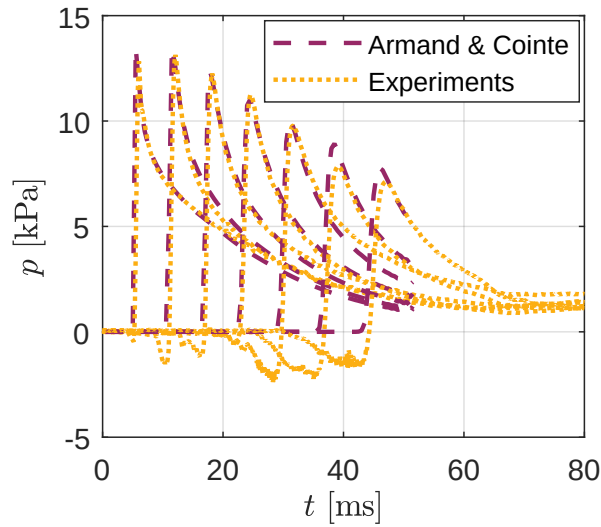


Figure 3.11: Pressure-time history estimated by the hybrid approach from re-derived Armand & Cointe model compared with the experimental results for  $H = 7.9$  cm,  $V_0 = 1.219$  m/s.

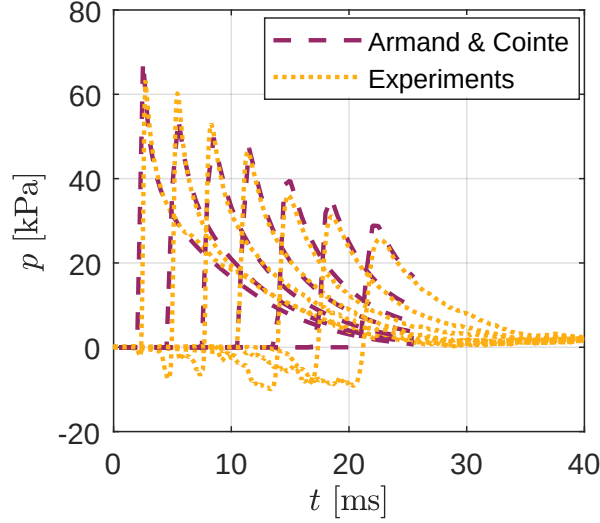


Figure 3.12: Pressure-time history estimated by the hybrid approach from re-derived Armand & Cointe model compared with the experimental results for  $H = 50.8$  cm,  $V_0 = 3.057$  m/s.

entry is computed from the pressure-spatial distribution. Fig. 3.13 shows the force-time history at the characteristic line estimated from the hybrid approach using Armand & Cointe model. As can be seen, results show a good agreement with the experimental measurements. The hydrodynamic force increases upon the penetration of the wedge into the water and reaches the maximum at  $t = 14.5$  ms due to the increase in the wetted length, on which the hydrodynamic pressure is exerted. Even though the wetted length continuously increases even after this moment, the hydrodynamic force decreases due to the loss of the wedge speed and therefore the reduction in the hydrodynamic pressure. Given the force-time history, the acceleration-, velocity- and position-time history can be obtained, subsequently. The kinematics of the wedge estimated from the Armand & Cointe model with the hybrid approach is reported in Fig. 3.14 for the impact velocity of  $V_0 = 2.175$  m/s. Since the kinematics are estimated using experimental spray root data, they are plotted until the chine-wetted moment.

The vertical acceleration of the wedge estimated from the Armand & Cointe model is illus-

trated in Fig. 3.14a along with the experimental measurements. The acceleration starts with a negative value of  $-8.95 \text{ m/s}^2$ , which is close to the gravitational acceleration of the falling body. The wedge experiences a sharp rise in the acceleration following the impact moment and reaches the maximum of acceleration of  $39.58 \text{ m/s}^2$  at  $t = 14.5 \text{ ms}$ . This magnitude is 5% higher than the corresponding experimental peak for the acceleration. This moment corresponds to the time, in which the hydrodynamic force reaches the maximum (Fig. 3.13). The wedge decelerates after this moment due to the reduction in the hydrodynamic force.

The vertical velocity of the wedge is shown in Fig. 3.14b. The estimated velocity that starts with the impact velocity of  $2.175 \text{ m/s}$ , matches very well with the experiments for the entire time. The rate of the loss of the speed at early stages of penetration is small due to the small wetted length and thus, the small hydrodynamic force. Considering the hydrodynamic force due to impact and the upward acceleration, the magnitude of the velocity decreases tangibly as the wedge penetrates into the water.

Lastly, Fig. 3.14c depicts the vertical position-time history of the wedge for the impact velocity of  $V_0 = 2.175 \text{ m/s}$ . As the wedge penetrates into the water, the position curve is nearly linear. The comparison shows a very good agreement between reconstructed position and the experiments except the very small deviation at the final time steps.

### 3.4.6 Structural Response

Fig. 3.15 indicates the maximum out-of-plane deflection of the bottom panel of the wedge for different impact velocities. The deflection is measured using Stereoscopic-Digital Image Correlation, S-DIC ([4]). For each curve, as time elapses, the deflection increases with the increase in the hydrodynamic force. The maximum deflection occurs with a short delay after the maximum hydrodynamic force is reached. For example, the maximum deflection of the

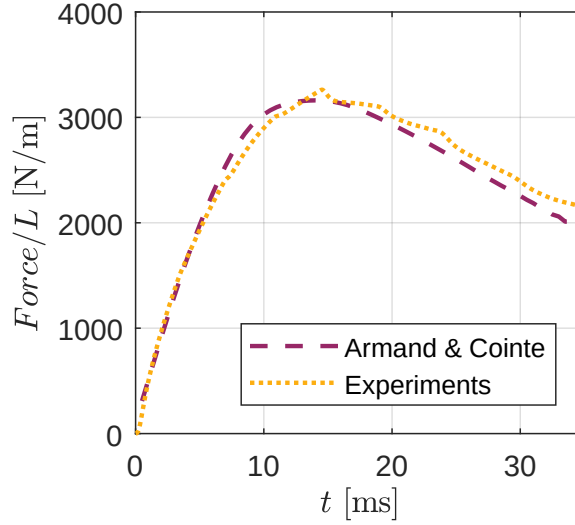


Figure 3.13: Force-time history at the characteristic line estimated from the hybrid approach using Armand & Cointe model for  $V_0 = 2.175$  m/s. Results are compared with the experimental measurements.

wedge panel for the impact speed of  $V_0 = 2.175$  m/s happens at  $t = 21$  ms and is equal to  $w_{max} = 0.88$  mm. This moment is 6 ms delayed compared with the moment in which the maximum hydrodynamic force occurs ( $t = 16$  ms as shown in Fig. 3.13). In other words, for this impact speed, it takes 6 ms for the structure to respond to the maximum hydrodynamic force and reach the maximum deflection.

As was expected, Fig. 3.15 shows that the higher impact velocity yields to a higher maximum deflection in the panel. In fact, the higher the impact velocity, the higher the hydrodynamic force, and thus, the higher the deflection. The maximum deflection for the impact speed of  $V_0 = 3.057$  m/s is  $w_{max} = 1.43$  mm, which is 100% and 226% greater than the corresponding values for impact speeds of  $V_0 = 2.175$  and  $V_0 = 1.219$  m/s, respectively. Furthermore, this figure demonstrates that as the impact velocity increases, the maximum deflection in the panel occurs earlier. This is attributed to the fact that higher impact velocity results in the faster spray root propagation (Fig. 3.4) and therefore, faster pressure wave traveling on the bottom of the panel. The hybrid approach will be extended in future to also predict the

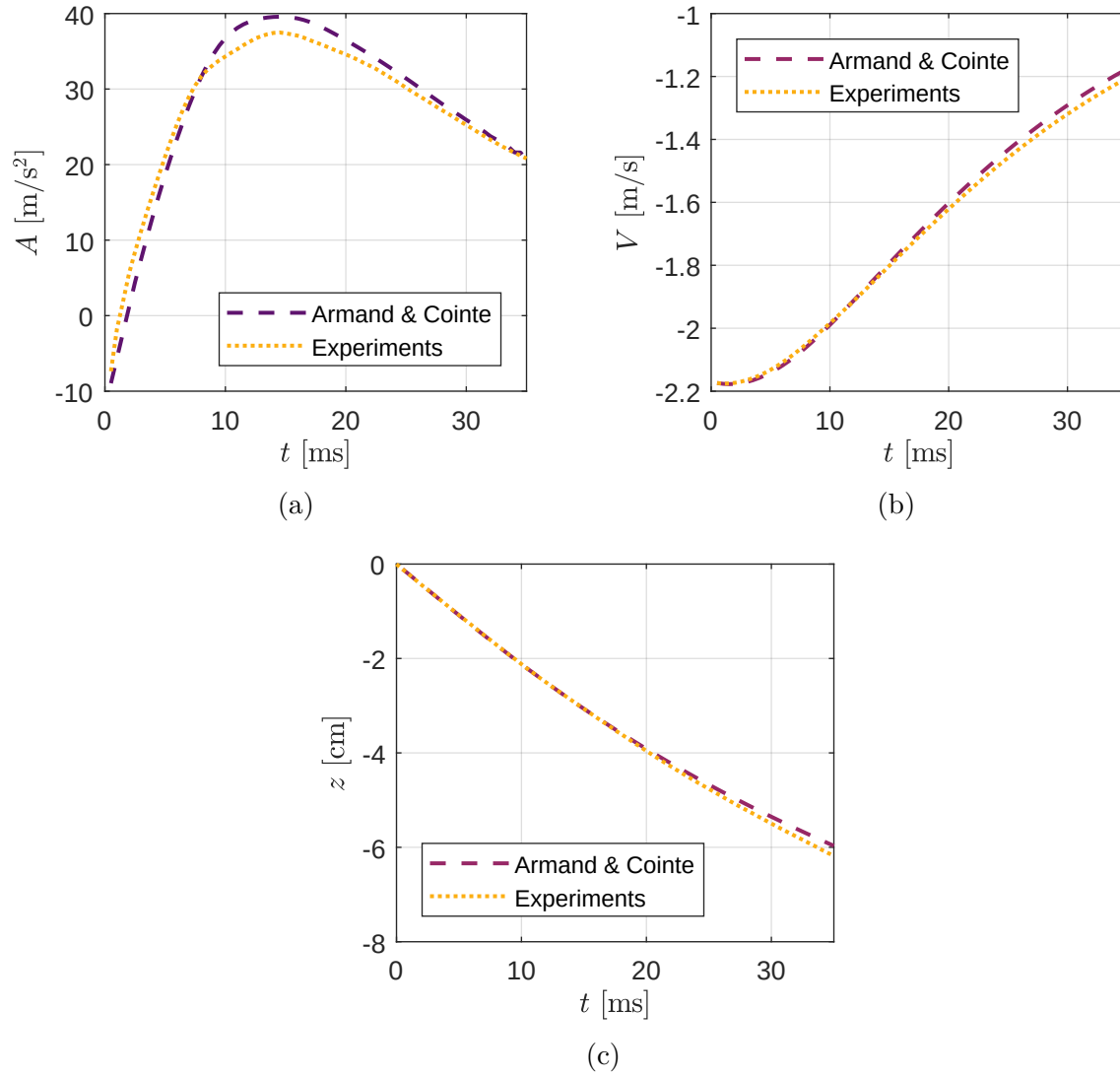


Figure 3.14: The (a) acceleration, (b) velocity and (c) position of the wedge estimated from the hybrid approach using re-derived Armand & Cointe model for  $V_0 = 2.175$  m/s. Results are compared with the experimental measurements.

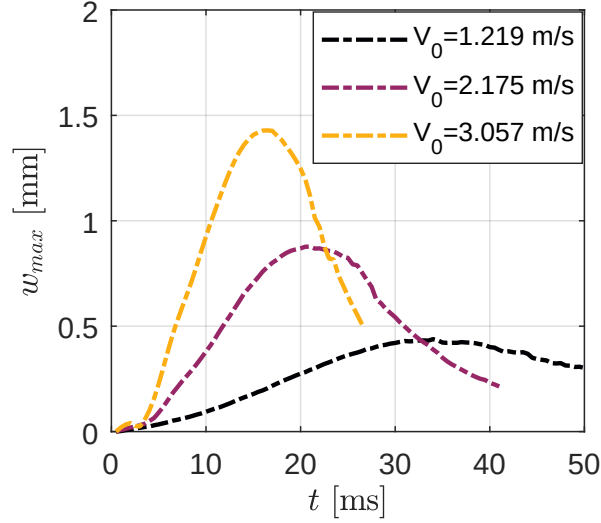


Figure 3.15: Maximum out-of-plane deflection of the wedge bottom panel during the experiments. Results are shown for three different impact velocities.

structural response of the panel in water entry of flexible structures.

### 3.5 Spray Root Prediction

It was mentioned in section 3.2.1 that most of the theoretical models use Eq. 3.1 to predict the spray root location. Since the location of spray root is one of the main inputs for these models, it is necessary to evaluate the accuracy of this equation. For a better comparison, the spray root location is also calculated using Vorus model, which is a non-linear hydrodynamic model [45]. Since these models are available for a rigid wedge, the comparison is conducted using the experimental results for the spray root propagation of a rigid wedge. This wedge has the same dimensions as that for the flexible wedge, except that the bottom panel is made of a 12.7 mm thick aluminum sheet in this wedge. Fig. 3.16 shows the spray root propagation from Wagner and Vorus model compared with the experimental measurements for three different impact velocities. Results show that the experimental measurements fall

between the Vorus and Wagner estimations. In fact, Vorus model underestimates the spray root propagation compared to experiments. On the other hand, Wagner model over-predicts the spray root propagation. This is important since, as noted before, the spray root location is the main input for predicting the pressure distribution. An uncertainty in predicting the location of spray root will result in an inaccurate estimation of the hydrodynamic pressure. The kinematics of the model then also will be impacted by this uncertainty.

To further study the accuracy of prediction, results are non-dimensionalized using the following definitions. The non-dimensional time, which represents the time it takes for the chine of the wedge to reach the calm water level after the impact, is defined as:

$$t^* = \frac{\int V(t)dt}{z_{ch}}, \quad (3.21)$$

where  $W$  is the wedge instantaneous vertical velocity and  $z_{ch}$  is the vertical distance of the chine with respect to the keel. Non-dimensional spray root position is also defined as the vertical position of spray root with respect to the keel normalized by the vertical distance from the keel to the chine ( $z_{ch}$ ), or:

$$z^* = \frac{z_{sp}}{z_{ch}} \quad (3.22)$$

Results presented in Fig. 3.16 are non-dimensionalized using the above-mentioned equations and are reported in Fig. 3.17. The non-dimensional curves are shown for three different impact velocities. The comparison shows that the results of each approach collapse nearly into one curve. Again, the non-dimensionalized spray root propagation derived from experiments falls between Vorus and Wagner estimations. The slope of the non-dimensional curves of Vorus and Wagner models shows a difference of -4% and 9% compared to the experiments, respectively. Therefore, a correction factor of 0.91 is suggested for the Wagner spray root equation presented in Eq. 3.1. It should be noted that this factor is obtained only from the

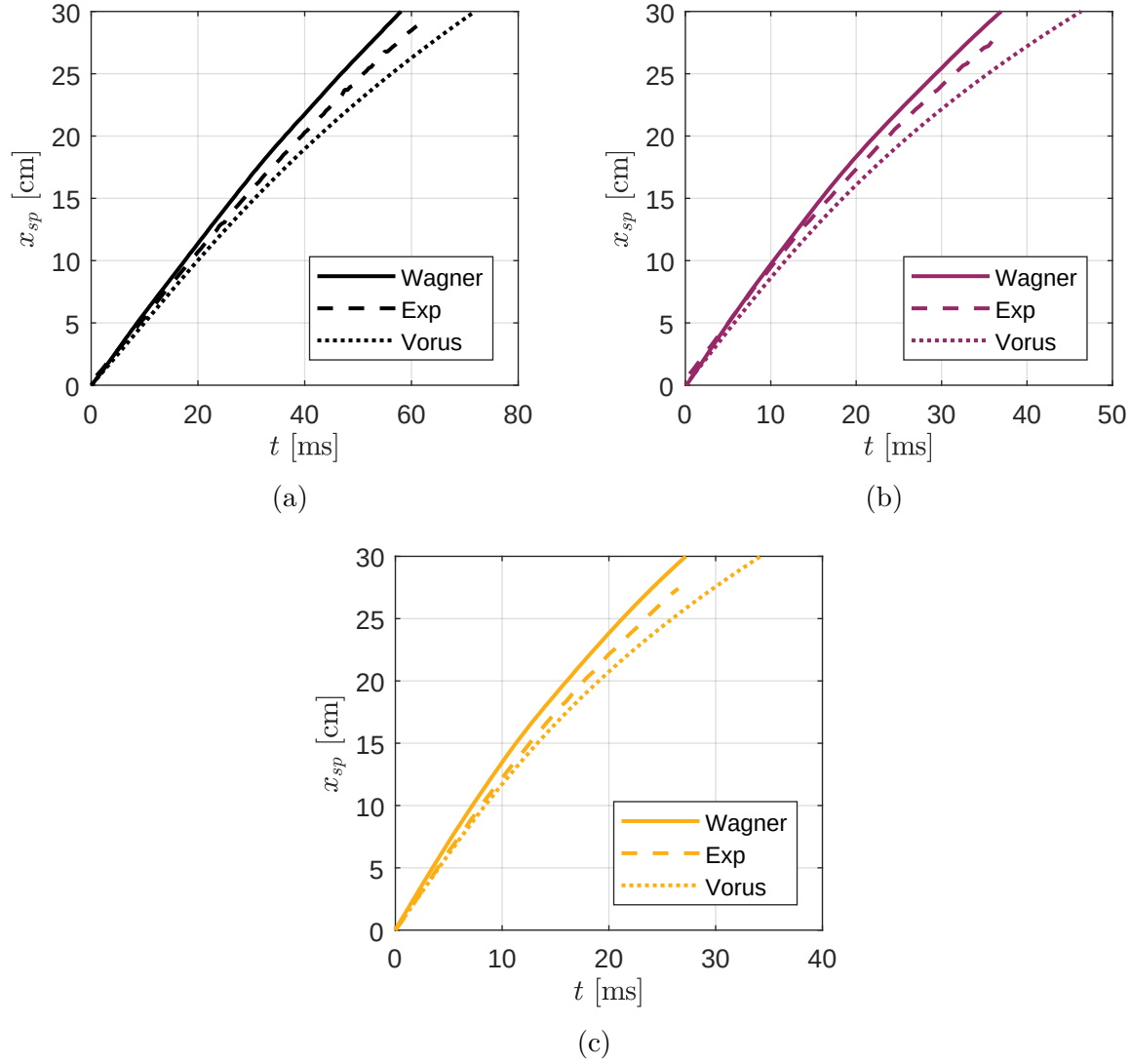


Figure 3.16: The spray root propagation calculated from Wagner [10] and Vorus [45] models for (a)  $V_0 = 1.221$  m/s, (b)  $V_0 = 2.175$  m/s and (c)  $V_0 = 3.057$  m/s. Results are compared with the experimental measurements.

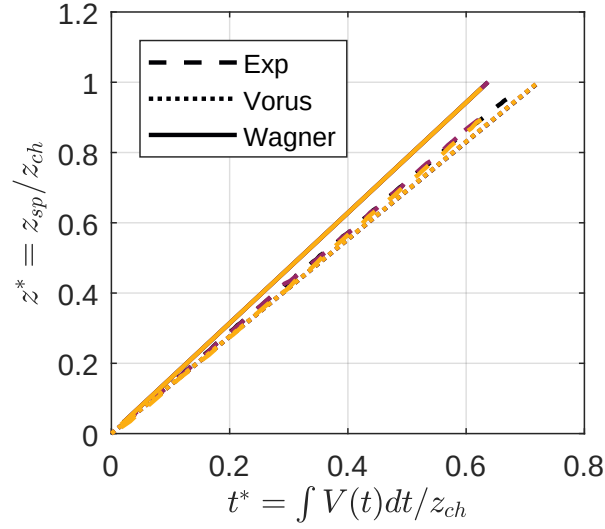


Figure 3.17: Non-dimensional spray root location plotted on top of each other. Results are shown for three different impact velocities.

experiments conducted in this study and should be verified for other deadrise angles.

## 3.6 Conclusions

In this paper, a new hybrid approach was proposed to estimate the hydrodynamic loading and kinematics of the wedge during water entry. This approach incorporates the spray root measurements into the traditional theoretical models. In fact, the spray root location is used as the input for the traditional theoretical models to estimate the hydrodynamic pressure and kinematics. This approach was evaluated by studying free-fall water entry of a flexible wedge with different impact velocities. A set of experiments conducted by Javaherian et al. [4] was considered as the case study. To have a better comparison with the experimental data, the existing theoretical models are re-derived so that the loss of the velocity during the impact is taken into account in the formulation. The experimental spray root propagation was then used to estimate the pressure distribution via the hybrid approach using five

different theoretical models. Results were then compared with the experimental pressure measurements. The comparison showed that the hybrid approach in conjunction with the Armand & Cointe model has the best prediction for pressure distribution. The location and the magnitude of the peaks were predicted within 2% and 7% of error, respectively.

To verify the proposed hybrid approach, the study was repeated for the case study using the traditional theoretical models (pure theory). The computed pressure distribution from traditional approach were then compared with those of the hybrid approach. The comparison demonstrated that the hybrid approach could predict the peak magnitudes better than the pure theory. As an instance, the error of estimating the first peak, corresponding to  $P_1$  for the experiments, could improve from 35% for the Traditional Approach to 10% for the hybrid approach using Generalized Wagner model.

It was also shown that the hybrid approach can appropriately predict the kinematics of the wedge. Acceleration, velocity and the position of the wedge estimated from the Armand & Cointe model were compared with the experimental results that showed a very good agreement.

This proposed hybrid approach will be extended in future to take into account the structural response. The current study was the first step to develop this technique for predicting the hydrodynamics and structural response in water entry of highly-flexible structures.

## 3.7 Acknowledgements

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# Chapter 4

## Structural Response due to the Slamming of High-Speed Craft by Water Entry and Towing Tank Experiments

(Most of the content in this chapter is reprinted with permission from “Structural Response due to the Slamming of High-Speed Craft by Water Entry and Towing Tank Experiments” by M. Javad Javaherian, Zhongshu Ren, and Christine Gilbert, 2019, Proceedings of the 33<sup>rd</sup> Symposium on Naval Hydrodynamics)

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(ABSTRACT) Small high-speed craft endure repeated slams into waves in the ocean. In an effort to better design these craft, it is important to understand this fluid-structure interaction when materials such as composites are used in their construction. In this paper, the slamming of composite high-speed craft is experimentally investigated in two phases. In the first phase, nearly free-falling water entry experiments are conducted on a wedge with flexible bottom panels. Kinematics, hydrodynamics, structural deformation, and spray-root propagation on the bottom of the wedge are measured and analyzed. It was found that for

a composite panel, small outward (toward the water) deflections are observed at very early times in the impact event and decaying vibrations are observed after chinewetting. The second phase of this paper is on analysis of a tow-tank test of a rigid composite planing-hull model performed at the U.S. Naval Academy. Experiments conducted in regular waves were examined in terms of their kinematics and pressure loads. It was found that heave velocity is independent of the towing speed for a range of speeds from 7.6 to 9.0 m/s. The goal of this analysis is to begin planning of non-traditional towing-tank tests that will be conducted at the VT Advanced Towing Tank Facility. These future VT experiments, Phase III, will combine the flexible panel of Phase I with the hull form and motions, which are analyzed in Phase II to study the fluid-structure interaction in the slamming of a flexible-planing hull.

## 4.1 Introduction

In recent years, there has been increased composite use in the construction of high-speed planing craft. One of the critical design parameters of high-speed craft is slamming into waves, which occurs when the hull repeatedly becomes airborne and impacts the wavy water surface. Slamming can cause structural failure, crew injury and may compromise the overall performance of the vessel. To this end, computational, analytical and experimental approaches have been conducted by many researchers, yet there are some challenges in understanding the behavior of these vessels against slamming when structural response becomes nonnegligible.

Water entry of a wedge is one of the methods by which researchers study slamming of high-speed planing craft. In this case, the wedge represents a cross section of these vessels and examines the problem from a 2-D plane when this cross section becomes airborne and falls vertically into the water. Early mathematical models on this topic were formulated for

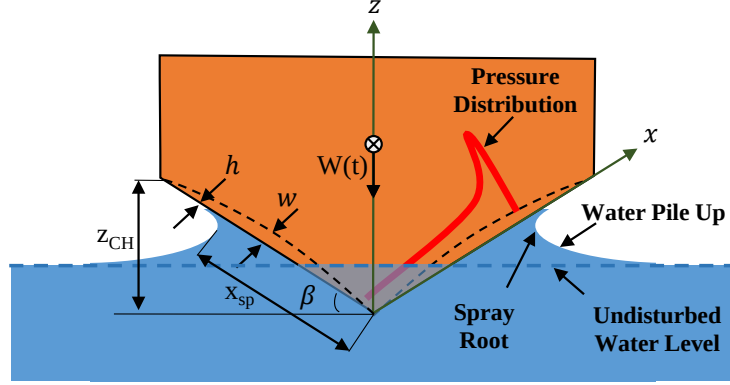


Figure 4.1: A schematic depicting the vertical water entry of a flexible wedge.

seaplane landing. The first was presented by von Karman [9], whose mathematical model was later developed by Wagner [10] who added asymptotic theory. The Wagner model assesses a reasonable impact pressure for a wide range of deadrise angles, yet a singularity at the spray-root exists in the model due to the fact that this model did not consider the jet domain and was focused only on the outer domain. Researchers further developed the Wagner model by considering the jet domain (inner domain) in the model. They could solve the singularity by matching the inner and outer domains (Armand & Cointe [14]). These models are now used to study pressure distribution (Korobkin [13]), spray and jet formation (Vorus [7]), oblique and asymmetric impacts (xu et al. [39], Judge et al. [8]) as well as the hydroelasticity of a composite beam (Qin and Batra [40]).

As a wedge model with the deadrise angle  $\beta$ , passes through a still-water surface, the resulting disturbance forms water pile-up and a spray jet, as shown in Fig. 4.1. The intersection between the spray jet and water pile-up is called the spray-root. The peak pressure lags behind the location of the spray-root, and if the velocity of the wedge is constant, the distance between the peak pressure and the zero pressure point is constant. When the velocity is nonconstant, the distance between the zero pressure and peak pressure scales with the instantaneous velocity (Vorus [7] and Judge et al. [8]). For water entry of a wedge

with a deformable bottom, the hydrodynamic loading is also dependent on the structural response of the wedge because the deformed surface alters the fluid dynamics. Because of this dependence, the flexible-wedge-drop problem becomes more challenging to predict with classical hydrodynamic models.

There have been many experimental investigations on wedge-water entry, and a few are summarized here. Lewis et al. conducted some experiments on water entry of a rigid wedge with a deadrise angle of 25 degrees [29]. Pressure and acceleration time-histories of the wedge were measured during the experiments. Lewis also investigated the jet formation to correlate it with the pressure distribution. Cooper and McCue investigated highly flexible, open wedges made of rubber [49]. A pneumatic actuator was used to drive the wedge into the water. They used a simplified version of digital image correlation to measure the strain on the rubber panels. They found that vibrations resulting from the slam are dominated by the structural properties of the membrane. Panciroli and Porfiri used particle image velocimetry (PIV) and high-speed video recording to assess the effects of flexural rigidity on hydrodynamic loading, location of maximum and minimum pressure, as well as deflection of the tip of the open wedge [34]. They compared their experimental results with their semi-analytical model. This model incorporates the experimental loading with the Kirchhoff-Love linear plate model to study the multimodal vibration. Shamset al. also experimentally investigated water entry and exit of a flexible wedge [35]. They used PIV to compute the velocity and pressure fields around the wedge. Free-surface elevation was also studied in their experiments. Using experimental results for kinematics, hydrodynamic loading, and water pile-up, they validated parts of their semi-analytical model for slamming. In this model, Wagner theory is adopted to study the fluid flow while the Euler-Bernoulli beam theory is used to model the elastic wedge.

The interaction between impact pressure and structural response of a flexible wedge was

experimentally studied by Ren et al. [36, 44] and Javaherian et al. [48]. They implemented Stereoscopic-Digital Image Correlation (S-DIC) to measure the displacement and out-of-plane deflection of the wedge during the impact. They also proposed a one-way coupled semi-theoretical model that used the Vorus model as the loading condition on a dynamic, linear finite element plate model. However, they found better agreement in comparisons to experimental results when using experimentally measured pressures in their model in place of the Vorus model (Ren et al. [37]). Javaherian et al. introduced a novel method for tracking and measuring spray-root propagation along the bottom of a wedge [48]. They used simultaneous side- and bottom-view cameras to map the spray-root. [43] proposed a two-way coupled, low fidelity, semi-analytical approach to simulating wedge-water entry. This method works when deflections are small and fall into the linear regime. The coupling relates the spray-root location in the fluid to the deflection on the wedge panel in contact with the water. This method showed very good agreement with experimental results presented in Javaherian et al. [48] and Ren et al. [44].

Towing-tank experiments are an extension on water-entry experiments. In addition to heave motions, the hull is free to pitch and is towed at a prescribed forward velocity. This approach involves measuring the kinematics, pressure, etc on a hull due to slamming events, which are caused by incoming waves ([50], [51] and [52]). Garne et al. conducted towing-tank experiments to get a detailed picture of the pressure distribution carrying a planing craft in calm water and in waves [53]. They used 14 pressure sensors to measure the slamming pressure on the bottom of the hull. They showed that the hydrodynamic pressure is a rapidly changing and propagating distribution. Thus, using point pressure transducers do not give a true picture of the instantaneous distribution. They concluded that reconstruction techniques could be a solution to this problem. Different researchers ([54, 55, 56, 57, 58, 59]) also studied a sequence of slamming events on a rigid high-speed planing boat. They

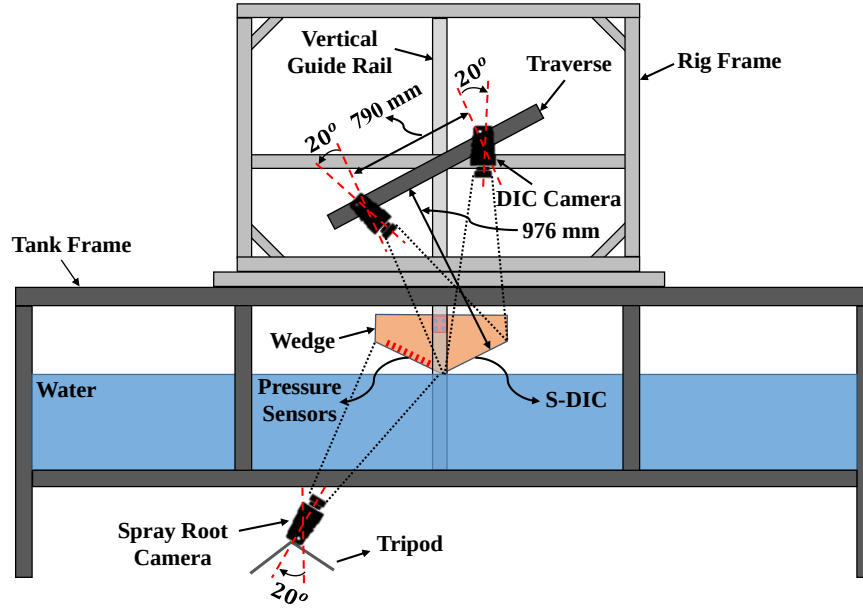


Figure 4.2: Schematic of the experimental setup showing reinforced transparent tank, rig frame, wedge and location of high-speed cameras.

conducted a set of towed scale-model experiments in a towing-tank to measure heave, pitch and slamming pressure on the bottom of the hull in calm water, and in regular and irregular waves. They used 12 pressure sensors as well as pressure pad instruments to gauge point and map pressure measurements due to slamming events. They also compared these experimental results to existing CFD tools.

As previously mentioned, the structural response of the vessel and the slamming loads are coupled when the structure is flexible. This is of great interest due to the fact that most modern, high-speed planing-hulls are manufactured from composite materials that are more prone to large deflections and vibrations than metals. Therefore, a more complete physics-based understanding of slamming-impact loading on flexible composite planing-hulls is crucial for designers to design the modern high-speed planing boats. This knowledge will directly enhance the structural design, operational envelope, and human factors, which are the main motivation for this body of work.

In this paper, the slamming of high-speed planing craft will be studied in two different

phases. In the first phase, nearly free-fall water entry of wedges with bottom plates of varying flexural rigidities are experimentally investigated with a goal of understanding the role of structural deformation on spray-root propagation. The flexural rigidity, defined as

$$D = \frac{Et^3}{12(1 - \nu^2)}, \quad (4.1)$$

can be varied easily by changing the plate thickness,  $t$ , or the material (elastic modulus,  $E$ , and Poisson ratio,  $\nu$ ). As the wedge penetrates the calm water surface, kinematics and hydrodynamic loading are measured. Two high-speed cameras are employed to measure the displacement and deflection on the bottom of the wedge using Stereoscopic-Digital Image Correlation (S-DIC). Another high-speed camera is used to record the spray-root propagation on the bottom of the wedge. Having fully synchronized measurements, the correlation between impact loading, structural response and spray-root propagation is carefully quantified. Experiments are also carried out on a rigid wedge of similar dimensions and mass for comparison. To evaluate the effect of impact velocity, the tests are repeated from different drop heights.

In the second phase of this paper, towing-tank experiments that were previously conducted at the U.S. Naval Academy (presented in [54, 55, 58, 59]) are analyzed *in preparation for non-traditional towing-tank experiments to be performed in the VT Advanced Towing Tank Facility*. Made of reinforced fiberglass composites, the model is an eight-foot prismatic hull (called GPPH, which stands for generic prismatic planing-hull), which is towed in calm water and in waves. A set of experimental measurements were taken. Those of interest in this paper are: heave, pitch, acceleration, wave elevation, and slamming pressures. Also of interest in this paper are two carriage speeds and experiments performed in regular waves.

As mentioned previously, the ultimate goal of this research is to study the slamming of

prismatic planing hulls with flexible structure in VT Advanced Towing Tank Facility. The experiments in the first phase, wedge water entry, are carried out to provide greater insight into the problem of fluid-structure interaction in water impact. Not only do these experiments help with establishing the measurement techniques, but the results are used to give insight into the design of future VT towing-tank tests. The second phase of this paper provides an analysis on rigid tow-tank experiments performed on GPPH at the U.S. Naval Academy. The goal of the current work is to replicate slamming events from the USNA experiments on a hull with flexible panels. Future phase III experiments will be a combination of phases I and II described herein. Ultimately, comparing the results of the VT hull with flexible panels with that of the rigid-planing hull in USNA experiments will shed light on the effect of fluid-structure interaction in the slamming of planing hulls that are free to heave and pitch in waves.

## 4.2 Phase I: Water Entry Experiments

A total of three wedge models are used in this experimental study. Each model is constructed with bottom plate panels of different flexural rigidities. Kinematics, hydrodynamics and structural responses of the wedge due to water entry are measured. Additionally, the spray-root propagation on the bottom of the wedge is quantified.

### 4.2.1 Experimental Setup

Fig. 4.2 is a schematic of the experimental setup located in the Hydroelasticity Laboratory housed in the Kevin T. Crofton Department of Aerospace and Ocean Engineering at Virginia Tech. The transparent water tank is made with thick acrylic sheets and is supported by a

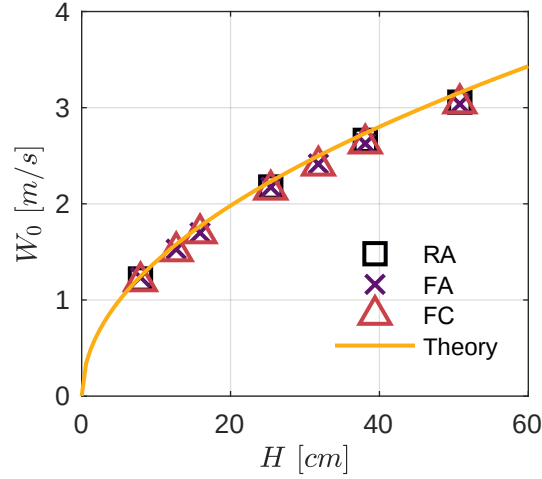


Figure 4.3: Comparison between theoretical impact velocity and measured impact velocity.

steel frame. The overall dimensions of the tank are 4.4 m long, 2.4 m wide, and 1.2 m deep. The nominal water depth for this experiment is 0.83 m. The tank frame raises the tank off the ground by 0.7 m to allow for optical access to the tank bottom. Composed of 80/20 T-slot aluminum, the rig frame is mounted on top of the water tank and includes two vertical guide rails to allow the wedge model to vertically impact the water. Linear bearings are used on both guide rails to constrain the motion of the model in the vertical direction during the fall with only a small amount of friction. The maximum deviation between free-fall impact velocity and the ideal velocity in vacuum is observed to be 4% (Javaherian et al. [48]) and is shown in Fig. 4.3. In this plot, the curve shows the theoretical terminal velocity for an object being dropped from a height,  $H$ , due to gravitational acceleration. The symbols show the measured impact velocity, where black squares indicate the rigid aluminum model, the purple x indicates the flexible aluminum model, and the red triangles indicate the flexible composite model. Electromagnets are used in a quick release system to hold and release the wedge from the required drop height to initiate the experiment. The setup allows for any drop heights up to 51 cm, where the drop heights are measured from the apex of the wedge to the calm water surface.

### 4.2.2 Models

In this study, three wedge models with different bottom plate flexural rigidities are examined. They are: Flexible Aluminum (FA), Flexible Composite (FC) and Rigid Aluminum (RA), and their respective flexural rigidities are reported in Table 4.1. These models are of comparable size, mass, and deadrise angle. Each model has a deadrise angle of 20 degrees and is 63.5 cm long, 57.27 cm wide, and 40.89 cm high. For the RA model, all side and bottom plates are made of aluminum alloy 6061-T6 and have a thickness of 12.7 mm. These thick bottom plates result in an adequate stiffness for the rigid wedge. While the side plates on the FA model are 12.7 mm thick, the bottom plates have a thickness of only 3.17 mm. Therefore, the FA wedge has a lower structural rigidity than the RA wedge. The FC model has aluminum side plates similar to RA and FA wedges, except that the bottom plates are manufactured from 3.3-mm thick glass-reinforced laminate. The laminate consists of four unidirectional layers with the stacking sequence of  $[0/90/90/0]$  and is fabricated by hand layup using E-glass fibers and an epoxy matrix. Due to the stiff plates at the sides and the keel stiffener, the boundary conditions are expected to be fixed on all edges. Additional mass is added to the flexible wedges, FA and FC, to keep mass constant and equal to 40.6 kg in this experimental study. Engineering properties of the bottom panels are shown in Table 4.2.

A cross-section view of the wedge is shown in Fig. 4.4. Fig. 5.2 shows the top view of the wedge along with the detailed dimensions. Pressure sensors are mounted on the left side of the wedge, and the S-DIC measurement area is on the right side. The string potentiometer is connected to the center of the keel. The horizontal red dashed line, seen in Fig. 5.2, represents the characteristic line ( $x$ ), which is the midline of the wedge section.. The DC accelerometer,  $A_{DC}$ , is also located next to one of the AC accelerometers,  $A_1$ , on top of the left chine plate. The DC accelerometer is implemented to record the free-fall acceleration, while the AC accelerometers are used to capture the impact acceleration. All wedge models

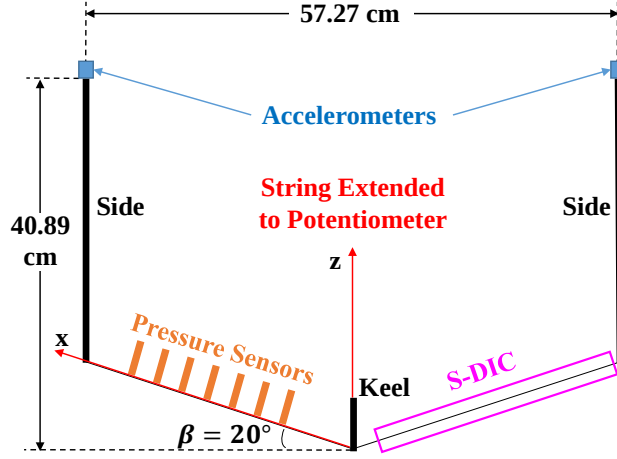


Figure 4.4: Section view of the wedge mid-plane with the sensor layout.

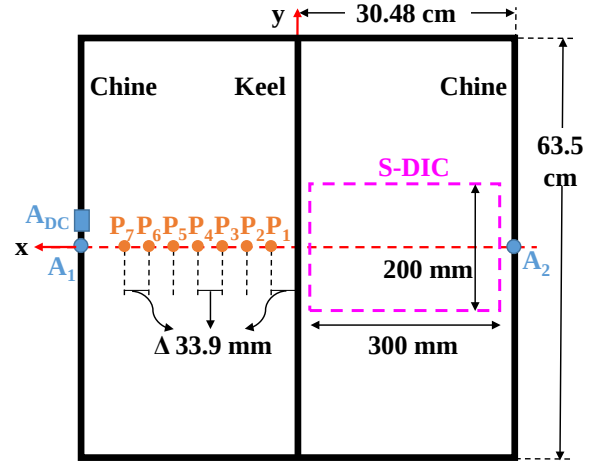


Figure 4.5: Plan view of the wedge showing the sensor layout.

have the same instrument configurations.

### 4.2.3 Measurement Instrumentation

The hydrodynamic loading due to the impact event is measured using an array of seven PCB-CA102B18 pressure transducers that are mounted on the bottom plate of the wedge. These sensors measure the dynamic pressure with a resolution of 0.007 kPa. There exists a thermal ablative coating applied on the diaphragm of the sensor to make it resistant against thermal shock in the measurement range of 344.7 kPa. The first sensor,  $P_1$ , is installed 33.9 mm away from the keel and all other pressure sensors are located at increments of 33.9 mm along the characteristic line as seen in Figs. 4.4 and 5.2. Two PCB-482C05 signal conditioners with 4 channels per unit are utilized to power, amplify and condition the pressure sensors. The instantaneous impact pressure on the bottom of the wedge was captured at 200 kHz during the experiments. To ensure the flushness of the pressure sensors with the bottom plating, a sensitive LIDAR system is used. In this process, the wedge bottom is illuminated with laser light, and the reflected light is analyzed by a transducer that records the distance between

the wedge bottom and the transducer. The pressure sensors are then adjusted so that the diaphragm of each sensor precisely aligns with the wedge bottom plate with the order of 0.01 mm.

Two miniature PCB-352A24 accelerometers,  $A_1$  and  $A_2$ , are used to measure the vertical impact acceleration of the wedge. These accelerometers are located on top of the plate at both sides of the wedge (Figs. 4.4 and 5.2). The measurement range for these accelerometers is  $\pm 50g$ . The signal of these accelerometers are amplified by a PCB-482C05 signal conditioner. Additionally, a DC response accelerometer PCB-3741B122G is also utilized to measure the low-frequency acceleration during the the free-fall. The measurement range of the DC accelerometer is  $\pm 2g$ . All AC and DC accelerometers record the acceleration with a sampling rate of 2 kHz.

The position time-history of the wedge apex during the impact event is recorded with an SGD-120-3 Resistive Linear Position String Potentiometer by Measurement Specialties. The string potentiometer is attached to the keel of the wedge so that the location of the keel during the experiments can be recorded (Fig. 4.4). This sensor has a measurement range up to 305 cm.

#### 4.2.4 High Speed Cameras

In these experiments, the Stereoscopic-Digital Image Correlation (S-DIC) technique is implemented to measure the structural response of the bottom of the wedge. This technique consists of two stationary high-speed Phantom VEO 710S cameras that are mounted on top of the rig on a traverse, which is located parallel to the bottom of the wedge at the distance of 976 mm (see Fig. 4.2). The location and angle of the cameras are set so that they can record overlapping images from the bottom plate of the moving wedge, which is coated

by a high-density speckle pattern. These images are recorded at 2000 fps. Two powerful GSVITEC LED lights facilitate the recording by illuminating the measurement area. The S-DIC images are post-processed using LaVision DaVis 8.0. The setup and calibration are optimized so that the displacement accuracy of 0.02 pixel is achieved.

During the wedge water entry, a spray-root is formed on either side of the wedge bottom that propagates from the keel to the chine. The spray-root location is highly correlated with the pressure distribution on the wedge and thus, the structural response of the wedge. There have not been many experimental measurements of spray-root propagation as it is difficult to measure. These studies, such as Lewis et al. [29] and Judge [60], have been performed on rigid wedges. Another Phantom VEO 710S high-speed camera is used in these experiments to measure spray-root evolution on a flexible wedge bottom. This camera is located underneath the transparent tank (as shown in Fig. 4.2) to visualize the formation and evolution of the spray-root on the bottom of the wedge during the impact. Similar to the S-DIC cameras, the spray-root camera also records the event at 2000 fps with resolution of 1280 x 800 pixels. The orientation of this camera is set so that it has a nearly perpendicular view to the bottom of the wedge. Images from this camera are synchronized with the data from other instruments with a delay of less than 100 ns. These fully synchronized measurements are meant to help study the instantaneous correlation between spray-root propagation, structural response, and hydrodynamics of the wedge.

#### 4.2.5 Data Acquisition

A National Instruments PXIe-8821 Controller DAQ conducts the data acquisition for experiments. All instruments are connected to this DAQ through different modules, each of which acquires samples simultaneously. A pulse generator, Berkley Nucleonics 577 Digital

Table 4.1: Experimental Matrix

| Drop Height[cm] | Wedge Type | Impact Vel.[m/s] | D [Pa.m <sup>3</sup> ] | Line Spec. |
|-----------------|------------|------------------|------------------------|------------|
| 7.9             | RA         | 1.219            | 13198                  | ---        |
|                 | FA         | 1.236            | 206.2                  | ....       |
|                 | FC         | 1.199            | 67.46                  | —          |
| 12.7            | FA         | 1.528            | 206.23                 | ....       |
|                 | FC         | 1.505            | 67.46                  | —          |
| 15.9            | FA         | 1.701            | 206.23                 | ....       |
|                 | FC         | 1.695            | 67.46                  | —          |
| 25.4            | RA         | 2.175            | 13198                  | ---        |
|                 | FA         | 2.174            | 206.2                  | ....       |
|                 | FC         | 2.151            | 67.46                  | —          |
| 31.8            | FA         | 2.415            | 206.23                 | ....       |
|                 | FC         | 2.401            | 67.46                  | —          |
| 38.1            | RA         | 2.660            | 13198                  | ---        |
|                 | FA         | 2.628            | 206.2                  | ....       |
|                 | FC         | 2.623            | 67.46                  | —          |
| 50.8            | RA         | 3.057            | 13198                  | ---        |
|                 | FA         | 3.038            | 206.2                  | ....       |
|                 | FC         | 3.07             | 67.46                  | —          |

Delay, is also used to trigger and synchronize all the measurement instrumentation with the electromagnets. In fact, by pressing the run button, the pulse generator cuts the power of electromagnets and triggers all the instruments at the same time. It is worthwhile to mention that there is a very short delay between the trigger and the actual time of wedge release by the electromagnets. This short delay is captured by an oscilloscope and is used in the post-processing of the experimental measurements. This procedure guarantees a precise synchronization of the system.

Table 4.2: Engineering properties of the bottom panels investigated in this study

| Property                    | RA    | FA     | FC    |
|-----------------------------|-------|--------|-------|
| $D$ [Pa.m <sup>3</sup> ]    | 13198 | 206.22 | 67.26 |
| $E_1$ [GPa]                 | 68.9  | 68.9   | 23.6  |
| $E_2$ [GPa]                 | 68.9  | 68.9   | 10.2  |
| $\nu_{12}$ [-]              | 0.33  | 0.33   | 0.23  |
| $G_{12}$ [GPa]              | 26    | 26     | 3.6   |
| $h$ [mm]                    | 12.7  | 3.17   | 3.3   |
| $\rho$ [g/cm <sup>3</sup> ] | 2.7   | 2.7    | 1.6   |

### 4.2.6 Experimental Matrix

The experimental matrix for the experiments is reported in Table 4.1. In the absence of the legend in the following figures, please refer to this table for the line specification that is defined for each condition. The FC and FA wedges are tested from seven drop heights while the RA wedge is studied in four drop heights. Each condition is investigated by three repeated runs. Unless stated otherwise, all the results reported for each condition is the average of the three repeated runs.

For each run, the wedge is hooked to a winch and is attached to the electromagnets that are leveled at the desirable drop height. Once the wedge is held by electromagnets, it will be unhooked so that it is only held by the electromagnets. The test begins by sending a signal from the pulse generator to cut the power from the electromagnets, which causes the wedge to fall and the measurement equipment to begin recording data. After each run, the bottom of the wedge is carefully wiped using compressed air, and the water surface of the tank is allowed to calm.

### 4.2.7 Results and Analysis

Experimental measurements for the wedge-water entry tests include kinematics of the wedge motion, pressures on the bottom plate, deflection and structural response of the plate, and the spray-root propagation. Ren et al. [37], [44] and Javaherian et al. [48] provide a summary of some of these types of measurements for the rigid and flexible aluminum models. In this body of work, a composite wedge has been added to the experiment and compared with these past experiments. Additional details on the physics of the water-entry experiments for the rigid and flexible aluminum models can be found in [44] and [48].

#### Kinematics

Fig. 4.6 shows the position, acceleration, and velocity time-histories of one representative experimental run for a drop height of 50.8 cm for the FC model. In this figure, the start time of the data signals are before the wedge is released. The impact moment when the apex touches the calm water surface occurs at time  $t = 0$ , and the chine impact moment is marked as a vertical dashed line. The vertical position curve, Fig. 4.6(a), shows a nearly free-fall behavior of the wedge until the impact moment. After that time, the position curve is nearly linear. The position time-history of the wedge recorded by the potentiometer is compared with the displacement of the wedge after the impact measured by S-DIC technique and shows very good agreement.

The vertical acceleration curve, Fig. 4.6(b), starts with a value of zero. When the wedge is released, there is a constant, negative acceleration of about  $9.81 \text{ m/s}^2$ . Just following the impact moment, there is a sharp rise in acceleration followed by a gradual decrease. After the chine impact, there are oscillations in the signal, associated with both the structural vibration of the composite panel and the rigid body motions due to the waves generated in

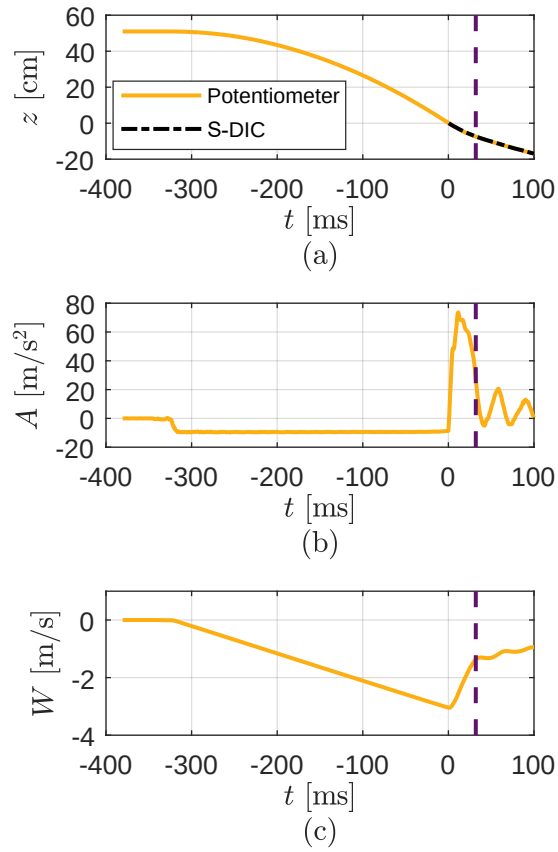


Figure 4.6: Plots of (a) position-, (b) acceleration- and (c) velocity time-histories of one representative drop from 50.8 cm of the FC model. The vertical dashed line shows the chine-wetted moment, which is acquired from the high-speed video.

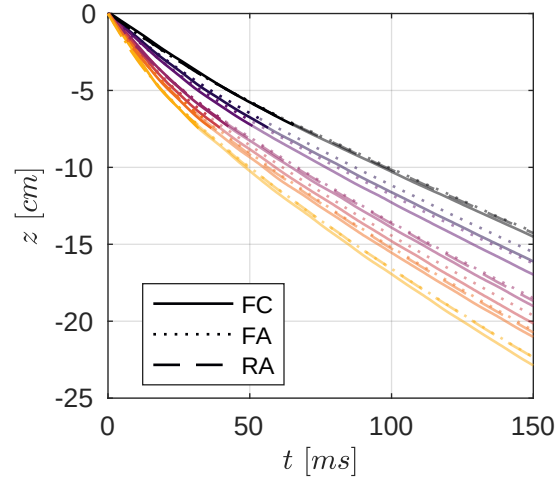


Figure 4.7: Summary of the position time-history of all the runs described in Table 4.1. Each curve represents the average results for three repeated runs. The color gradient goes from dark to light as the drop height increases.

the tank. The vertical velocity curve, Fig. 4.6(c), is computed by integrating the acceleration-time curve. Here, there is a linearly varying negative velocity, consistent with free-fall motion after the model is released. After the impact moment, the magnitude of the velocity decreases due to buoyancy and hydrodynamic forces.

Figs. 4.7 to 4.9 are summary plots of the wedge kinematic motions for all experimental conditions. The solid lines indicate measurements from the wedge model with flexible composite (FC) panels, the dotted lines correspond to the measurements of the wedge model with flexible aluminium panels, and the dashed lines show the measurements from the wedge model with rigid aluminum panels. The color gradient indicates the drop height, where the darkest colors correspond to the lowest drop height and the lightest colors correspond to the highest drop height. Please see Table 4.1 for more details. In each figure,  $t = 0$ , references the impact moment, which is the time that the apex of the wedge touches the undisturbed free surface.

Fig. 4.7 shows a summary of the position time-history of all the experimental conditions.

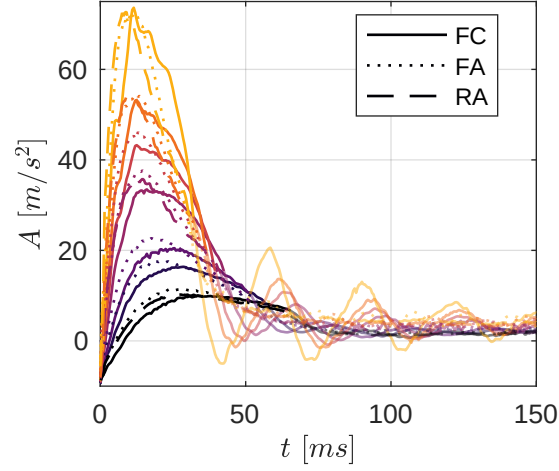


Figure 4.8: Summary of the acceleration time-history of all the runs described in Table 4.1. Each curve represents the average results for three repeated runs. The color gradient goes from dark to light as the drop height increases.

In each experimental run, the wedge continues to descend after the impact moment. For instance, the apex of the wedge is 10.2 cm lower than the free surface for  $H=50.8$  cm 100 ms after the impact. The vertical position from S-DIC method (presented in Fig. 4.6) is not shown here for sake of clarity. For each model, the chine-wetting time decreases with increasing  $H$  due to the increase in the impact velocity. For any drop height,  $H$ , the position time-histories agree well among the three different models. The drop height of 12.7 cm has the largest discrepancy of  $z$  between the three models. The maximum deviation between the absolute value of  $z$  for the FA model is 3.92% smaller than that for the FC model. The kinematics of the different models are very similar, to within 4%, and therefore, it is concluded that the model type does not influence the kinematic motion of the experiment by more than 4%.

A summary of the vertical acceleration of all experimental runs is shown in Fig. 4.8. In general, the acceleration signals for the RA and FA models agree quite well. The FC model shows slightly different behavior in acceleration, namely, the peak of acceleration occurs a

bit later in time. The small deviation in acceleration history of the composite wedge can be attributed to the deflection of this wedge during the impact. This will be discussed in more detail when the results of the plate deflection are presented. As is expected, the peak acceleration increases with increasing drop height and occurs later in time with decreasing flexural rigidity of the panel. The peak acceleration occurs at 8, 10, and 12 ms for the RA, FA, and FC models, respectively, for a drop height of 50.8 cm. For the drop height of 38.1 cm, there is the largest difference in the magnitude of the peak acceleration across the different models. This difference is  $2.1 \text{ m/s}^2$  (about 5.9%). It is worthwhile to mention that the FC model has noticeable oscillation after chine-wetting, while RA and FA do not. The frequency of this oscillation is  $\approx 35 \text{ Hz}$  for all drop heights.

Fig. 4.9 is a summary of the computed velocity time-histories for all experimental conditions. Each color represents a drop height, which is increased from the top to the bottom. At each drop height, the impact speed for the different wedge models are very close to each other. The maximum difference between the impact speeds for a given drop height is  $0.075 \text{ m/s}$  or a 2.9% relative difference. This occurs for the highest drop height, and demonstrates the fact that the difference in the wedge model does not have a significant effect on the impact velocity. It is noted that there is visible oscillation in the velocity due to vibration of the FC model after impact. The amplitudes and frequencies are more visible for the conditions of higher drop height due to the larger impulsive force applied to the structure.

### Pressure Time-History

The pressure time-history of the wedge is captured using an array of pressure sensors mounted in the bottom panel of the wedge. The raw data was filtered using a Lowpass Butterworth filter with a cutoff frequency of 2 kHz. The filtering process was monitored to make sure that the physical peak pressures, magnitudes and time of arrival, are not altered by using a similar

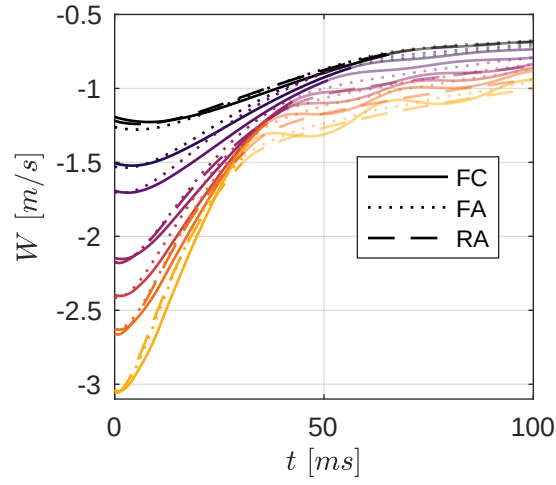


Figure 4.9: Summary of the velocity time-history of all the runs described in Table 4.1. Each curve represents the average results for three repeated runs. The color gradient goes from dark to light as the drop height increases.

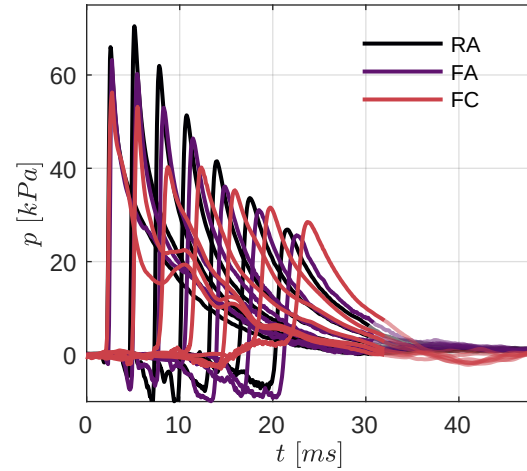


Figure 4.10: Pressure time-history of seven pressure sensors for RA, FA, and FC dropped from  $H=50.8$  cm. Each curve represents an average of 3 repeated runs.

methodology to that discussed in McCue et al. [61]. The point-pressure time-histories for the drop height of  $H = 50.8$  cm are depicted in Fig. 4.10 for all wedge models. Representing the average of three repeated runs, pressure data from  $P_1$  to  $P_7$  are plotted. The peak pressures occur in their order of arrival at each sensor,  $P_1$  to  $P_7$ , from left to right. At first glance, it is observed that the FC wedge has experienced lower peak pressures compared to FA and RA wedges.

The maximum peak pressure for FC is recorded by  $P_1$  and is 56.25 kPa. This magnitude is 10% lower than the peak-pressure recorded by the same sensor for FA (63.21 kPa). These magnitudes are still lower than the peak of  $P_1$  for RA wedge, 65.97 kPa. In addition to a decrease in the peak pressure magnitude, the time of arrival of the peaks at each sensor is in the following order: first the RA model, then the FA model, and finally the FC model. The decrease in magnitude and this delay are attributed to the structural response of the bottom panel. Due to the fact that the plate-flexural rigidity is lower, the deformation of the structure, due to the same loading condition, increases. When this larger, inward deformation occurs, a local pressure relaxation occurs in the fluid. This is because the volume that the fluid occupies becomes larger (Ikeda et al. [62]). As a result, for models with lower flexural rigidities, the peak pressure is lower at later times when the deformations become more significant. Furthermore, because of the deformed shape of the bottom panel, the inward curvature requires that the spray-root must travel a longer distance to reach each pressure sensor. This causes the delay in the peak pressure arrival. It should be noted that the last pressure sensor of the FC wedge records a peak pressure higher than the other models. This behavior is attributed to the increase in speed of the spray-root due to the deflection of the panel. This will be discussed in more detail later.

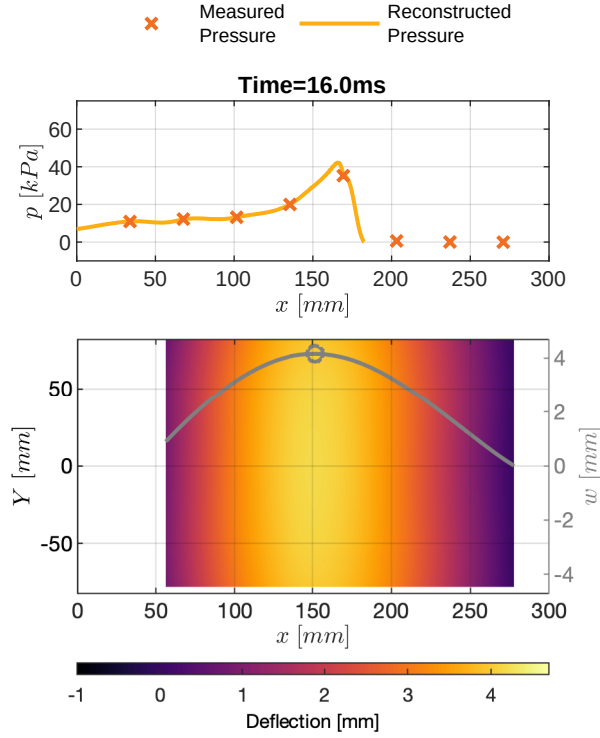


Figure 4.11: Spatial pressure distribution and the contour of deflection computed by S-DIC method for FC wedge dropped from  $H = 50.8$  cm at  $t = 16$  ms. This time corresponds to the moment that the peak pressure reaches  $P_5$ .

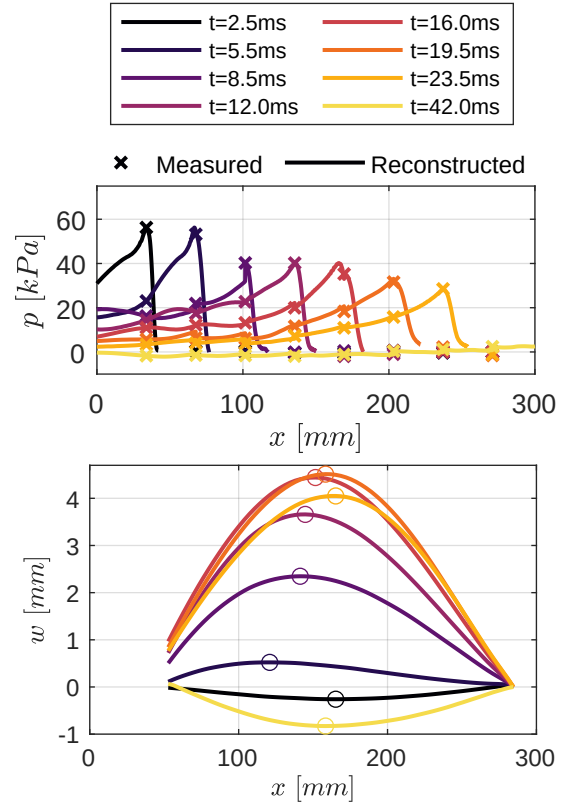


Figure 4.12: The pressure distribution and deflection of the characteristic line at eight times. The circles show the location of maximum deflection at each time step.

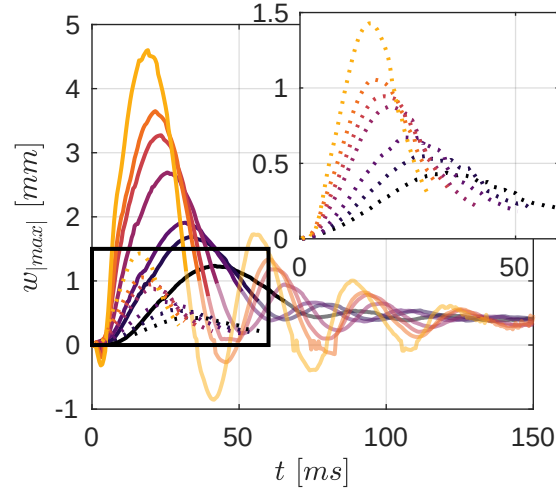


Figure 4.13: A summary of the maximum deflection of the wedge bottom plate vs time for the FA and FC wedge models and all drop heights. Color and line styles are given in Table. 4.1.

### Structural Response

In this study, the S-DIC method is implemented to compute the full-field out-of-plane deflection of the bottom panel of the wedge due to impact. A static test shows the displacement-noise floor of 0.01 mm for the wedge, meaning that any deformation less than 0.01 mm is within the error of the measurement. Fig. 4.11 shows the spatial pressure distribution and deflection contour of the bottom panel of the FC wedge for a drop height of 50.8 cm. A reconstruction technique is used to infer the pressure distribution using discrete point pressure measurements assuming the wave speed is constant between sensors (Ren et al. [36]). The deflection contour and pressure distribution are shown for a time of 16 ms after the apex impact, which corresponds to the time that the peak pressure reaches the fifth sensor,  $P_5$ . At this time, the maximum deflection is calculated to be  $w = 4.1$  mm and occurs at  $x = 152$  mm. The gray line in the figure shows the deflection of the characteristic line.

The pressure distribution and corresponding deflection of the characteristic line are shown in Fig. 4.12 for eight times. The first seven times correspond to the times when the peak

pressure arrives at each of the seven pressure sensors. The last time is well after the chine impact time to illustrate the outward deflection that is resulting from vibration of the plate. The deflection curve at  $t = 2.5$  ms reveals that the wedge bottom experiences some downward deflection at early stages of penetration. This behavior can be attributed to the inertia of the wedge during the impact. The forces acting on the wedge bottom plates are gravity-acting downward and the hydrodynamic-force acting upward. At early times, only a small area of the plate is loaded with the hydrodynamic force while the dry portion of the plate has a downward force of gravity acting on it. This causes a small negative, or outward, deflection. Similar results have been reported by Panciroli et al. [63] and Qin and Batra [40] *from simulations only*. As the wedge penetrates into the water, the wet portion of the wedge bottom becomes larger and eventually overcomes the gravitational force so that an upward deflection will occur. The deflection of the wedge increases upward until it reaches the maximum deflection at  $t = 19.5$  ms. At this time, the deflection is  $w = 4.4$  mm at  $x = 157$  mm. The wedge then starts to go back to its original configuration; however, due to the momentum and low flexural rigidity of the panel, the plate continues to deform until it results in a convex shape,  $t = 42$  ms. The plate continues to vibrate until the momentum is completely dampened. It is noted that this behavior is not observed for the flexible aluminum model due to higher flexural rigidity of the bottom panel.

Fig. 4.13 is a plot of the time history of the maximum deflection along the characteristic line. The maximum deflection of the FA model is much lower than that of the FC model, and is shown in the inset figure for clarity. At early times for the FC model, the maximum deflection is negative, or outward. Also of note is the vibration of the maximum deflection at later times for the FC model. The frequency of oscillations is approximately 32 Hz, which is close to the frequency of oscillations of acceleration data, 35 Hz, that was discussed previously.

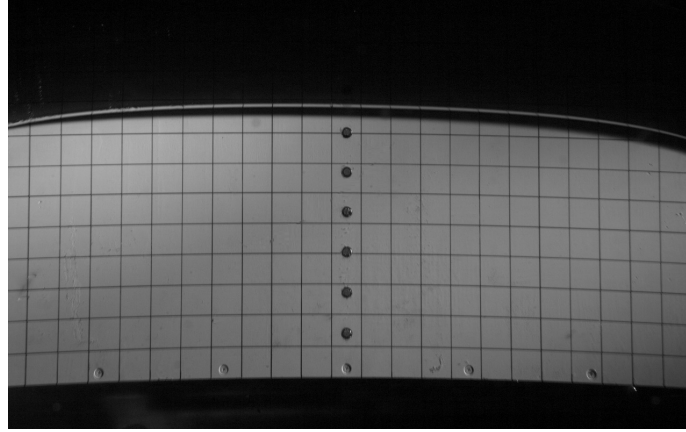


Figure 4.14: A typical image recorded by the spray-root camera showing the submerged area of the FC wedge bottom. The top bright line shows the spray-root (Javaherian et al. [48]). The waterline is moving from the bottom (keel) to the top of the image (chine).

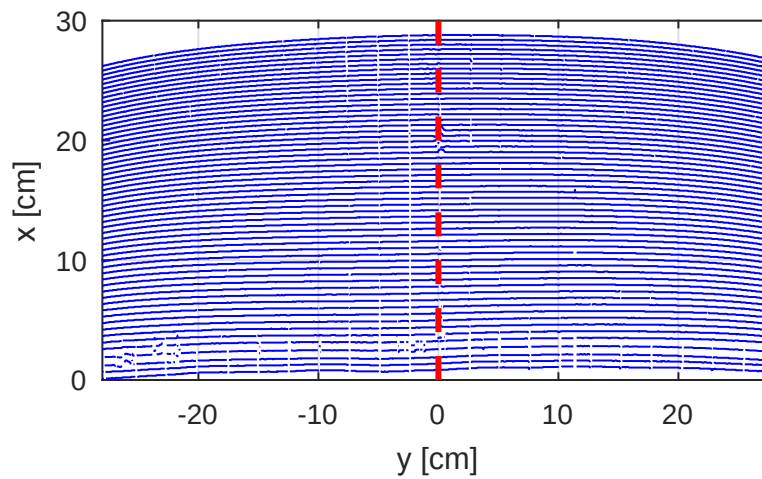


Figure 4.15: Calibrated waterline curves extracted using a gradient-based edge detection technique. The points of intersection between the red dashed line, indicating the characteristic line, and the blue water line are collected and shown in subsequent figures.

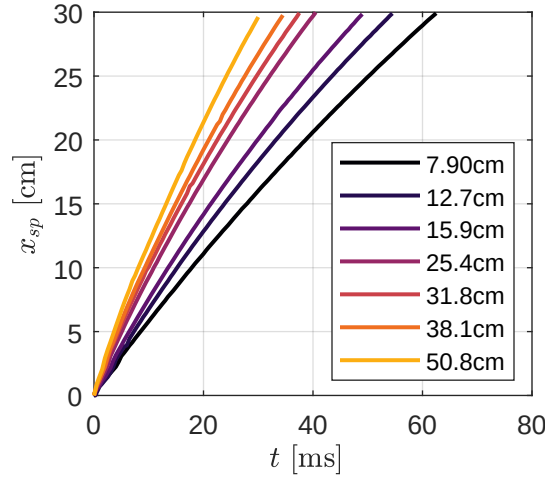


Figure 4.16: Spray-root propagation for the FC model summarized for all drop heights.

### Spray-Root Propagation

As shown in Fig. 4.2, another high-speed camera is located underneath the tank to capture the spray-root propagation. Fig. 4.14 illustrates a frame of the recorded image by this camera. The wedge is descending in the image and the water line (top bright line) is moving from the bottom of the image (keel) to the top (chine). Six of pressure sensors are also visible in the image. The images are processed using a gradient-based edge detection technique, and the upper line, defined as the spray-root in [48], has been captured for all frames captured by the high speed camera between the keel impact and the chine impact. These lines are calibrated using the grid lines on the bottom of the wedge and are plotted for different time steps in Fig. 4.15. It should be noted that for clarity, the waterline curves are decimated by a factor of 2 so that the time between curves is 1 ms.

The spray-root data are extracted from the waterlines shown in Fig. 4.15 on the characteristic line and are plotted in Fig. 4.16. This figure shows the spray-root propagation for FC wedge from different drop heights. As the drop height increases, the impact velocity also increases and consequently, the spray-root propagates faster.

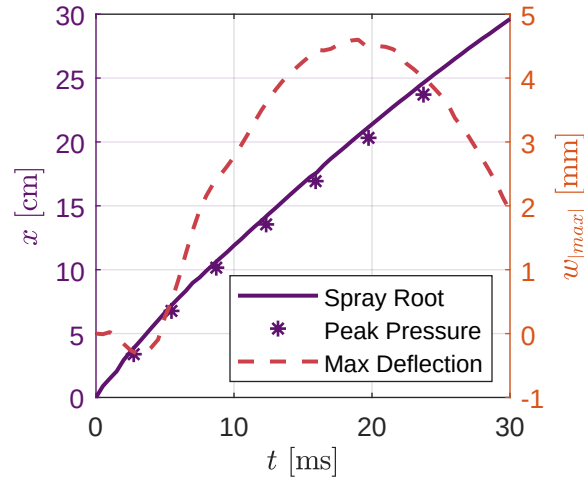


Figure 4.17: Correlation between the spray-root, peak pressure captured by each sensor, and maximum deflection of the FC model for a drop height of 50.8 cm

### Correlation between Hydrodynamics, Spray-Root and Structural Response

Fig. 4.17 shows the correlation between spray-root propagation, peak pressure of each sensor, and maximum deflection for the FC wedge for a drop height of 50.8 cm. Seven peaks are plotted that correspond to the peak of sensors  $P_1$  to  $P_7$ , sequentially. It is observed that the peak pressure always occurs with a short delay after the spray-root. The delay is increased from  $P_1$  to  $P_7$ . It can be seen that although  $P_1$  records the peak at  $t = 3$  ms, the maximum deflection in the wedge is still downward prior to  $t = 5$  ms due to the small hydrodynamic force acting on the plate at early times in the wedge penetration. At this time, the spray-root has traveled 6 cm from the keel. Thus, approximately 20% of the wedge bottom is wet and the remaining 80% of the bottom plate is still dry. For  $t > 5$  ms, the hydrodynamic force becomes larger than the self-weight of the model, and the maximum deflection increases inward. The maximum deflection,  $w_{max} = 4.6$  mm, is then reached at  $t = 19$  ms. At this time, the spray-root is located at  $x_{sp} = 20.49$  cm and is about to reach the 6th sensor,  $P_6$ . As a matter of fact, the maximum deflection of the wedge occurs when the spray-root has traveled approximately 2/3 of the distance on the bottom of the wedge. Eventually,

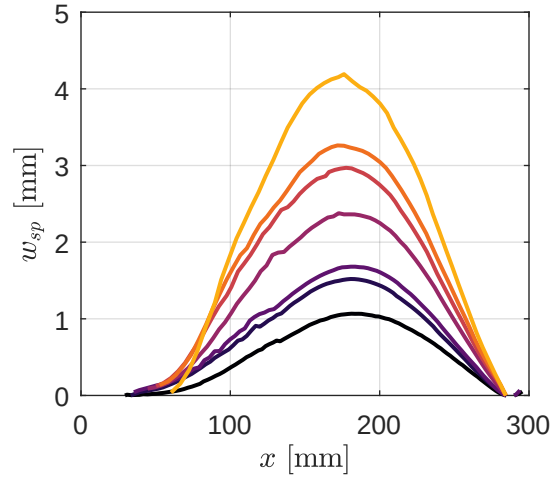


Figure 4.18: Approximate trajectory of the spray-root for all drop heights, for the FC model. The colors of the curves go from dark to light as the drop height is increased.

the maximum deflection decreases after the maximum peak is reached. This reduction is attributed to the decrease in magnitude of the pressure wave and the fact that this wave is approaching the rigid-end supports.

The approximate trajectory of the spray-root for the FC wedge dropped from all drop heights is presented in Fig. 4.18. The vertical component of the trajectory of the spray-root is simply the deflection at the location of the spray-root. The reason this is only approximate is due to the fact the spray-root is defined by most researchers as the point of maximum curvature of the free surface, which does not directly intersect the bottom panel (Javaherian et al [48]). Results show that with the increase of the drop height from 7.9 cm to 50.8 cm, the highest point of the spray-root trajectory increases from 1.07 mm to 4.19 mm. Moreover, the location of the highest point varies in the range of 172 cm to 183 cm with respect to the keel. This means that the maximum deflection occurs at a location that is located approximately 7-10% farther from the midpoint of the bottom plate with respect to the keel.

The vertical component of the trajectory of the FC and FA wedge dropped from  $H=50.8$  cm is extracted from Fig. 4.18 and is plotted versus time in Fig. 4.19. According to this Figure,

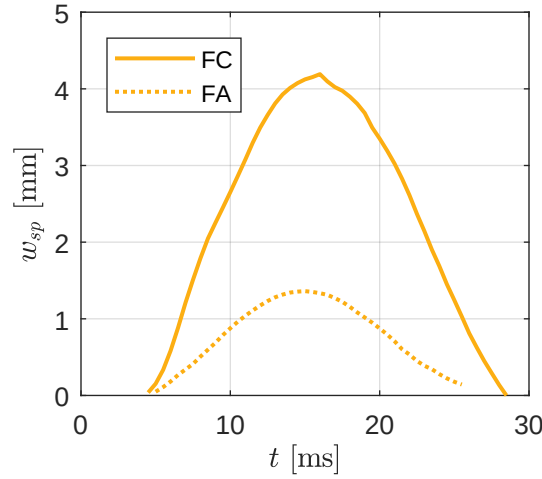


Figure 4.19: A comparison of the time-history of the vertical component of the spray-root trajectory for the two flexible wedge models for a drop height of 50.8 cm.

the highest point of the spray-root of the FC wedge is 4.19 mm, which is 208% greater than that for the FA wedge. This maximum for the FC wedge occurs at  $t = 16$  ms, which is 1 ms later than that for the FA wedge. This is consistent with the trend of  $w_{max}$  that was shown in Fig. 4.13.

Computed from the spray-root propagation, the spray-root velocity for the different wedge models dropped from a height of 50.8 cm are shown in Fig. 4.20. Results show that the spray-root velocity for all wedge models start with a similar magnitude, about  $12.6 \text{ m/s} \pm 1\%$ . This is due to the fact that the impact velocity is within 3% for the different wedge models (see Fig. 4.9). However, the spray-root velocity for flexible wedges decreases  $\approx 2.5$  ms after the impact. At this time, the spray-root has to travel upward due to the bottom deflection as seen in Fig. 4.19. When  $t = 16$  ms for the FC model, the slope of the spray-root velocity changes so the velocity curve follows that for the rigid model. This corresponds to the time that the spray-root is at top of its path. Hereafter, the spray-root travels slightly downward to follow the plate deflection and is slightly accelerated. At time  $t = 20$  ms, the spray-root velocity for FC wedge becomes larger than the RA wedge. This increase in speed explains

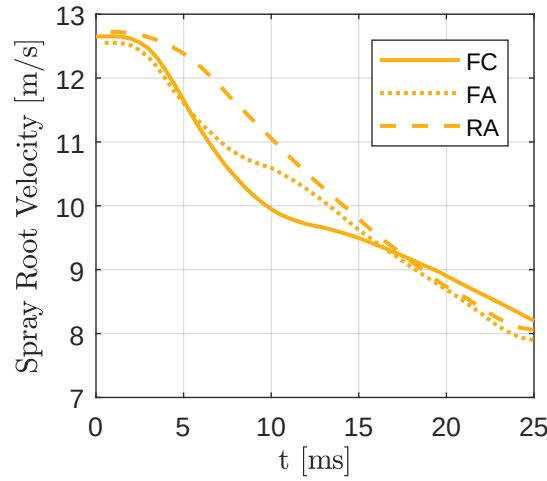


Figure 4.20: A comparison of the spray-root speed for all wedge models dropped from  $H = 50.8$  cm.

the increase in the pressure recorded by the last sensor,  $P_7$ , in Fig. 4.10.

The following sections of this paper include a discussion of experiments conducted at USNA. Slamming events from these experiments will be replicated at VT in the near future to consider fluid-structure interactions. Analysis of important events and behavior will be determined to help design the VT experimental matrix. These experiments (Phase III) are a natural extension and combination of the phase I and II experiments described in this paper.

### 4.3 Phase II: Towing-Tank Experiments

An extension to the wedge-water entry experiments are towing-tank experiments. Controlled-motion in the vertical plane slamming experiments will be conducted with the new vertical planar motion mechanism (VPMM) in the VT Advanced Towing Tank (still under development). The prescribed motions will simulate slamming events from the GPPH model that was tested at the U.S. Naval Academy (USNA) and the Naval Surface Warfare Center Carderock Division. The VT model tests will use flexible panels to measure the structural

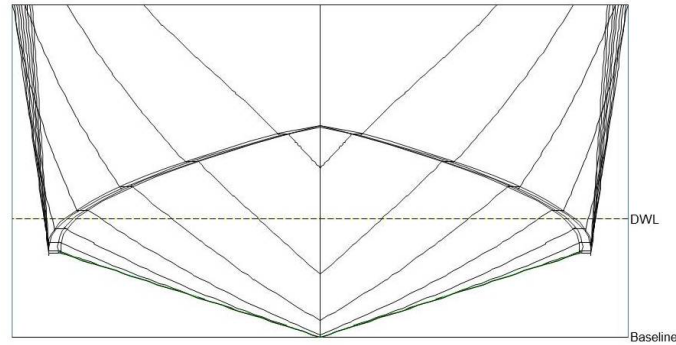


Figure 4.21: Section lines for the GPPH hull.

response due to the slam events, and the results will be compared to the USNA rigid-hull experiments. The first step toward planning these experiments is to analyze the motions and measurements from some of these rigid-model experiments. What follows in this section are the results from the towing-tank tests conducted at USNA for the 8-foot GPPH model in regular waves at two towing speeds. These experiments are discussed in more detail in Judge et al. [59].

### 4.3.1 Experimental Setup

The towing-tank experiments were conducted to study the effect of slamming on a rigid-planing hull at model scale. The body plan of the original hull is depicted in Fig. 4.21. The main characteristics of the full-scale hull and the scaled model are also reported in Table 5.2. The model is towed in a 115.8 m long water channel in both calm water and waves, allowing for a sequence of individual impact events. Heave, pitch, accelerations, encountered wave profile, and pressure-time history at single points of interest on the model during the run were measured.

Table 4.3: Full-scale and model-scale hull characteristics for GPPH.

| Characteristic      | Full-Scale Hull | Model |
|---------------------|-----------------|-------|
| Length Overall [m]  | 13.0            | 2.41  |
| Max Beam [m]        | 4.0             | 0.72  |
| Draft [m]           | 0.77            | 0.15  |
| Displacement [tons] | 15.9            | 0.10  |
| LCG [m]             | 4.6             | 0.86  |
| VCG [m]             | 1.5             | 0.13  |
| Deadrise [deg]      | 18              | 18    |

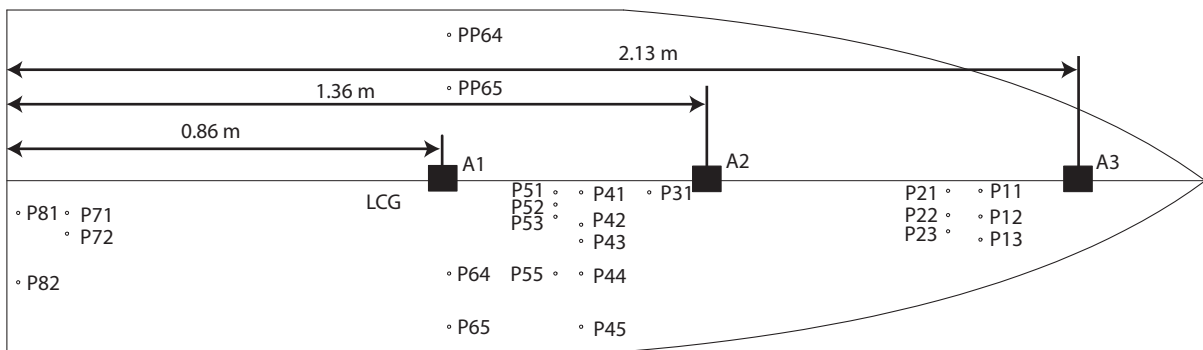


Figure 4.22: A plan view of the GPPH hull showing the pressure sensor and accelerometer placement.

Table 4.4: Experimental runs analyzed in this paper for the 8-foot model.

| Run<br>No. | Speed<br>[m/s] | Froude<br>No. |
|------------|----------------|---------------|
| 21         | 9              | 1.84          |
| 22         | 9              | 1.84          |
| 57         | 7.6            | 1.56          |
| 60         | 7.6            | 1.56          |

### 4.3.2 Instruments and Measurements

The kinematic motions of the hull were recorded with pitch and heave potentiometers that are located at the longitudinal center of gravity (LCG). A triaxial accelerometer was placed on the LCG to measure the vertical (heave), longitudinal (pitch) and horizontal (sway) accelerations. The vertical acceleration from the LCG model is designated as  $A1$ . Two single-axis accelerometers were also mounted near the bow ( $A3$ ) and halfway between LCG and the bow ( $A2$ ) to measure the vertical accelerations at these points. Fig. 4.22 illustrates the location of the accelerometers as viewed from above.

Pressure measurements on the bottom of the hull were taken using 12 point-pressure sensors for any given run. As shown in Fig. 4.22, there are 24 available locations to mount the sensors. Based on the requirement of each test, 12 of these locations were used to mount the pressure transducers.

Regular waves are generated by a wavemaker that exists on one side of the tank. The incoming water surface was recorded by a sonic-wave probe, which is located 142.2 cm in front of the model along the model centerline. Another wave probe is placed at a stationary location in the tank, 30.5 m downstream of the wavemaker.

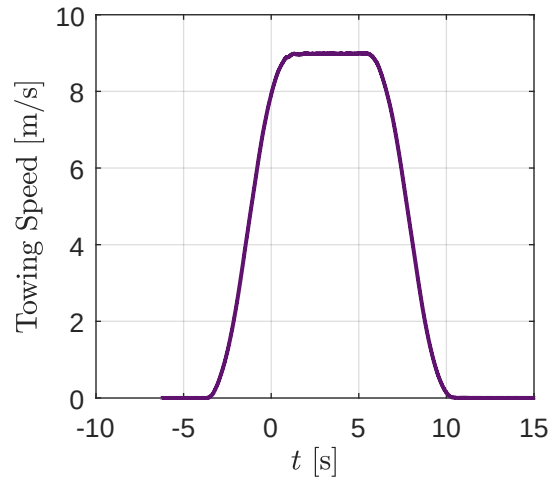


Figure 4.23: Towing speed of the carriage for run 21.  $t=0$  is the moment in which the trigger activates the DAQ to record data.

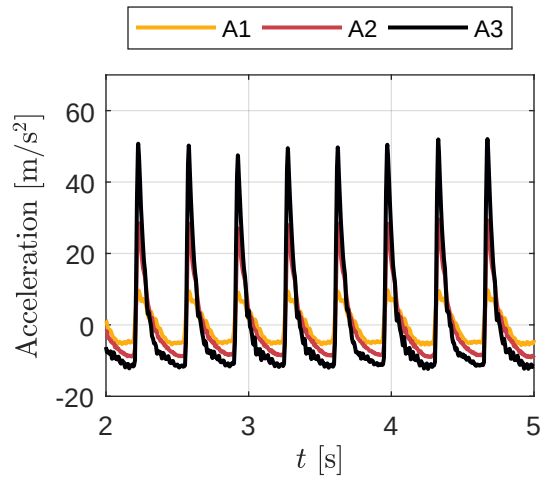


Figure 4.24: Vertical acceleration time-history from run 21 during the steady-state speed of the carriage motion. Each peak shows an individual slamming event.

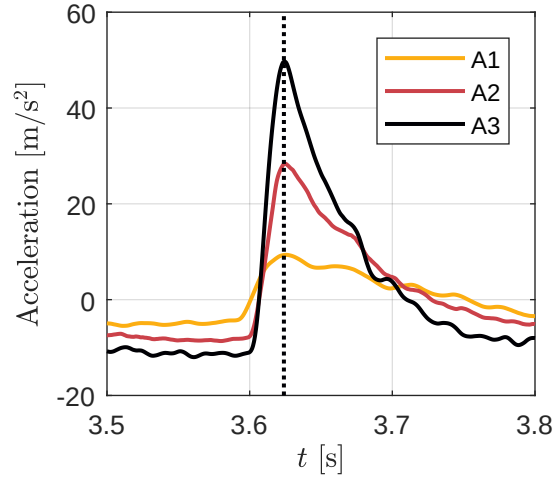


Figure 4.25: Vertical acceleration from a single slamming event observed during run 21. This event is the 5th individual impact that was presented in Fig. 4.24. The vertical dotted line shows the time that the peak acceleration occurs.

### 4.3.3 Experimental Matrix

In this analysis, two different towing speeds were used to examine the change in the encounter frequency on the loads and motions on the model. The intended regular-wave height and period are 12 cm and 1.6 s, respectively. These characteristics were chosen to represent the most probable wave occurring in Sea State 3 based on a Bretschneider spectrum. Table 4.4 shows the test configurations that are considered in this paper. During each experimental run, the model was free in heave and pitch, was constrained to the motion of the carriage in surge, but fixed in sway, yaw, and roll.

### 4.3.4 Results and discussions

Fig. 4.23 shows the towing speed of the carriage for Run 21. The motion of the carriage begins at  $t = -3.7$  s.  $t = 0$  corresponds to the moment, in which the trigger activates the data acquisition system. The carriage reaches its steady-state speed after 5 s at  $t = 1.3$  s

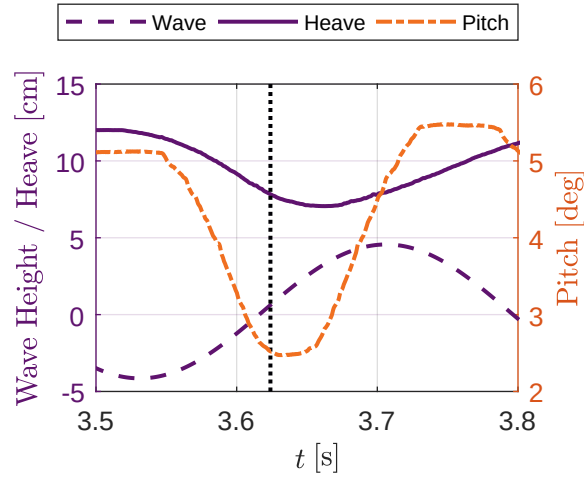


Figure 4.26: Wave profile, heave and pitch motion of the planing-hull model for run 21 during the individual slamming event shown in Fig. 4.25. Note: an adjustment was made to show the wave profile at the bow.

and again decelerates at  $t = 5.3$  s. The time period  $t = 2$  to  $5$  s is chosen as the steady-state range. The following analysis will cover this range.

The vertical acceleration of the planing-hull model during the steady-state motion is depicted in Fig. 4.24. Each acceleration peak corresponds to the maximum acceleration in an individual slamming event. As the impact occurs, the model experiences a sharp rise in vertical acceleration and reaches the peak of acceleration, which is similar to the wedge-water entry experimental results. Then, the model decelerates until the next impact arrives. The model is attached to the towing carriage at its center of gravity, so the model pitches about this point. The bow, being farthest away from the center of gravity, will record the largest vertical acceleration due to rigid-body dynamics.

To further investigate the kinematics of a single slamming event, a representative individual slamming is selected. Vertical acceleration and the motions of the model during this event are shown in Figs. 4.25 and 4.26, respectively. The dotted line in both figures, at  $t = 3.62$  s, indicates the moment that the peak acceleration is measured. At this time, the maximum

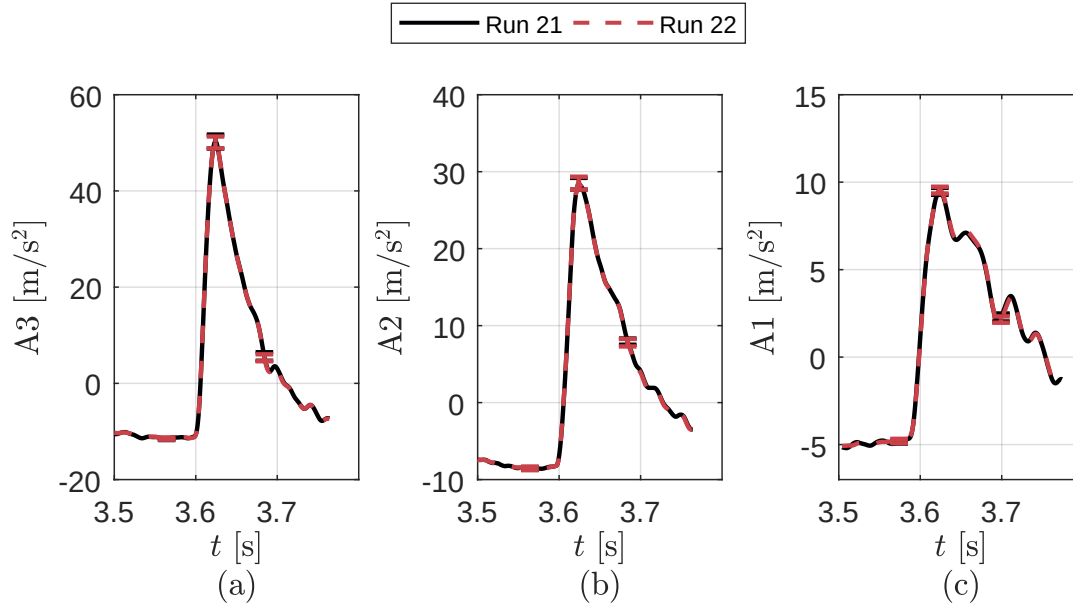


Figure 4.27: The average vertical acceleration resulting from 18 individual slamming events recorded by (a)  $A3$  (bow), (b)  $A2$  (“mid”), and (c)  $A1$  (LCG). The error bars show the standard deviation. The tow speed was 9 m/s.

peak is  $49.72 \text{ m/s}^2$ , is recorded by  $A3$ , and is 75% greater than that of  $A1$ .  $A2$  and  $A1$  show peak values equal to  $28.34 \text{ m/s}^2$  and  $9.32 \text{ m/s}^2$ , respectively. Fig. 4.26 reveals that the model encounters the slamming event at a point that the wave profile is 0.62 cm higher than the undisturbed water level. The heave of the vessel is 7.79 cm at this time. The hull descends into the water for a while after the impact and reaches the lowest position 7.06 cm. Then, it will again move upward due to the increase in the wave height. The plot also shows that at the impact, the hull possesses the lowest pitch angle, 2.5 deg. The pitch angle increases right after the impact, especially during the time that the hull proceeds toward the crest of the wave. This behavior is nearly identical to that shown in Ikeda and Judge [54].

To examine whether an individual slamming event can be a representative of the USNA experiments, a study has been conducted on a number of the slamming events. In this study, the accelerations in 9 consecutive impact events during the steady-state motion are extracted for two different runs, 21 and 22, and are plotted on top of each other. It was

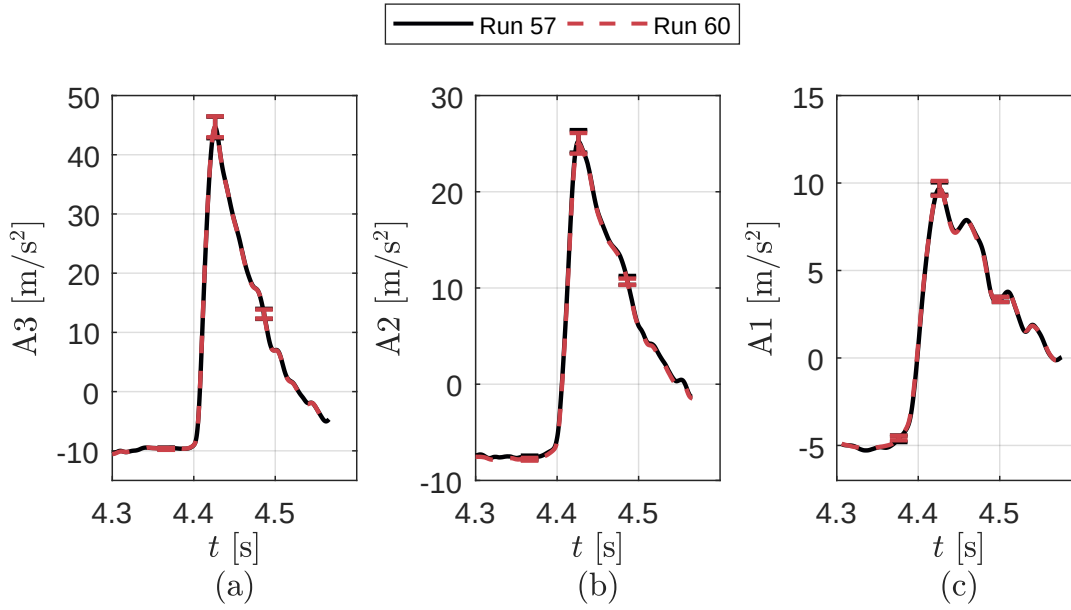


Figure 4.28: The average vertical acceleration resulting from 20 individual slamming events recorded by (a) A3 (bow), (b) A2 (“mid”), and (c) A1 (LCG). The error bars show the standard deviation. The tow speed is 7.6 m/s.

seen that the curves are very close. The average curve for each sensor is shown in Fig. 4.27. The error bars represent the standard deviation. The error bars can help to define lower and upper bounds for the acceleration results that can be considered in future VT controlled-motion slamming tests. The worst standard deviation is for A3 at its peak and is equal to  $1.23 \text{ m/s}^2$ .

The same analysis has been carried out for runs 57 and 60, which are conditions when the towing speed was 7.6 m/s. Ten individual slamming events are considered per run. The average acceleration results from all accelerometers are illustrated in Fig. 4.28. Results demonstrate a very good repeatability in different individual impacts for this lower towing speed. This confirms that there is good repeatability in a sequence of slamming events in regular waves at the U.S. Naval Academy towing-tank.

The acceleration data from the three accelerometers are summarized in Fig. 4.29 for both

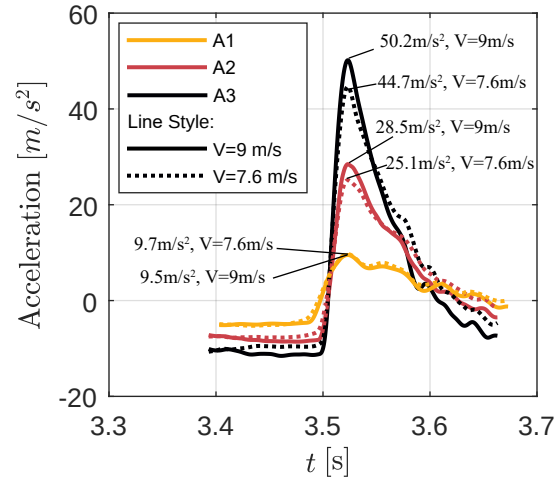


Figure 4.29: The average vertical accelerations plotted in Figs. 4.27 and 4.28 compared here with peak accelerations lined up in time.

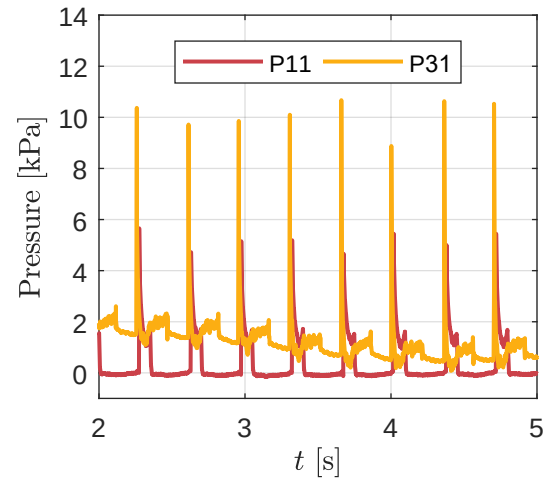


Figure 4.30: A sequence of slamming pressures from run 21 for two pressure sensors. The pressure sensor  $P11$  is located at bow while the  $P31$  is installed around midspan of the keel (see Fig. 4.22).

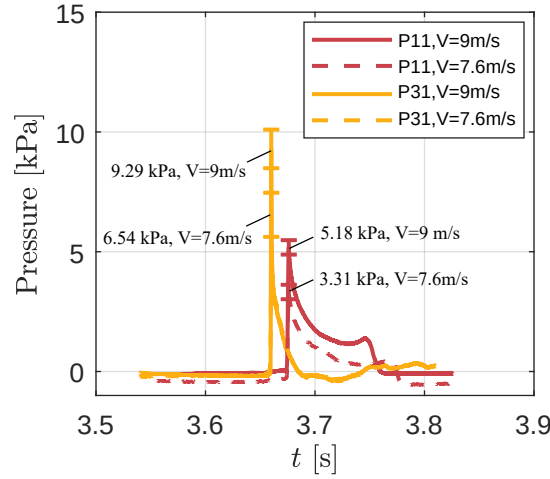


Figure 4.31: A comparison between the average pressure-time history for  $P11$  and  $P31$  for both towing speeds.

towing speeds, 9 and 7.6 m/s. Each curve is the average of two runs, each of which consists of several individual slamming events (shown in Figs. 4.27-4.28). The curves are lined up at their peaks. Results show that the maximum acceleration among the three sensors is captured by the bow accelerometer,  $A3$ , and is  $50.2 \text{ m/s}^2$  for  $V = 9 \text{ m/s}$ . This magnitude is 12% higher than the corresponding acceleration for  $V = 7.6 \text{ m/s}$ . Comparing the peaks for  $A2$  for both towing speeds provide a peak of  $28.5 \text{ m/s}^2$  at  $V = 9 \text{ m/s}$  that is 13% higher than that for the lower towing speed. The peak vertical acceleration at  $A1$  (LCG) was measured to be 9.7 and  $9.5 \text{ m/s}^2$  for the higher and lower towing speeds, respectively. The difference is just 2% in this case. The comparison concludes that a 20% increase in the Froude number will result in approximately a 12% increase in the peak of the bow and midpoint accelerations. However, it does not have any significant influence on the results of the LCG accelerometer.

A similar analysis has been conducted for the pressure data. A sequence of slamming pressures captured by sensors  $P11$  and  $P31$ , shown in Fig. 4.30, is considered. The pressure data from different runs for both towing speeds has been analyzed. In total, 38 slamming

events have been covered from four different runs and two towing speeds. The summary of the pressure data in these runs are plotted and compared with each other in Fig. 4.31. According to this figure, higher pressure loads are seen near the midspan of the model than the bow. As previously mentioned, the model has a positive pitch angle of 2.5 deg while the impact occurs. Therefore, the impact mainly happens at an area closer to the midspan of the model rather than the bow of the hull. Later and due to the motion of the hull, the bow also hits the water surface. This justifies the short time delay between peak of  $P31$  (at the midspan area) and  $P11$  (at bow). As shown in Fig. 4.30, there was a drift in the voltage change in the  $P31$  sensor that appears in the pressure signal. This curve is shifted to eliminate the drift in the signal so it would be comparable with the pressure data measured by this sensor for  $V=7.6$  m/s. By making this shift, the change in pressure magnitude is explored in Fig. 4.31. The zero pressure before and after the impact for the  $P11$  ( $t < 3.67$  s and  $t > 3.76$  s) confirms that the bow is above the water at these times. The peak pressure for  $P31$  is 6.54 and 9.29 kPa when the model is towed at  $V = 7.6$  m/s and  $V = 9$  m/s, respectively. This corresponds to a 42% increase in pressure for the higher speed. The peak pressure recorded by  $P11$  is 5.18 kPa for  $V=9$  m/s that shows a 56% increase compared to the peak of 3.31 kPa for  $V = 7.6$  m/s.

The average heave and pitch motions of the planing-hull model in regular waves with the towing speed of  $V=7.6$  m/s (runs 57 and 60) are presented in Fig. 4.32. Each curve represents the average of 20 individual slamming events that are lined up at their peaks. The error bars show the standard deviation of the data in different locations, including the peak of the curves. The small error bars indicate that these individual slams are highly repeatable. The heave curve in this figure is extracted and compared with the average heave results from the other towing speed,  $V=9$  m/s, and these are both shown in Fig. 4.33. Again, each curve is the average of 20 individual events with an error bar that shows the standard deviation.

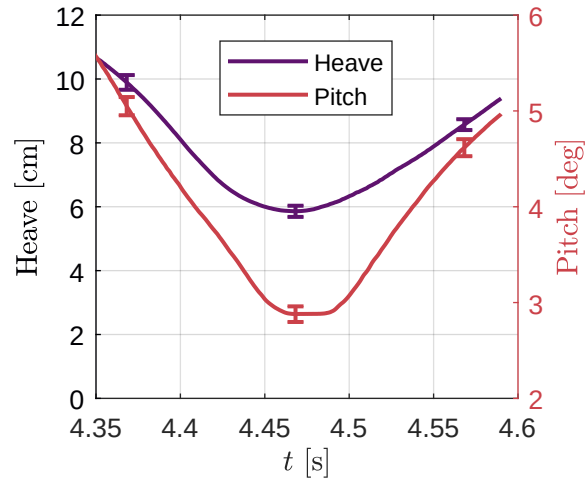


Figure 4.32: The average heave and pitch motion for heave and pitch for the towing speed of 7.6 m/s. Each curve is the average of the motion of 20 individual slamming events that are lined up at their local minima. The error bars show the standard deviation.

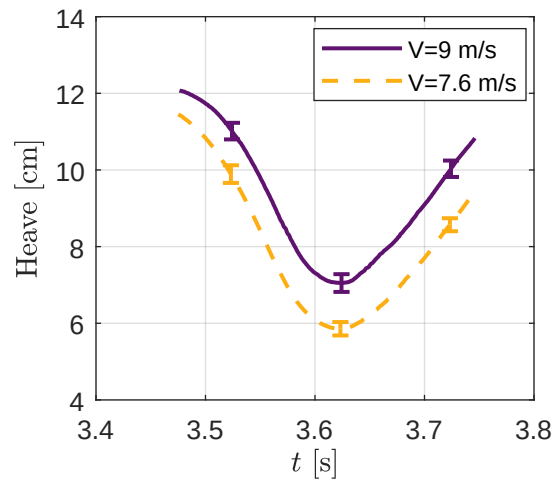


Figure 4.33: A comparison of the average heave motion for both speeds. The error bars show the standard deviation of the data.

These curves show that increasing the towing speed will influence the minimum of the heave motion; however, the slope of the heave curve, or heave velocity, seems to remain unchanged. It should be noted that the frequency of the heave oscillation is different for different towing speeds (please see Discussion #2 for more information). In Fig. 4.33 it may appear that the heave periods are identical for both towing speeds, but that is due to the truncation of the curves shown in the plot.

Fig. 4.34 shows a calculation of the heave velocity taken by differentiating the heave motion from Fig. 4.33. Interestingly, this shows that the vertical-slamming velocity is not a function of the towing speed, but instead is dependent only on the incoming wave characteristics. This is another justification for the fact that the acceleration at LCG ( $A_1$ ) is almost identical for the two towing speeds, as was shown in Fig. 4.29. It should be noted that this only holds true for the two towing speeds examined here,  $V=7.6$  and  $9$  m/s, and this should be verified for a wider range of towing speeds. This could broaden the application of wedge-water entry, presented in phase I. In fact, the results of wedge-water entry for a certain drop height could be generalized and used for planing-hulls with a wide range of cruising speeds in a certain regular wave. Yet, the role of the wave characteristics would be crucial since it will define the relative-impact angle between the planing-hull and the wave surface (inclination angle or degree of asymmetry for the wedge-water entry).

### 4.3.5 Predicting Slamming Pressure

In Chapter 3, a hybrid approach was presented to predict the pressure distribution on a wedge during water entry. In this chapter, the findings of the proposed approach is used to predict the slamming pressure on the bottom of a planing vessel encountering regular waves. As mentioned earlier, the GPPH model tested in towing tank has a prismatic hull

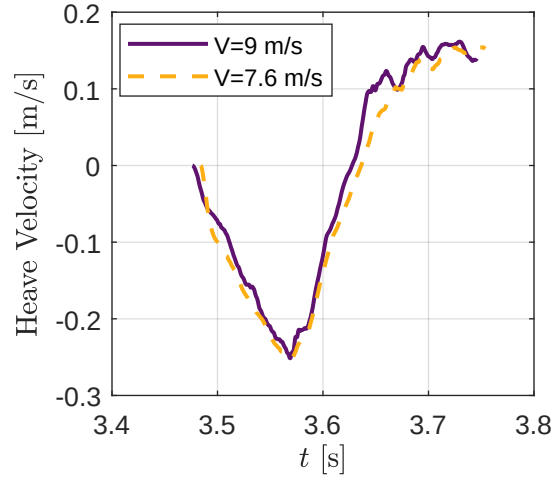


Figure 4.34: A comparison of the heave velocity for both experimental conditions. Each curve is derived by differentiating the average heave motion of the corresponding tow speed that was presented in Fig. 4.33

with a deadrise angle of  $\beta=18^\circ$ . Knowing the submergence of the keel from the heave motion and wave profile (shown in Fig. 4.26), the location of spray root at each time step can be estimated using Wagner spray root formulation (Eq. 3.1). As proposed in Section 3.5, a correction factor of 0.91 is also considered for this equation. The re-derived formulation of Modified Logvinovich model (presented in Section 3.3.1) is used for this analysis. Fig. 4.35 shows the slamming pressure on the bottom of the hull at the location of pressure sensor *P64* for the tow speed of 7.6 m/s. This sensor is located at the same section as the LCG of the vessel (see Fig. 4.22). Results show that the hybrid approach can predict the trend and location of the peak pressure very well. But the peak magnitudes are mostly lower than that from the experimental measurements. Compared to the experiments, the predicted slamming pressure seem to be more repeatable for the individual impacts. It is worthwhile to mention that for the hybrid results, the negative pressure data are excluded due to the singularity problem of the Modified Logvinovich model (discussed in Section 3.2.3). Also, the wave profile at LCG is assumed to be similar as that recorded by the wave probe in front of the model. This could potentially be a source of uncertainty in the analysis. Nonetheless,

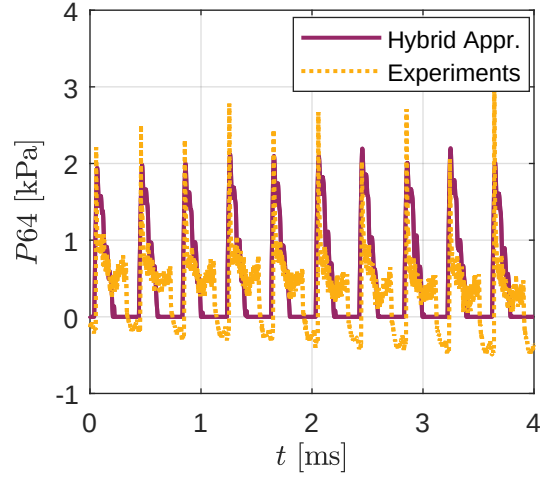


Figure 4.35: The slamming pressure on the bottom of the hull at the location of pressure sensor  $P_{64}$ . The hull is encountering regular waves with the forward speed of  $7.6 \text{ m/s}$ . The predicted results from the hybrid approach is obtained using the re-derived Logvinovich Model and are compared with that of the experimental measurements in the towing tank tests.

the hybrid approach seems to be promising for predicting the hydrodynamics of the planing hulls. This will be further investigated in the next chapter.

## 4.4 Conclusions

In this paper, slamming of high-speed planing craft was investigated experimentally in two phases. In Phase I, wedge-drop experiments were carried out on different wedge models with bottom panels having different flexural rigidity. Phase II involved preliminary analysis that will be used to help design a controlled-motion slamming experiment for a towed model in future VT towing-tank tests.

#### 4.4.1 Phase I: Water-Entry Experiments

The wedge models include a flexible composite wedge (FC), a flexible aluminum wedge (FA), and also a rigid aluminum wedge (RA). For each wedge, kinematics were measured. A set of seven pressure sensors was used to measure the impact pressure distribution on the bottom of the wedge. The Stereoscopic-Digital Image Correlation Technique, S-DIC, was implemented to measure the full-field out-of-plane deflection of the bottom of the wedge during impact. Experiments were repeated from different drop heights to evaluate the role of impact velocity on slamming measurements. Results show that the global kinematics are nearly identical for the different wedge models, no matter their flexural rigidity. This is reassuring since steps were taken to ensure that the mass/weight of each model was constant. However, the hydrodynamic loading, i.e. the impact pressure, is tangibly influenced by the flexural rigidity of the bottom panel. The maximum measured peak pressure drops 4% for the FA model, and 10% for the FC model compared to the RA model. The time interval between peak pressure of the sensors also eventually increases as the flexural rigidity of the wedge decreases. This delay accounts for the deflection of the bottom plate that causes the pressure wave and waterline to travel a longer distance to reach the chines.

Interestingly, S-DIC results show that the FC model experiences some downward deflections at early times of impact due to only a small area of the plate being loaded hydrodynamically. The opposing downward load is the self-weight of the model. This negative deflection exists in the bottom plating prior to  $t = 5$  ms for  $H = 50.8$  cm. After this moment, the hydrodynamic force increases so that it conquers the downward inertia and causes the upward deflection until the maximum deflection is reached. With the decrease of the pressure wave strength, the deflection again decreases and the plate vibrates. It is noted that due to the lower flexural rigidity of the composite material, this behavior was not observed for the aluminum model at the same loading conditions.

The comparison between maximum deflection between the two wedges with flexible bottom panels reveals that the maximum deflection for the FC wedge for each drop height is almost 68% higher than that for the FA wedge. This increase relates with the change in flexural rigidity between wedges. The FC wedge has flexural rigidity of  $D = 67.26 \text{ Pa/m}^3$ , which is 67% smaller than the FA wedge,  $D = 206.2 \text{ Pa/m}^3$ .

Spray-root propagation was also visualized using a high-speed camera, and the spray-root position was extracted using image-processing. As was expected, the higher drop height, which corresponds to a higher impact velocity, results in a higher speed for spray-root propagation.

#### 4.4.2 Phase II: Towing-Tank Experiments

The slamming experiments at the U.S. Naval Academy were conducted in towing-tank with regular waves. The model was free to heave and pitch. The accelerations at the three locations along the length of the hull were recorded. Two towing speeds were considered to study the effect of towing speed on slamming characteristics. Several individual slamming events were extracted during the steady-state speed and the above-mentioned measurements were compared.

Results show that for each towing speed, the heave and pitch motions, acceleration data, and pressure results are very repeatable during different slamming events. The average curve of each measurement was reported as the representative of that quantity. The representative measurements are proposed for future use in replicating the slamming events in VT Advanced Towing Tank.

Furthermore, the heave velocity for the two towing speeds was computed from the heave motion. Results reveal that the heave velocity from the two different towing speeds is nearly identical during an individual impact. In fact, the heave velocity is independent of the towing

speed and is a function of the incoming waves. In the future, the presented representative average acceleration and motion results recorded for the rigid hull will be used as inputs for the motion-controlled slamming experiments. These experiments will be carried out in the VT Advanced Towing Tank facility to further study the effect of structural response on the slamming characteristics of high-speed planing craft.

The main goal of this paper is to motivate the future experiments (Phase III) that will be conducted in the VT Advanced Towing Tank Facility. Phase I provides experiments conducted on wedge-water entry of a flexible panel. These experiments give insight on this fluid-structure interaction problem where the only motion is heave. In Phase II, a discussion of rigid-planing hull experiments are conducted at the USNA Tow Tank. In these experiments, the model is towed at a prescribed, controlled, forward speed and is free to heave and pitch. The authors analyzed this experimental data to understand a single slamming event. The future VT experiments shall combine the flexible panel of Phase I with the hull form and motions of Phase II. Controlled-motion experiments at VT will be performed to replicate slamming events observed from the USNA experiments to measure structural response. These two seemingly separate experiments presented in this paper form the basis of the next set of novel experiments that will be conducted.

## 4.5 Acknowledgements

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# Chapter 5

## Hydrodynamics of High-Speed Planing Craft: Savitsky Method and 2D+t Approach

(Most of the content in this chapter is from the draft of a paper that is submitted to International conference on FAST Sea Technology.)

M. Javad Javaherian, Richard Royce, Raju Datla and Christine Gilbert

(ABSTRACT) The progressive interest in high-speed planing craft has made it crucial to conduct more accurate assessments of the behavior of these vessels in motion. In this paper, a 2D+t approach is employed to predict the resistance, trim and wetted length of a prismatic planing craft cruising in calm water. Although this approach is based on original Zarnick 2D+t model, the hydrodynamic force is estimated using experimental wedge drop experiments in conjunction with the Logvinovich wedge water entry model. The analysis is repeated employing Savitsky prediction method and results are compared with that of towing tank measurements of Naples series. The comparison shows that the Savitsky prediction results match very well with the experimental data. The 2D+t approach also shows reasonable outcomes for the trim and wetted length. However, this approach under-predicts the resistance of the craft at very low Froude numbers.

## NOTATION

|               |                                                        |                 |                                                 |
|---------------|--------------------------------------------------------|-----------------|-------------------------------------------------|
| $A$           | area of the vessel section                             | $S$             | total wetted area of the craft                  |
| $a$           | distance between $D_f$ and $CG$                        | $T$             | thrust of propeller                             |
| $B$           | beam of craft                                          | $T_x$           | thrust of propeller in x direction              |
| $b$           | wetted half beam of each section                       | $T_z$           | thrust of propeller in z direction              |
| $C_f$         | frictional drag coefficient                            | $t$             | time                                            |
| $C_{L_0}$     | lift coefficient at zero deadrise angle                | $V$             | forward velocity of craft                       |
| $C_{L_\beta}$ | lift coefficient at deadrise angle $\beta$             | $VCG$           | vertical center of gravity                      |
| $C_p$         | distance from center of pressure to transom along keel | $V_c$           | vertical velocity of the section                |
| $c$           | distance between $N$ and $CG$                          | $V_m$           | mean velocity over bottom of planing surface    |
| $D_f$         | frictional drag along keel                             | $x$             | horizontal coordinate                           |
| $d$           | draft                                                  | $\ddot{x}$      | surge acceleration                              |
| $F_r$         | beam Froude number                                     | $y$             | lateral coordinate                              |
| $F_x$         | total hydrodynamic force in x direction                | $y_{sp}$        | horizontal distance between spray root and keel |
| $F_z$         | total hydrodynamic force in z direction                | $z$             | vertical coordinate                             |
| $f$           | distance between T and CG                              | $\ddot{z}$      | heave acceleration                              |
| $f_b$         | two dimensional hydrostatic force                      | $z_c$           | chine immersion height                          |
| $f_h$         | two dimensional hydrodynamic force                     | $z_{sp}$        | vertical distance from spray root to keel       |
| $g$           | gravitational acceleration                             | $z_{ch}$        | vertical distance from chine to keel            |
| $I$           | pitch moment of inertia                                | $\beta$         | deadrise angle                                  |
| $L$           | length of craft                                        | $\Delta$        | displacement                                    |
| $LCG$         | longitudinal center of gravity                         | $\epsilon$      | angle between $T$ and keel                      |
| $L_m$         | mean wetted length                                     | $\zeta$         | body coordinate normal to keel                  |
| $m_a$         | sectional (two-dimensional) added mass                 | $\lambda$       | mean wetted length-beam ratio                   |
| $M$           | mass of craft                                          | $\theta$        | trim                                            |
| $N$           | normal component of resistance                         | $\ddot{\theta}$ | pitch acceleration                              |
| $p$           | hydrodynamic pressure                                  | $\xi$           | body coordinate parallel to keel                |
| $R$           | total resistance                                       | $\rho$          | density of water                                |
| $Re$          | Reynolds number                                        |                 |                                                 |

## 5.1 Introduction

The interest of employing recreational, military and commercial high speed planing craft is progressively increasing nowadays due to their speed and performance. As the attention is

drawn to these vessels, attempts are ongoing in order to provide more accurate assessments of the behavior of the craft in motion. To this end, several empirical, numerical and analytical approaches have been proposed by researchers. Some of these efforts can be reviewed in Yousefi et al. [64].

Among the outstanding researches on this topic, one can note the Savitsky prediction method, in which he proposed some empirical formulae to predict the drag, lift and center of pressure for a prismatic planing craft [65]. These relationships were obtained based on extensive experimental investigations on V-type planing hulls. Due to the simplicity and accuracy, this method is still used in early design phase of high-speed planing craft. This method will be further elaborated in the following subsections.

With the development of computational tools, a modified version of strip theory, also known as 2D+t approach, was also proposed by researchers. Zarnick presented his non-linear mathematical model for a prismatic planing hull and used a combination of theoretical and empirical expressions to define the coefficients in the equations of motion [5]. His mathematical model predicted a very reasonable results for the heave, pitch and acceleration of the planing craft in regular waves. The 2D+t method was further developed and verified by other researchers ([66, 67, 68, 69, 70]).

In this paper, the 2D+t method proposed by Zarnick [5] is employed to predict the resistance, trim and wetted length of a prismatic planing craft with a nearly fixed deadrise angle in calm water. Despite the Zarnick method that estimates the hydrodynamic force using Wagner water entry model [10], the hydrodynamic force exerted on V-type cross sections of the vessel is imported from Logvinovich model [12] and Javaheian et al. wedge water entry experiments [48]. Using this combination, the total hydrodynamic force and pitch motion of the vessel in different speeds are then calculated and compared with those from the Savitsky prediction method.

### 5.1.1 2D+t Method

As mentioned previously, Zarnick developed a non-linear mathematical model for a prismatic craft planing in regular waves based on a 2D+t method [5]. For each section of the vessel, he determined the hydrodynamic forces through added mass planing theory and momentum variation incorporating the Wagner [10] two-dimensional wedge water entry theory. The total hydrodynamic force is then calculated through integration over the length of the vessel in conjunction with the buoyancy force.

Considering the coordinate system shown in Fig. 5.1, Zarnick restricted the motion of the craft to pitch  $\theta$ , heave  $z_{CG}$  and surge  $x_{CG}$  and wrote the equations of motion:

$$\begin{cases} M\ddot{x}_{CG} = T_x - N \sin(\theta) - D_f \cos(\theta), \\ M\ddot{z}_{CG} = T_z - N \cos(\theta) + D_f \sin(\theta) + \Delta, \text{ and} \\ I\ddot{\theta} = Nc - D_fa + Tf, \end{cases} \quad (5.1)$$

where  $M$  is mass of the craft,  $I$  is the pitch moment of inertia,  $N$  is the hydrodynamic force normal to the keel,  $D_f$  is the friction drag along the keel,  $\Delta$  is weight of craft,  $T_x$  and  $T_z$  are thrust components in x and z directions,  $c$  is the distance from CG to center of pressure for normal force,  $a$  is the distance from CG to center of action for friction drag force, and  $f$  is the moment arm of thrust about CG. Eq. 5.1 is exact; however, defining the hydrodynamic force and moments is challenging. Zarnick [5] showed that the hydrodynamic force per unit length,  $f_h$ , acting at a cross section of the craft can be written as:

$$f_h = - \left[ \frac{D}{Dt}(m_a V_c) + \rho b V_c^2 \cos(\beta) \right], \quad (5.2)$$

where  $V_c$  is the velocity in plane of cross section normal to the keel,  $m_a$  is the added mass

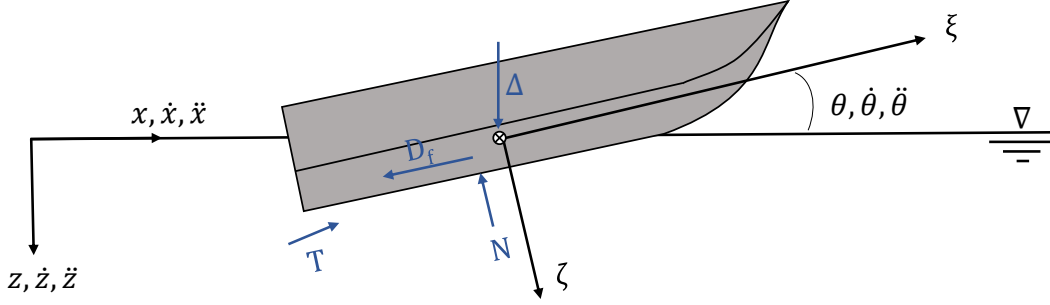


Figure 5.1: The coordinate system and free body diagram of the vessel used by Zarnick. (Figure regenerated from Zarnick [5]).

associated with the cross section,  $\rho$  is the density of the flow and  $b$  is the half beam of the section (wetted length). The added mass of each section up to the chine-wetted moment is assumed based on the Wagner wedge water entry model:

$$m_a = \pi/2 \rho b^2. \quad (5.3)$$

For the chine-wetted moment, the sectional added mass is assumed to be constant and equal to

$$m_a = \pi/2 \rho b_{max}^2. \quad (5.4)$$

In addition to the hydrodynamic force, the buoyancy force,  $f_B$ , acting in the vertical direction is assumed as:

$$f_B = -a\rho g A, \quad (5.5)$$

where  $A$  is the area of the cross section and  $g$  is the gravitational acceleration. A correction factor of  $a = 1/2$  is used that reflects the fact that the buoyancy of the sections is not fully realized since the water separates from the transom and chine at planing speeds. The total

vertical hydrodynamic force can now be formulated as:

$$-N \cos(\theta) = F_z(t) = \int_l f_h \cos(\theta) d\xi + \int_l f_B d\xi, \quad (5.6)$$

where  $l$  is the wetted length of the vessel. The total horizontal hydrodynamic force can similarly be written as:

$$F_x(t) = \int_l f_h \sin(\theta) d\xi. \quad (5.7)$$

The details of calculation of all the above-mentioned terms in the hydrodynamic force is well elaborated in Zarnick [5]. In this paper, the hydrodynamic force at each section is implemented from the experimental measurements. The process; however, is similar to that implemented by Zarnick. It should also be noted that Zarnick [5] proposed his method for a vessel encountering regular waves. But in this paper, a simplified configuration of calm water is investigated.

Once the hydrodynamic forces are calculated, the frictional drag can be computed as:

$$D_f = 1/2 \rho C_f V^2 S, \quad (5.8)$$

where  $C_f$  is the ITTC 78 frictional drag coefficient and  $S$  is the wetted surface of the planing hull. It is worth mentioning that in this study, the frictional drag due to spray area is neglected compared to the drag of the pressure area. Given the frictional and hydrodynamic drag, the resistance of the vessel can be obtained:

$$R = D_{fx} + F_x = D_f \cos(\theta) + F_x. \quad (5.9)$$

### 5.1.2 Experiments

Wedge water entry experiments have been considered in order to provide the wetted length and hydrodynamic force, which are used as the inputs for the 2D+t method. The details of these experiments can be found in Javaherian et al. [48] and Ren et al. [37]. A rigid aluminum wedge model with a deadrise angle of  $\beta = 20^\circ$  is used in free-fall water entry experiments.

The top view of the wedge along with the locations of all the sensors used in the experiment is shown in Fig. 5.2. A string potentiometer was implemented to record the position time-history of the wedge during the experiments. Three accelerometers were mounted on both sides of the wedge ( $A_1$ ,  $A_2$  and  $A_{DC}$ ) to measure the vertical impact acceleration of the wedge.

Seven pressure transducers were installed on one side of the wedge bottom to measure the hydrodynamic loading during the impact. The pressure measurements were sampled at 200 kHz. A reconstruction technique was then implemented to calculate the pressure over the span of the wedge (Ren et al. [36]). The hydrodynamic force per unit length can then be obtained by integrating the pressure-distribution over the span of the wedge.

The spray root was quantified using high-speed visualization with 2000 fps and digital image processing. The wetted beam, as shown in Fig. 5.3, can be calculated by measuring the distance from the spray root to the keel. Fig. 5.4 demonstrates the spray root position on the wedge bottom for different drop heights,  $H = 7.9, 25.4$  and  $50.8$  cm (corresponding to impact velocity of  $V_{c0} = 1.219, 2.175$  and  $3.057$  m/s, respectively). A non-dimensional analysis on the spray root location, shown in Fig. 5.5, revealed that the non-dimensional spray root location for different impact velocities collapse nearly onto one curve (Javaherian et al. [48]). This infers that the spray root location on any arbitrary section with  $20^\circ$  can

be obtained from the non-dimensional curve.

Fig. 5.6 also illustrates the hydrodynamic force corresponding to the above-mentioned impact velocities. Based on these experiments, a new approach was proposed by Javaherian et al. [71] to predict the pressure distribution on a wedge using experimental spray root location in conjunction with theoretical models, such as Logvinovich [12]. In this approach, the spray root location derived from experiments is used as the input in the pressure distribution formulation (Javaherian et al. [71, 72]). The simplified version of the Logvinovich formulation for the hydrodynamic pressure in the contact region is presented by Korobkin [13]:

$$p(y) = \frac{1}{2}\rho V_c^2 \left[ \frac{\pi}{\tan \beta} \frac{y_{sp}}{\sqrt{y_{sp}^2 - y^2}} - \frac{y_{sp}^2}{y_{sp}^2 - y^2} \right], \quad (5.10)$$

where  $y_{sp}$  is horizontal distance from keel to spray root and  $0 \leq y \leq y_{sp}$ . Knowing the spray root location (from non-dimensional curves, Fig. 5.5), the pressure distribution on any arbitrary section could be computed from Eq. 5.10. It should be noted that Eq. 5.10 is valid until the chine-wetting. For the chine-wetted phase, Payne [73] presented a correction factor, which can be incorporated into the original Zarnick's model in order to implement the effect of chine immersion on the added mass. Since Eq. 5.10 is directly employed in this paper instead of Zarnick's added mass formulations, the pressure distribution in the chine-wetted phase is approximated using the Payne's factor:

$$p(y) = p(b_{max}) \left( 1 + k \frac{z_c}{2b_{max}} \right), \quad (5.11)$$

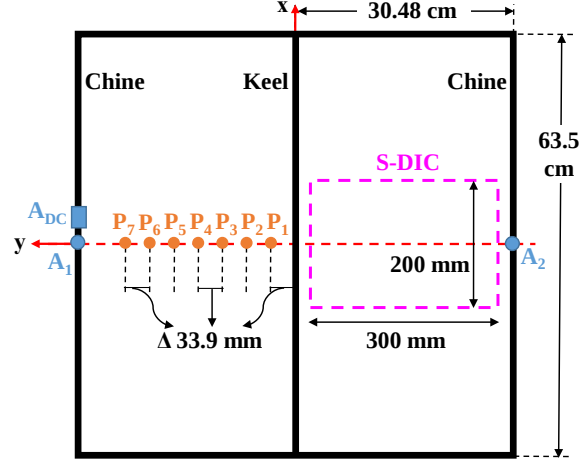


Figure 5.2: Plan view of the wedge showing the dimensions of the wedge and locations of the sensors.

where  $z_c$  is the chine immersion depth. The  $k$  factor here is proposed by Payne [73] as:

$$k = 2.05 \left[ 1 - \left( \frac{2\beta}{\pi} \right)^{9/2} \right]. \quad (5.12)$$

Once the pressure distribution is calculated on the section, the hydrodynamic force can then be calculated by integrating the pressure distribution on the wetted beam of the section using:

$$f_h = 2 \int_0^{y_{sp}} p(y) dy. \quad (5.13)$$

The force resulted from Eq. 5.13 at each time step will be used as the input for the 2D+t method. Again, it should be noted that all the relationships derived from the experiments are validated only for deadrise angle of  $20^\circ$ .

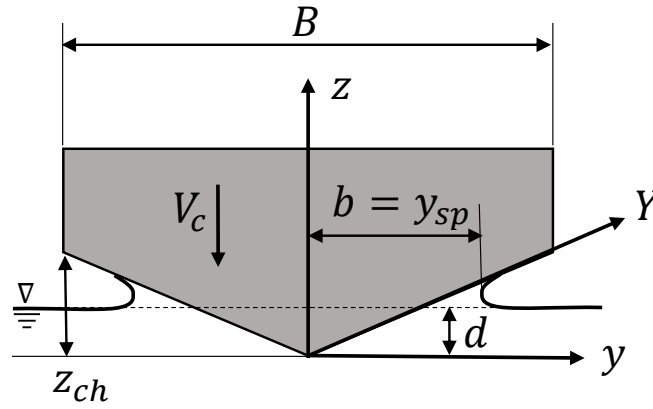


Figure 5.3: the coordinate system and parameters view of the wedge water entry.

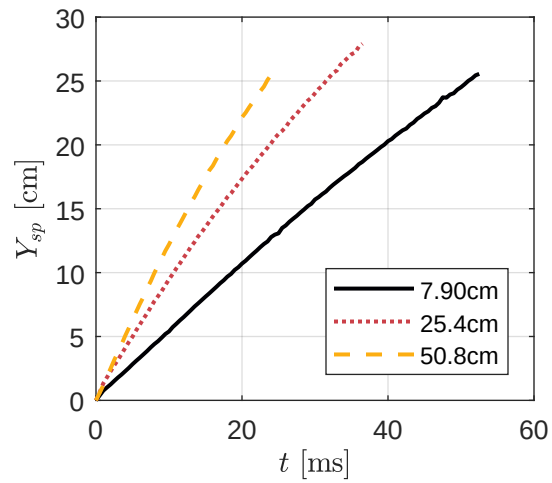


Figure 5.4: Spray root location on the bottom of the wedge for three different drop heights. (Figure regenerated from Javaherian et al. [48]).

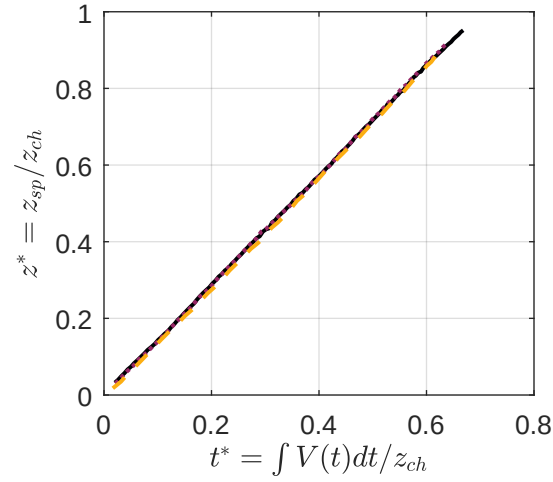


Figure 5.5: Non-dimensional spray root location on the bottom of the wedge for different drop heights. (Figure regenerated from Javaherian et al. [48]).

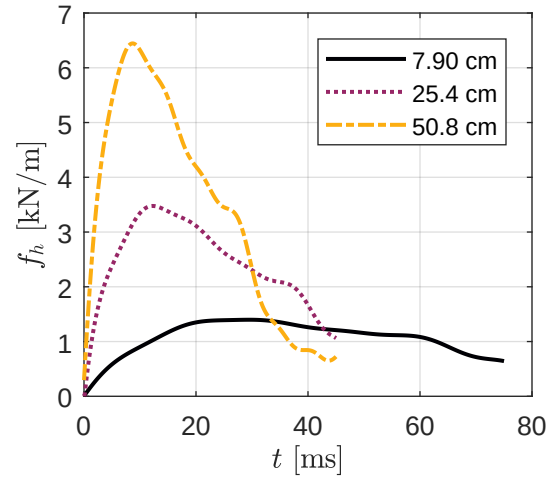


Figure 5.6: Hydrodynamic force exerted on the bottom of the wedge during the free fall for three different drop heights.

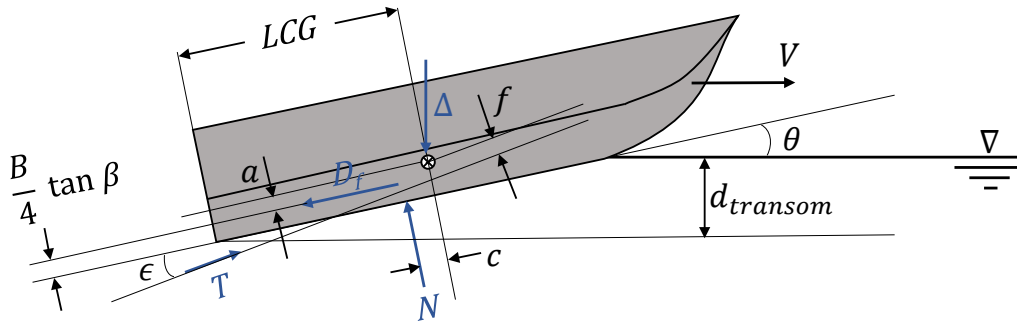


Figure 5.7: The schematic of the vessel and the parameters defined by Savitsky. (Figure regenerated from Savitsky [65]).

### 5.1.3 Savitsky Method

Savitsky proposed empirical planing equations for a prismatic planing craft in order to estimate the resistance, trim, draft and porpoising stability of the vessel [65]. These empirical equations include lift, drag, wetted area and center of pressure of the vessel as a function of speed, trim angle, deadrise angle and also loading of the craft. The calculated parameters at each time are used in a non-linear system of equations, derived from the equilibrium of the vessel, to solve for the required unknowns.

Fig. 5.7 shows a vessel and the parameters defined by Savitsky [65]. At this state, a vessel with a displacement of  $\Delta$  is cruising forward with the speed of  $V$  and the trim angle of  $\theta$ . The hydrodynamic force normal to the keel,  $N$ , is applied to an arbitrary point which has a distance of  $c$  from the LCG. Please see the Nomenclature for a detailed explanation of each parameter.

From the equilibrium of the vessel for vertical and horizontal forces and the pitching moment,

it can be written:

$$\begin{cases} \Delta = N \cos(\theta) + T \sin(\theta + \epsilon) - D_f \sin(\theta) & \text{Vertical Force Equilibrium} \\ T \cos(\theta + \epsilon) - D_f \cos(\theta) + N \sin(\theta) & \text{Horizontal Force Equilibrium} \\ Nc + D_f a - Tf = 0 & \text{Pitching Moment Equilibrium} \end{cases} \quad (5.14)$$

The following solution for this non-linear system of equations is reported by Savitsky:

$$\Delta \left[ \frac{[1 - \sin(\theta) \sin(\theta + \epsilon)]c}{\cos(\theta)} - f \sin(\theta) \right] + D_f(a - f) = 0. \quad (5.15)$$

In order to employ this solution, a value for  $\theta$  is first assumed. Other parameters are all calculated and the Eq. 5.15 is checked to find whether the assumed  $\theta$  satisfies the equation or not. This will result in an iteration procedure that will be adequately elaborated in the following steps.

The first step is to calculate the beam Froude number of the vessel:

$$Fr = V / \sqrt{gB}. \quad (5.16)$$

The lift coefficient of deadrise surface is then calculated as:

$$C_{L_\beta} = \Delta / \rho V^2 B^2. \quad (5.17)$$

The lift coefficient of zero deadrise can then be numerically solved by iterating  $C_{L_0}$  until  $C_{L_\beta}$  matches Eq. 5.17 using:

$$C_{L_\beta} = C_{L_0} - 0.0065 \beta C_{L_0}^{0.6}. \quad (5.18)$$

At this point, an arbitrary value of  $\theta$  is assumed. The validity of the Savitsky method is restricted to  $2 \leq \theta \leq 15$ , which is the common trim angle for the planing vessels. The mean wetted length-beam ratio,  $\lambda = L_m/B$  can then be derived solving the following equation:

$$C_{L_0} = \theta^{1.1} (0.012 \lambda^{0.5} + 0.0055 \lambda^{2.5} / Fr^2). \quad (5.19)$$

The mean velocity on the bottom of the hull can then be calculated using factor of k,

$$V_m = V \sqrt{1 - \frac{k - 0.0065 \beta k^{0.6}}{\lambda \cos(\theta)}}, \quad (5.20)$$

where  $k = 0.012 \lambda^{0.5} \theta^{1.1}$  and  $\beta$  is in degrees. Having the mean velocity known, the Reynolds number of the flow on the bottom of the vessel can be obtained:

$$Re = V_m \lambda b / \nu. \quad (5.21)$$

At this point, the frictional drag coefficient can be derived using the Shoenherr formulation:

$$0.242 / C_f^{0.5} - \log_{10}(Re \cdot C_f) = 0. \quad (5.22)$$

Now, the frictional drag force component along the bottom surface can be calculated:

$$D_f = \rho V_m^2 \lambda b^2 (C_f + \Delta C_f) / [2 \cos(\beta)], \quad (5.23)$$

where  $\Delta C_f$  is the ATTC standard roughness and is assumed to be  $\Delta C_f \approx 0.0004$ . The total drag can now be formulated as:

$$D = D_f / \cos(\theta) + \Delta \tan(\theta). \quad (5.24)$$

The location of pressure can now be estimated as:

$$C_p = 0.75 - \frac{\lambda^2}{5.21 Fr^2 + 2.39 \lambda^2}. \quad (5.25)$$

Knowing the location of applied hydrodynamic forces, the distance between  $D_f$  and CG (normal to  $D_f$ ) can be calculated:

$$a = VCG - B/4 \tan(\beta). \quad (5.26)$$

Similarly, the distance between  $N$  and  $CG$  (normal to  $N$ ) can be obtained as:

$$c = LCG - C_p \lambda B. \quad (5.27)$$

Having all the required parameters, the solution of the equilibrium equation can be controlled:

$$\Delta \left[ \frac{[1 - \sin(\theta) \sin(\theta + \epsilon)]c}{\cos(\theta)} - f \sin(\theta) \right] + D_f(a - f) = 0. \quad (5.28)$$

As mentioned earlier, if Eq. 5.28 is satisfied, the assumed trim,  $\theta$ , is the equilibrium trim and the corresponding drag is the final resistance at this state. If not, a new trim value is assumed and the solution is re-iterated from Eqs. 5.18 to 5.28 until the convergence is obtained. Once the final trim is achieved, the draft of the vessel at transom can be calculated:

$$d = \lambda B \sin(\theta). \quad (5.29)$$

The Savitsky [65] method can be implemented for different speeds of the vessel going from Eq. 5.16 to 5.28.

Table 5.1: Principal characteristics of the MONO vessel (Begovic and Bertorello [74])

| Parameter     | Magnitude |
|---------------|-----------|
| $L$ (m)       | 1.9       |
| $B$ (m)       | 0.424     |
| $\Delta$ (N)  | 319.7     |
| LCG (m)       | 0.697     |
| VCG (m)       | 0.143     |
| Fr (-)        | 1.37-3.66 |
| $\beta$ (deg) | 16.7      |

## 5.2 Case Study

A case study is presented to evaluate the presented 2D+t and Savitsky method. To this end, both approaches are used to predict the resistance, trim and wetted length of a specific prismatic planing craft running in calm water. To study the effect of speed, the analysis is repeated for different Froude numbers. Results are then compared with the experimental measurements recorded in towing tank tests on a similar model.

### 5.2.1 Model

The prismatic planing hull, which is studied as the case study is the MONO model given from the series of Naples presented by Begovic and Bertorello [74]. Table 5.1 shows the principal characteristics of this hull.

## 5.3 Results

Fig. 5.8 illustrates the wetted length of the vessel from 2D+t study and Savitsky prediction method. Results are compared with the experimental measurements recorded in the towing

tank experiments of Begovic and Bertorello [74]. As expected, the wetted length of the vessel is decreased with the increase in the Froude number. This is due to the fact that higher speeds yield to higher lift force exerted to the vessel. Thus, the draft and consequently the wetted length is decreased. It can be seen that the predicted values from Savitsky method and 2D+t approach are in good agreement with the experimental data. As reported in Table 5.2, the maximum error for 2D+t study and Savitsky method is 9.4 and 9.8 percent, respectively. In both cases, the maximum error corresponds to the lowest Froude number,  $Fr=1.37$ .

The predicted results for the trim angle of the model are shown in Fig. 5.9. The comparison states that the Savitsky method possesses a better prediction for the trim angle. The maximum error for this method is seen at  $Fr=2.26$  equal to 5.8 percent. The 2D+t approach also demonstrates an acceptable prediction for the trim results. The trend of the trim is well captured in this method and results are mostly predicted within 10 percent of error.

The friction drag in different Froude numbers estimated from 2D+t study and Savitsky method are plotted on top of each other in Fig. 5.10. It can be seen that both approaches result in a very similar prediction for the friction drag. The difference in the magnitude of the predicted values is within 3 percent.

Finally, Fig. 5.11 illustrates the resistance of the vessel predicted from 2D+t and Savitsky methods. As expected, with the increase in Froude number, the resistance of the vessel is also drastically increasing. The comparison shows that the Savitsky method can reasonably predict the resistance of the craft in different speeds. The maximum error of this method is within 10 percent for the range of speeds covered in this study. The 2D+t method implemented in this study; however, has a lower accuracy compared with the Savitsky method. As shown in Table 5.2, the maximum error is determined to be 22 percent for the  $Fr=1.37$ . One possible reason for this error could be the uncertainty of using the experimental mea-

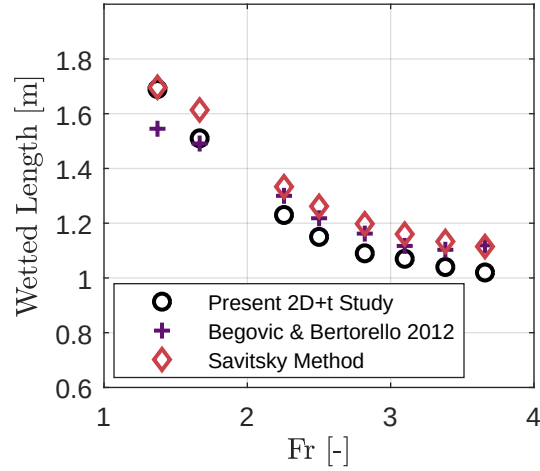


Figure 5.8: Wetted length of the vessel in different Froude numbers predicted from 2D+t study and Savitsky Method. Results are compared with the experimental measurements of Begovic and Bertorello [74].

measurements of the spray root location derived from a wedge with  $\beta = 20^\circ$ , while the deadrise angle of the hull is  $16.7^\circ$ . Another source of uncertainty could be attributed to the fact that the pressure distribution on wedge cross sections was derived from the Logvinovich [12] model, which is valid before the chine-wetted moment. However, incorporating the pressure data for the chine-wetted phase was an approximate approach using Payne [73] added mass correction factor. Yet, as the Froude number increases, the accuracy of prediction is enhanced. Other than the two lower Froude numbers, the 2D+t approach could predict the resistance of the vessel mostly within 7 percent.

## 5.4 Conclusions

In this paper, a 2D+t approach was implemented to predict the resistance, trim and wetted length of a prismatic planing craft running in calm water. The 2D+t approach presented by Zarnick [5] was modified in that the hydrodynamic force was estimated using Logvinovich

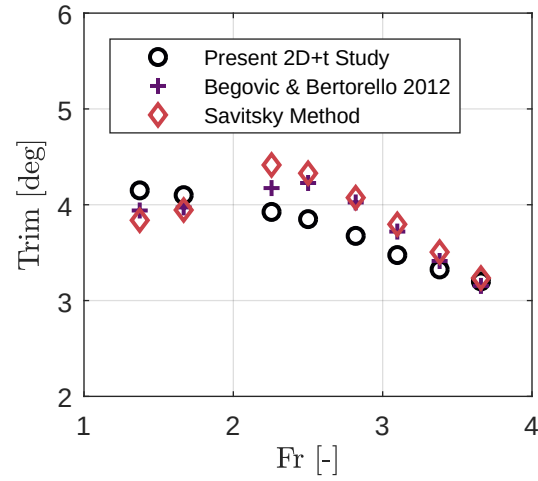


Figure 5.9: Trim of the vessel in different Froude numbers predicted from 2D+t study and Savitsky Method. Results are compared with the experimental measurements of Begovic and Bertorello [74].

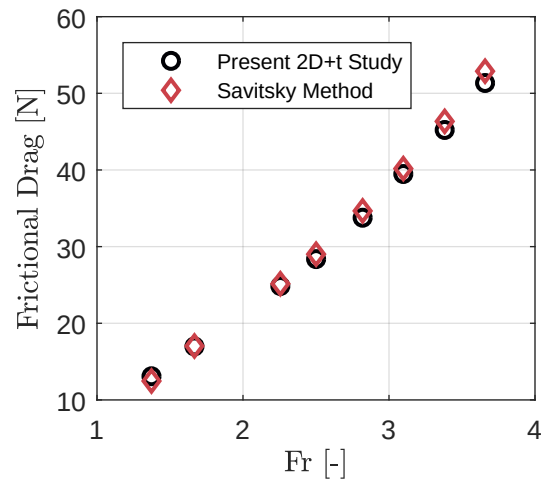


Figure 5.10: Frictional drag of the vessel in different Froude numbers predicted from 2D+t study and Savitsky Method.

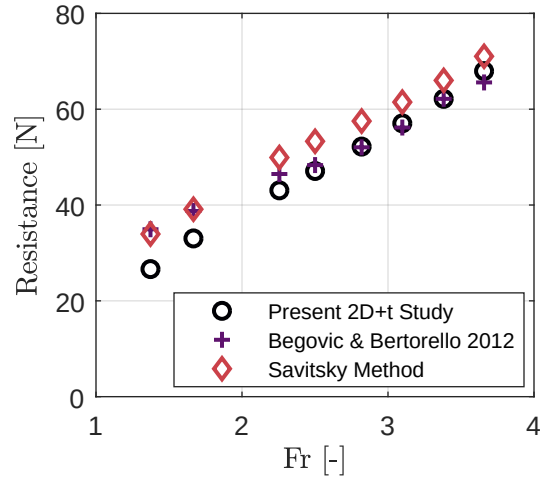


Figure 5.11: Resistance of the vessel in different Froude numbers predicted from 2D+t study and Savitsky method. Results are compared with the experimental measurements of Begovic and Bertorello [74].

Table 5.2: Predicted results from Savitsky method and 2D+t approach compared with the experimental towing tank results of Begovic and Bertorello [74].

| Froude No. | Resistance |           |          | Wetted Length |           |          | Trim       |           |          |
|------------|------------|-----------|----------|---------------|-----------|----------|------------|-----------|----------|
|            | Exp. [N]   | Error [%] |          | Exp. [m]      | Error [%] |          | Exp. [deg] | Error [%] |          |
|            |            | 2D+t      | Savitsky |               | 2D+t      | Savitsky |            | 2D+t      | Savitsky |
| 1.37       | 34.94      | -22.86    | -2.86    | 1.55          | 9.39      | 9.81     | 3.94       | 5.33      | -2.59    |
| 1.67       | 38.76      | -15.59    | 0.94     | 1.49          | 1.27      | 8.22     | 3.97       | 1.96      | -0.68    |
| 2.26       | 46.47      | -7.24     | 7.40     | 1.30          | -5.38     | 2.56     | 4.17       | -6.56     | 5.77     |
| 2.50       | 48.34      | -3.28     | 10.26    | 1.22          | -3.94     | 3.59     | 4.23       | -11.28    | 2.41     |
| 2.82       | 52.04      | -0.35     | 10.52    | 1.16          | -5.34     | 3.08     | 4.02       | -10.54    | 1.24     |
| 3.10       | 56.17      | 1.63      | 9.44     | 1.12          | -4.21     | 3.88     | 3.72       | -7.26     | 2.04     |
| 3.38       | 62.16      | 0.07      | 6.19     | 1.10          | -5.71     | 2.75     | 3.41       | -3.31     | 2.72     |
| 3.66       | 65.59      | 2.94      | 8.31     | 1.12          | -8.85     | -0.38    | 3.15       | -0.10     | 2.60     |

[12] theory and experimental investigations conducted on a wedge with  $20^\circ$  deadrise angle. Results were also obtained using Savitsky [65] prediction method for a similar planing hull and were compared with the towing tank measurements presented by Begovic and Bertorello [74]. The comparison showed that the Savitsky method demonstrates a very good prediction for the above-mentioned measurements in different speeds. Although this method is simple and straightforward, it can accurately predict the resistance, trim and wetted length of the vessel. Using Savitsky [65] prediction method, the resistance, wetted length and the trim angle of the planing hull could be predicted within 10, 6 and 10 percent, respectively.

The proposed 2D+t approach also exhibited promising outcomes for the planing hull, which was considered as the case study. Except for the low Froude numbers, this approach could predict the resistance of the vessel mostly within 7 percent. However, this approach underestimated the resistance for the low Froude numbers up to 22 percent. The implemented 2D+t approach also showed a reasonable prediction for the wetted length and the trim of the vessel. In most cases, wetted length and the trim of the vessel were predicted within 10 percent. This paper is the very first step toward developing an accurate 2D+t approach using experimental 2-D wedge water entry results.

## 5.5 Acknowledgements

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# Chapter 6

## Conclusions and Future Work

In this work, the effects of fluid-structure interaction on the slamming of high-speed planing craft were experimentally investigated. The problem was studied in a few different stages.

In Stage I, free-fall water entry experiments were conducted and kinematics were measured. An array of pressure sensors were installed to the wedge bottom to record hydrodynamic pressure during impact. Structural response of the wedge bottom was measured through the use of the Stereoscopic-Digital Image Correlation Technique, or S-DIC. Additionally, the spray root propagation during the impact was visualized with high-speed cameras. Given a fully synchronized measurement, instantaneous correlation between peak pressure, spray root propagation, and structural response of the wedge was studied. To assess the effect of hydroelasticity, experiments were repeated for a wedge with three different bottom panels including a flexible composite wedge (FC), a flexible aluminum wedge (FA), and also a rigid aluminum wedge (RA). The experiments with varying different drop heights were taken into account to gauge the role of impact velocity.

Results show that the global kinematics are nearly identical for the different wedge models, no matter their flexural rigidity. This is reassuring since steps were taken to ensure that the mass/weight of each model was constant. However, the hydrodynamic loading, i.e. the impact pressure, is tangibly influenced by the flexural rigidity of the bottom panel. The maximum measured peak pressure drops 4% for the FA model, and 10% for the FC model compared to the RA model. The time interval between peak pressure of the sensors also

eventually increases as the flexural rigidity of the wedge decreases. This delay accounts for the deflection of the bottom plate that causes the pressure wave and waterline to travel a longer distance to reach the chines.

Interestingly, S-DIC results show that the FC model experiences some downward deflections at early times of impact due to only a small area of the plate being loaded hydrodynamically. The opposing downward load is the self-weight of the model. This negative deflection exists in the bottom plating prior to  $t = 5$  ms for  $H = 50.8$  cm. After this moment, the hydrodynamic force increases so that it conquers the downward inertia and causes the upward deflection until the maximum deflection is reached. With the decrease of the pressure wave strength, the deflection again decreases and the plate vibrates. It is noted that due to the lower flexural rigidity of the composite material, this behavior was not observed for the aluminum model at the same loading conditions. The comparison between maximum deflection between the two wedges with flexible bottom panels reveals that the maximum deflection for the FC wedge for each drop height is almost 68% higher than that for the FA wedge. This increase relates with the change in flexural rigidity between wedges. The FC wedge has flexural rigidity of  $D = 67.26$  Pa/m<sup>3</sup>, which is 67% smaller than the FA wedge,  $D = 206.2$  Pa/m<sup>3</sup>. When evaluating the structural response of the wedge, it is seen that the deflection increases at a higher rate than it decreases. One possible reason is when the structure returns to its original shape, the wedge bottom is backed up by the water that was not there when the deformation was first occurring.

As was expected, the higher drop height, which corresponds to a higher impact velocity, results in a higher speed for spray-root propagation. Results show that there is a delay between the spray root and the peak pressure propagation. The distance between these two remains almost constant and equal to 6% of the height of the chine,  $z_C$ . More specifically, the peak reaches each pressure sensor when the spray root has already passed that sensor

as predicted by theory. Despite being nearly identical in early stages of penetration, the evolution of the spray root on a flexible wedge lags compared to the rigid one at a critical time when the deformation becomes large enough. In fact, once the deformation becomes important, the spray root of the flexible wedge has to travel a longer distance due to the curvature in the bottom of the wedge. The additional distance seemingly hinders the spray root propagation. The larger the deflection, the longer the delay. Results show that for drop height of  $H = 25.4$  cm, the non-dimensional speed of spray root propagation is reduced up to 4%. Regarding the dimensional spray root propagation, higher impact velocity increases the speed of propagation of spray root for both rigid and flexible wedge, as expected. However, non-dimensional analysis reveals that for the rigid wedge, impact velocity has no significant effect on non-dimensional spray root propagation. On the other hand, as far as the flexible wedge is concerned, impact velocity slightly reduces the non-dimensional speed of spray root propagation. In other words, higher impact velocity leads to higher hydroelasticity effects that influence the non-dimensional spray root propagation at later times during the impact event.

In Stage II, a new hybrid approach was proposed to estimate the hydrodynamic loading and kinematics of the wedge during water entry. This approach incorporates the spray root measurements into the traditional theoretical models. In fact, the spray root location is used as the input for the traditional theoretical models to estimate the hydrodynamic pressure and kinematics. This approach was evaluated by studying free-fall water entry of a flexible wedge with different impact velocities that was presented in Phase I. To have a better comparison with the experimental data, the existing theoretical models are re-derived so that the loss of the velocity during the impact is taken into account in the formulation. The experimental spray root propagation was then used to estimate the pressure distribution via hybrid approach using five different theoretical models. Results were then compared with

the experimental pressure measurements. The comparison showed that the hybrid approach in conjunction with the Armand & Cointe model has the best prediction for pressure distribution. The location and the magnitude of the peaks were predicted within 2% and 7% of error, respectively.

To verify the proposed hybrid approach, the study was repeated for the case study using the traditional theoretical models (pure theory). The computed pressure distribution from traditional approach were then compared with those of the hybrid approach. The comparison demonstrated that the hybrid approach could predict the peak magnitudes better than the pure theory. As an instance, the error of estimating the first peak, corresponding to  $P_1$  for the experiments, could improve from 35% for the Traditional Approach to 10% for the hybrid approach using Generalized Wagner model.

It was also shown that the hybrid approach can appropriately predict the kinematics of the wedge. Acceleration, velocity and the position of the wedge estimated from the Armand & Cointe model were compared with the experimental results that showed a very good agreement.

This proposed hybrid approach will be extended in future to take into account the structural response. The current study was the first step to develop this technique for predicting the hydrodynamics and structural response in water entry of highly-flexible structures. Also, this approach will be used to study the flow field changes due to shape reconfiguration of highly-flexible plates near the free-surface in order to reduce the drag on undulatory propulsors and vessels.

In Stage III, the towing-tank slamming experiments conducted at the U.S. Naval Academy were analyzed. The model was free to heave and pitch and was towed in regular waves. The accelerations at the three locations along the length of the hull were recorded. Two towing

speeds were considered to study the effect of towing speed on slamming characteristics. Several individual slamming events were extracted during the steady-state speed and the above-mentioned measurements were compared.

Results show that for each towing speed, the heave and pitch motions, acceleration data, and pressure results are very repeatable during different slamming events. The average curve of each measurement was reported as the representative of that quantity. The representative measurements are proposed for future use in replicating the slamming events in VT Advanced Towing Tank.

Furthermore, the heave velocity for the two towing speeds was computed from the heave motion. Results reveal that the heave velocity from the two different towing speeds is nearly identical during an individual impact. In fact, the heave velocity is independent of the towing speed and is a function of the incoming waves. In the future, the presented representative average acceleration and motion results recorded for the rigid hull will be used as inputs for the motion-controlled slamming experiments. These experiments will be carried out in the VT Advanced Towing Tank facility to further study the effect of structural response on the slamming characteristics of high-speed planing craft.

The experiments in Stage III were analyzed to motivate the future experiments that will be conducted in the VT Advanced Towing Tank Facility. The future VT experiments shall combine the flexible panel of Stage I with the hull form and motions of Stage II. Controlled-motion experiments at VT will be performed to replicate slamming events observed from the USNA experiments to measure structural response. These two seemingly separate phases analyzed in this dissertation form the basis of the next set of novel experiments that will be conducted.

In Stage IV, a 2D+t approach was proposed based on the wedge water entry experiments in

order to predict the hydrodynamics and motion of a prismatic planing craft in calm water. In this approach, the experimental measurements of the wedge water entry (Stage I) were used in conjunction with the presented hybrid approach (Stage II) to estimate the hydrodynamic loading on 2D V-type cross sections of a prismatic planing hull. Results were then used as the input in a 2D+t approach to predict the resistance, wetted length and trim of a planing craft cruising in calm water. The outcomes were compared with the well-known Savitsky prediction method and available towing tank experiments. It was shown that the proposed 2D+t approach can reasonably predict the wetted length and trim angle of the vessel in different Froude numbers. Results were predicted mostly within 10 percent for the wetted length and trim. However, the 2D+t approach underestimates the resistance of the vessel in lower Froude numbers ( $Fr=1.37$  and  $1.67$ ). For the higher Froude numbers ( $Fr=2.26-3.66$ ), the resistance can be fairly predicted with less than 7 percent of error.

All in all, this dissertation provided a comprehensive study for the spray root propagation as the key element in water-entry studies. In light of that, the effects of flexural rigidity on fluid-structure interactions due to water entry of compliant structures were discussed. Moreover, a hybrid approach was proposed that can adequately predict the hydrodynamic loading and kinematics of a wedge during water entry. The experimental measurements presented in this dissertation along with the hybrid approach could be employed in conjunction with a 2D+t method to predict the hydrodynamics and motions of a prismatic planing craft. This approach can be further studied and evaluated with the future VT towing tank tests on flexible planing hulls.

# Appendices

# Appendix A

## Re-derivation of Theoretical Models

### A.1 Introduction

As discussed in [3.3.1](#), the current theoretical models need to be re-derived to account for the acceleration terms so that the updated formulations can model the physics of the experiment better. This is due to the fact that most of the theoretical models are derived based on the constant-entry assumption while the experiments correspond to free-fall water entry.

### A.2 Re-derivation

Korobkin [\[13\]](#) showed that the hydrodynamic pressure in the water-entry problem can be obtained from:

$$p(x, t) = -\rho \left[ \phi_t + \frac{f_x \dot{z}(t)}{1 + f_x^2} \phi_x + \frac{1}{2} \frac{\phi_x^2 - \dot{z}^2(t)}{1 + f_x^2} \right], \quad (\text{A.1})$$

where  $\phi$  is the velocity potential function that can be estimated employing the asymptotic analysis:

$$\phi(x, t) \approx -\dot{z}(t) \sqrt{x_{sp}^2(t) - x^2} - \dot{z}(t) \left[ f(x) - z(t) - d(t) \right]. \quad (\text{A.2})$$

For the time derivative of the velocity potential function, one can write:

$$\begin{aligned} \phi_t(x, t) = & -\ddot{z}(t)\sqrt{x_{sp}^2(t) - x^2} - \dot{z}(t) \left( x_{sp}(t) \cdot \dot{x}_{sp}(t) [x_{sp}^2(t) - x^2]^{-1/2} \right) \\ & - \ddot{z}(t)[f(x) - z(t) - d(t)] - \dot{z}(t) [-\dot{z}(t) - \dot{d}(t)], \end{aligned} \quad (\text{A.3})$$

that can be simplified as:

$$\begin{aligned} \phi_t(x, t) = & -\ddot{z}(t) \left[ \sqrt{x_{sp}^2(t) - x^2} + f(x) - z(t) - d(t) \right] \\ & - \dot{z}(t) \left[ \frac{x_{sp}(t) \cdot \dot{x}_{sp}(t)}{\sqrt{x_{sp}^2(t) - x^2}} - \dot{z}(t) - \dot{d}(t) \right]. \end{aligned} \quad (\text{A.4})$$

For the derivative of the velocity potential with respect to  $x$ , the  $\phi_x$  can be easily derived:

$$\phi_x(x, t) = \dot{z}(t) \left[ \frac{x}{\sqrt{x_{sp}^2(t) - x^2}} - f_x \right]. \quad (\text{A.5})$$

Having the derivatives of the velocity potential function from Eq. A.4 and A.5, the hydrodynamic pressure distribution can be obtained:

$$p(x, t) = -\rho \left[ \underbrace{\phi_t}_{T1} + \underbrace{\frac{f_x \dot{z}(t)}{1 + f_x^2} \phi_x}_{T2} + \underbrace{\frac{1}{2} \frac{\phi_x^2 - \dot{z}^2(t)}{1 + f_x^2}}_{T3} \right]. \quad (\text{A.6})$$

The three terms  $T1$ ,  $T2$  and  $T3$  in the Eq. A.6 can be defined using the derivatives:

$$T1 = -\ddot{z}(t) \left[ \sqrt{x_{sp}^2(t) - x^2} + f(x) - z(t) - d(t) \right] - \dot{z}(t) \left[ \frac{x_{sp}(t) \cdot \dot{x}_{sp}(t)}{\sqrt{x_{sp}^2(t) - x^2}} - \dot{z}(t) - \dot{d}(t) \right], \quad (\text{A.7})$$

$$T2 = \frac{f_x \dot{z}^2(t)}{1 + f_x^2} \left[ \frac{x}{\sqrt{x_{sp}^2(t) - x^2}} - f_x \right], \quad (\text{A.8})$$

and

$$T3 = \frac{1}{2} \frac{\dot{z}^2(t)}{1 + f_x^2} \left[ \frac{x^2}{x_{sp}^2(t) - x^2} - f_x^2 - \frac{2 x f_x}{\sqrt{x_{sp}^2(t) - x^2}} - \dot{z}^2(t) \right]. \quad (\text{A.9})$$

Substituting Eqs. A.7 to A.9 in Eq. A.6, the hydrodynamic pressure can be obtained as:

$$p(x, t) = \rho \dot{z}^2(t) \left\{ \frac{\dot{x}_{sp}(t)}{\dot{z}(t)} \frac{x_{sp}(t)}{\sqrt{x_{sp}^2(t) - x^2}} - \frac{1}{2} \frac{1}{1 + f_x^2} \left[ f_x^2 + \frac{x^2}{x_{sp}^2(t) - x^2} + 1 \right] - \frac{\dot{d}(t)}{\dot{z}(t)} \right. \\ \left. + \frac{\ddot{z}(t)}{\dot{z}^2(t)} \left[ \sqrt{x_{sp}^2(t) - x^2} + f(x) - z(t) - d(t) \right] \right\}, \quad (\text{A.10})$$

which after simplification will be:

$$p(x, t) = \rho \dot{z}^2(t) \left\{ \frac{\dot{x}_{sp}(t)}{\dot{z}(t)} \frac{x_{sp}(t)}{\sqrt{x_{sp}^2(t) - x^2}} - \frac{1}{2} \frac{1}{1 + f_x^2} \frac{x_{sp}^2}{x_{sp}^2(t) - x^2} - \frac{1}{2} \frac{f_x^2}{1 + f_x^2} - \frac{\dot{d}(t)}{\dot{z}(t)} \right. \\ \left. + \frac{\ddot{z}(t)}{\dot{z}^2(t)} \left[ \sqrt{x_{sp}^2(t) - x^2} + f(x) - z(t) - d(t) \right] \right\}. \quad (\text{A.11})$$

The general formulation for hydrodynamic pressure expressed in Eq. A.11 can be used to obtain the pressure distribution for different theoretical models based on the main assumption of each model (as was elaborated in 3.3.1).

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